CHAPTER

8

INTERNAL FORCED CONVECTION

Liquid or gas flow through pipes or ducts is commonly used in heating and cooling applications. The fluid in such applications is forced to flow by a fan or pump through a flow section that is sufficiently long to accomplish the desired heat transfer. In this chapter, we pay particular attention to the determination of the *friction factor* and *convection coefficient* since they are directly related to the *pressure drop* and *heat transfer rate,* respectively. These quantities are then used to determine the pumping power requirement and the required tube length.

There is a fundamental difference between external and internal flows. In *external flow,* considered in Chap. 7, the fluid has a free surface, and thus the boundary layer over the surface is free to grow indefinitely. In *internal flow,* however, the fluid is completely confined by the inner surfaces of the tube, and thus there is a limit on how much the boundary layer can grow.

We start this chapter with a general physical description of internal flow, and the *average velocity* and *average temperature.* We continue with the discussion of the *hydrodynamic* and *thermal entry lengths, developing flow,* and *fully developed flow.* We then obtain the velocity and temperature profiles for fully developed laminar flow and develop relations for the friction factor and Nusselt number. Finally, we present empirical relations for developing and fully developed flows and demonstrate their use.

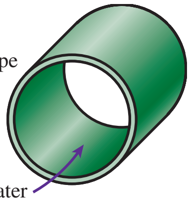
OBJECTIVES

When you finish studying this chapter, you should be able to:

- Obtain average velocity from a knowledge of velocity profile, and average temperature from a knowledge of temperature profile in internal flow,
- Have a visual understanding of different flow regions in internal flow, such as the entry and the fully developed flow regions, and calculate hydrodynamic and thermal entry lengths,
- Analyze heating and cooling of a fluid flowing in a tube under constant surface temperature and constant surface heat flux conditions, and work with the logarithmic mean temperature difference,
- Obtain analytic relations for the velocity profile, pressure drop, friction factor, and Nusselt number in fully developed laminar flow, and
- Determine the friction factor and Nusselt number in fully developed turbulent flow using empirical relations, and calculate the pressure drop and heat transfer rate.

8–1 ▪ INTRODUCTION

The terms *pipe, duct,* and *conduit* are usually used interchangeably for flow sections. In general, flow sections of circular cross section are referred to as *pipes* (especially when the fluid is a liquid) and flow sections of noncircular cross section as *ducts* (especially when the fluid is a gas). Small-diameter pipes are usually referred to as *tubes*. Given this uncertainty, we will use more descriptive phrases (such as *a circular pipe* or *a rectangular duct*) whenever necessary to avoid any misunderstandings.



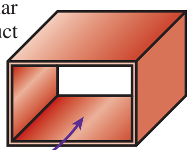


FIGURE 8–1
Circular pipes can withstand large pressure differences between the inside and the outside without undergoing any significant distortion, but noncircular pipes cannot.

You have probably noticed that most fluids, especially liquids, are transported in *circular pipes*. This is because pipes with a circular cross section can withstand large pressure differences between the inside and the outside without undergoing significant distortion. *Noncircular pipes* are usually used in applications such as the heating and cooling systems of buildings where the pressure difference is relatively small, the manufacturing and installation costs are lower, and the available space is limited for ductwork (Fig. 8–1). For a fixed surface area, the circular tube gives the most heat transfer for the least pressure drop, which explains the overwhelming popularity of circular tubes in heat transfer equipment.

Although the theory of fluid flow is reasonably well understood, theoretical solutions are obtained only for a few simple cases such as fully developed laminar flow in a circular pipe. Therefore, we must rely on experimental results and empirical relations for most fluid flow problems rather than closed-form analytical solutions. Noting that the experimental results are obtained under carefully controlled laboratory conditions and that no two systems are exactly alike, we must not be so naive as to view the results obtained as exact. An error of 10 percent (or more) in friction factors calculated using the relations in this chapter is the norm rather than the exception.

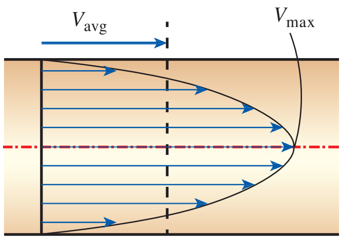


FIGURE 8–2
Average velocity V_{avg} is defined as the average speed through a cross section. For fully developed laminar pipe flow, V_{avg} is half of the maximum velocity.

The fluid velocity in a pipe changes from *zero* at the wall because of the no-slip condition to a maximum at the pipe center. In fluid flow, it is convenient to work with an *average* velocity V_{avg}, which remains constant in incompressible flow when the cross-sectional area of the pipe is constant (Fig. 8–2). The average velocity in heating and cooling applications may change somewhat because of changes in density with temperature. But in practice we evaluate the fluid properties at some average temperature and treat them as constants. The convenience of working with constant properties usually more than justifies the slight loss in accuracy.

Also, the friction between the fluid particles in a pipe does cause a slight rise in fluid temperature as a result of the mechanical energy being converted to sensible thermal energy. But this temperature rise due to *frictional heating* is usually too small to warrant any consideration in calculations and thus is disregarded. For example, in the absence of any heat transfer, no noticeable difference can be detected between the inlet and outlet temperatures of water flowing in a pipe. The primary consequence of friction in fluid flow is pressure drop, and thus any significant temperature change in the fluid is due to heat transfer. But frictional heating must be considered for flows that involve highly viscous fluids with large velocity gradients.

8–2 ▪ AVERAGE VELOCITY AND TEMPERATURE

In external flow, the free-stream velocity served as a convenient reference velocity for use in the evaluation of the Reynolds number and the friction coefficient. In internal flow, there is no free stream, and thus we need an alternative. The fluid velocity in a tube changes from zero at the surface because of the no-slip condition to a maximum at the tube center. Therefore, it is convenient to work with an **average** or **mean velocity** V_{avg}, which remains constant for incompressible flow when the cross-sectional area of the tube is constant.

The value of the average velocity V_{avg} at some streamwise cross section is determined from the requirement that the *conservation of mass* principle be satisfied (Fig. 8–2). That is,

$$\dot{m} = \rho V_{avg} A_c = \int_{A_c} \rho u(r)\, dA_c \tag{8–1}$$

where $\dot{m}$ is the mass flow rate, ρ is the density, A_c is the cross-sectional area, and $u(r)$ is the velocity profile. Then, the average velocity for incompressible flow in a circular pipe of radius R is expressed as

$$V_{avg} = \frac{\int_{A_c} \rho u(r)\, dA_c}{\rho A_c} = \frac{\int_0^R \rho u(r) 2\pi r\, dr}{\rho \pi R^2} = \frac{2}{R^2}\int_0^R u(r) r\, dr \tag{8–2}$$

Therefore, when we know the flow rate or the velocity profile, the average velocity can be determined easily.

When a fluid is heated or cooled as it flows through a tube, the temperature of the fluid at any cross section changes from T_s at the surface of the wall to some maximum (or minimum in the case of heating) at the tube center. In fluid flow, it is convenient to work with an **average** or **mean temperature** T_m, which remains constant at a cross section. Unlike the mean velocity, the mean temperature T_m *changes* in the flow direction whenever the fluid is heated or cooled.

The value of the mean temperature T_m is determined from the requirement that the *conservation of energy* principle be satisfied. That is, the energy transported by the fluid through a cross section in actual flow must be equal to the energy that would be transported through the same cross section if the fluid were at a constant temperature T_m. This can be expressed mathematically as (Fig. 8–3)

$$\dot{E}_{fluid} = \dot{m} c_p T_m = \int_{\dot{m}} c_p T(r) \delta \dot{m} = \int_{A_c} \rho c_p T(r) u(r) dA_c \tag{8–3}$$

where c_p is the specific heat of the fluid. Note that the product $\dot{m} c_p T_m$ at any cross section along the tube represents the *energy flow* with the fluid at that cross section. Then, the mean temperature of a fluid with constant density and specific heat flowing in a circular pipe of radius R can be expressed as

$$T_m = \frac{\int_{\dot{m}} c_p T(r) \delta \dot{m}}{\dot{m} c_p} = \frac{\int_0^R c_p T(r) \rho u(r) 2\pi r dr}{\rho V_{avg}(\pi R^2) c_p} = \frac{2}{V_{avg} R^2}\int_0^R T(r) u(r)\, r dr \tag{8–4}$$

Note that the mean temperature T_m of a fluid changes during heating or cooling. Also, the fluid properties in internal flow are usually evaluated at the *bulk*

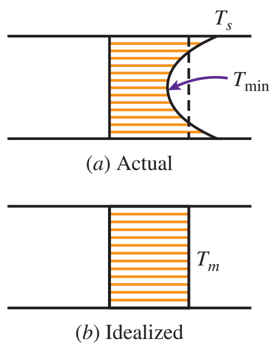


FIGURE 8–3
Actual and idealized temperature profiles for flow in a tube (the rate at which energy is transported with the fluid is the same for both cases).

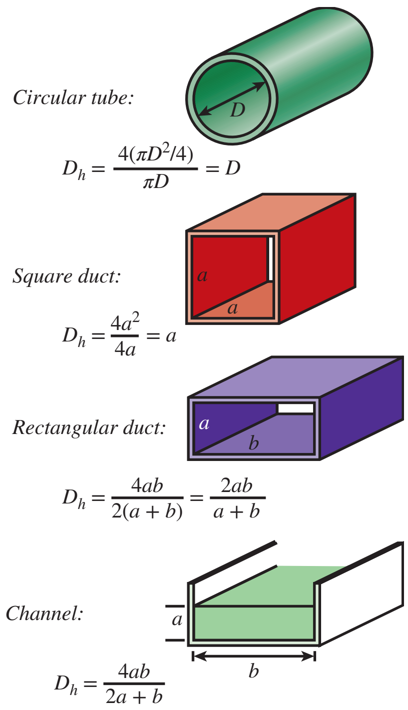


FIGURE 8–4
The hydraulic diameter $D_h = 4A_c/p$ is defined such that it reduces to ordinary diameter for circular tubes. When there is a free surface, such as in open-channel flow, the wetted perimeter includes only the walls in contact with the fluid.

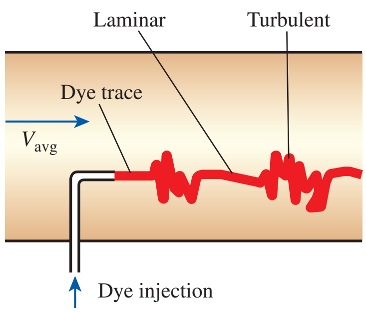


FIGURE 8–5
In the transitional flow region of the flow switches between laminar and turbulent somewhat randomly.

mean fluid temperature, which is the arithmetic average of the mean temperatures at the inlet and the exit. That is, $T_b = (T_i + T_e)/2$.

Laminar and Turbulent Flow in Tubes

Flow in a tube can be laminar or turbulent, depending on the flow conditions. Fluid flow is streamlined and thus laminar at low velocities, but turns turbulent as the velocity is increased beyond a critical value. Transition from laminar to turbulent flow does not occur suddenly; rather, it occurs over some range of velocity where the flow fluctuates between laminar and turbulent flows before it becomes fully turbulent. Most pipe flows encountered in practice are turbulent. Laminar flow is encountered when highly viscous fluids such as oils flow in small-diameter tubes or narrow passages.

For flow in a circular tube, the Reynolds number is defined as

$$\text{Re} = \frac{V_{avg}D}{\nu} = \frac{\rho V_{avg} D}{\mu} = \frac{\rho D}{\mu}\left(\frac{\dot m}{\rho \pi D^2/4}\right) = \frac{4\dot m}{\mu \pi D} \tag{8–5}$$

where V_{avg} is the average flow velocity, D is the diameter of the tube, and $\nu = \mu/\rho$ is the kinematic viscosity of the fluid.

For flow through noncircular tubes, the Reynolds number as well as the Nusselt number, and the friction factor are based on the **hydraulic diameter** D_h defined as (Fig. 8–4)

$$D_h = \frac{4A_c}{p} \tag{8–6}$$

where A_c is the cross-sectional area of the tube and p is its wetted perimeter. The hydraulic diameter is defined such that it reduces to ordinary diameter D for circular tubes since

Circular tubes: $$D_h = \frac{4A_c}{p} = \frac{4\pi D^2/4}{\pi D} = D$$

It certainly is desirable to have precise values of Reynolds numbers for laminar, transitional, and turbulent flows, but this is not the case in practice. This is because the transition from laminar to turbulent flow also depends on the degree of disturbance of the flow by *surface roughness, pipe vibrations,* and the *fluctuations in the flow.* Under most practical conditions, the flow in a tube is laminar for $\text{Re} < 2300$, fully turbulent for $\text{Re} > 10{,}000$, and transitional in between. But it should be kept in mind that in many cases the flow becomes fully turbulent for $\text{Re} > 4000$, as discussed in the Topic of Special Interest later in this chapter. When designing piping networks and determining pumping power, a conservative approach is taken, and flows with $\text{Re} > 4000$ are assumed to be turbulent.

In transitional flow, the flow switches between laminar and turbulent in a disorderly fashion (Fig. 8–5). It should be kept in mind that laminar flow can be maintained at much higher Reynolds numbers in very smooth pipes by avoiding flow disturbances and tube vibrations. In such carefully controlled experiments, laminar flow has been maintained at Reynolds numbers of up to 100,000.

8–3 ▪ THE ENTRANCE REGION

Consider a fluid entering a circular pipe at a uniform velocity. Because of the no-slip condition, the fluid particles in the layer in contact with the wall of the pipe come to a complete stop. This layer also causes the fluid particles in the adjacent layers to slow down gradually as a result of friction. To make up for this velocity reduction, the velocity of the fluid at the midsection of the pipe has to increase to keep the mass flow rate through the pipe constant. As a result, a velocity gradient develops along the pipe.

The region of the flow in which the effects of the viscous shearing forces caused by fluid viscosity are felt is called the **velocity boundary layer** or just the **boundary layer**. The hypothetical boundary surface divides the flow in a pipe into two regions: the **boundary layer region**, in which the viscous effects and the velocity changes are significant, and the **irrotational** (**core**) **flow region**, in which the frictional effects are negligible and the velocity remains essentially constant in the radial direction.

The thickness of this boundary layer increases in the flow direction until the boundary layer reaches the pipe center and thus fills the entire pipe, as shown in Fig. 8–6, and the velocity becomes fully developed a little further downstream. The region from the pipe inlet to the point at which the velocity profile is fully developed is called the **hydrodynamic entrance region**, and the length of this region is called the **hydrodynamic entry length** L_h. Flow in the entrance region is called *hydrodynamically developing flow* since this is the region where the velocity profile develops. The region beyond the entrance region in which the velocity profile is fully developed and remains unchanged is called the **hydrodynamically fully developed region**. The velocity profile in the fully developed region is *parabolic* in laminar flow and somewhat *flatter* or fuller in turbulent flow due to eddy motion and more vigorous mixing in a radial direction.

Now consider a fluid at a uniform temperature entering a circular tube whose surface is maintained at a different temperature. This time, the fluid particles in the layer in contact with the surface of the tube assume the surface temperature. This initiates convection heat transfer in the tube and the development of a *thermal boundary layer* along the tube. The thickness of this boundary layer also increases in the flow direction until the boundary layer reaches the tube center and thus fills the entire tube, as shown in Fig. 8–7.

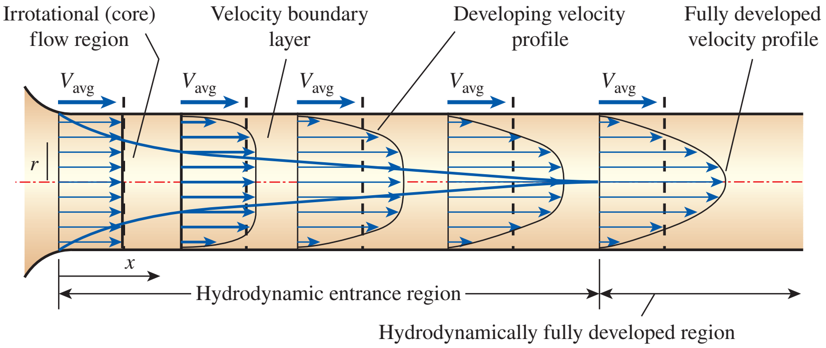


FIGURE 8–6
The development of the velocity boundary layer in a pipe. (The developed average velocity profile is parabolic in laminar flow, as shown, but much flatter or fuller in turbulent flow.)

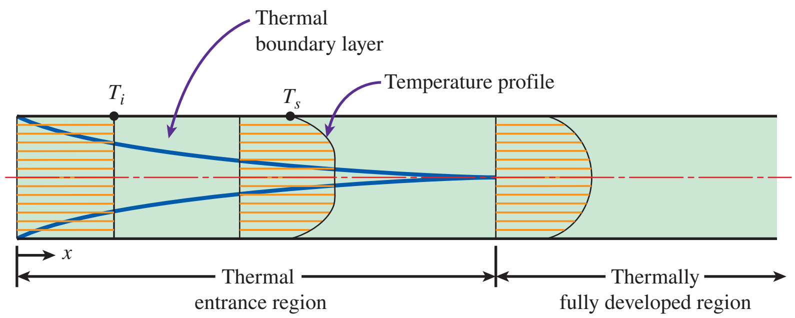


FIGURE 8–7
The development of the thermal boundary layer in a tube. (The fluid in the tube is being cooled.)

The region of flow over which the thermal boundary layer develops and reaches the tube center is called the **thermal entrance region**, and the length of this region is called the **thermal entry length** L_t. Flow in the thermal entrance region is called *thermally developing flow* since this is the region where the temperature profile develops. The region beyond the thermal entrance region in which the dimensionless temperature profile expressed as $(T_s - T)/(T_s - T_m)$ remains unchanged is called the **thermally fully developed region**. The region in which the flow is both hydrodynamically and thermally developed and thus both the velocity and dimensionless temperature profiles remain unchanged is called *fully developed flow.* That is,

Hydrodynamically fully developed: $$\frac{\partial u(r, x)}{\partial x} = 0 \longrightarrow u = u(r) \quad \textbf{(8–7)}$$

Thermally fully developed: $$\frac{\partial}{\partial x}\left[\frac{T_s(x) - T(r, x)}{T_s(x) - T_m(x)}\right] = 0 \quad \textbf{(8–8)}$$

The shear stress at the tube wall τ_w is related to the slope of the velocity profile at the surface. Noting that the velocity profile remains unchanged in the hydrodynamically fully developed region, the wall shear stress also remains constant in that region. A similar argument can be given for the heat transfer coefficient in the thermally fully developed region.

In a thermally fully developed region, the derivative of $(T_s - T)/(T_s - T_m)$ with respect to x is zero by definition, and thus $(T_s - T)/(T_s - T_m)$ is independent of x. Then, the derivative of $(T_s - T)/(T_s - T_m)$ with respect to r must also be independent of x. That is,

$$\frac{\partial}{\partial r}\left(\frac{T_s - T}{T_s - T_{\rm m}}\right)\bigg|_{r=R} = \frac{-(\partial T/\partial r)|_{r=R}}{T_s - T_{\rm m}} \neq f(x) \quad \textbf{(8–9)}$$

Surface heat flux can be expressed as

$$\dot{q}_s = h_x(T_s - T_m) = k\frac{\partial T}{\partial r}\bigg|_{r=R} \longrightarrow h_x = \frac{k(\partial T/\partial r)|_{r=R}}{T_s - T_m} \quad \textbf{(8–10)}$$

which, from Eq. 8–9, is independent of x. Thus, we conclude that *in the thermally fully developed region of a tube, the local convection coefficient is constant* (does not vary with x). Therefore, both *the local friction factor* f_x (*which is related to the local wall shear stress*) and the local convection coefficient h_x remain constant in the hydrodynamically and thermally fully developed regions, respectively, as shown in Fig. 8–8 for Pr > 1.

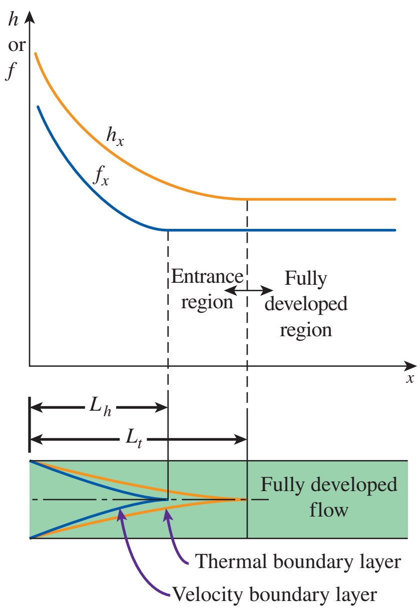


FIGURE 8–8
Variation of the friction factor and the convection heat transfer coefficient in the flow direction for flow in a tube (Pr > 1).

Note that the *temperature profile* in the thermally fully developed region may vary with x in the flow direction. That is, unlike the velocity profile, the temperature profile can be different at different cross sections of the tube in the developed region, and it usually is. However, the dimensionless temperature profile defined previously remains unchanged in the thermally developed region when the temperature or heat flux at the tube surface remains constant.

During laminar flow in a tube, the magnitude of the dimensionless Prandtl number Pr is a measure of the relative growth of the velocity and thermal boundary layers. For fluids with $\text{Pr} \approx 1$, such as gases, the two boundary layers essentially coincide with each other. For fluids with $\text{Pr} \gg 1$, such as oils, the velocity boundary layer outgrows the thermal boundary layer. As a result, the hydrodynamic entry length is smaller than the thermal entry length. The opposite is true for fluids with $\text{Pr} \ll 1$ such as liquid metals.

Consider a fluid that is being heated (or cooled) in a tube as it flows through it. The wall shear stress and the heat transfer coefficient are *highest* at the tube inlet where the thickness of the boundary layers is smallest, and they decrease gradually to the fully developed values, as shown in Fig. 8–8. Therefore, the pressure drop and heat flux are *higher* in the entrance regions of a tube, and the effect of the entrance region is always to *increase* the average friction factor and heat transfer coefficient for the entire tube. This enhancement can be significant for short tubes but negligible for long ones.

Entry Lengths

The hydrodynamic entry length is usually taken to be the distance from the tube entrance where the wall shear stress (and thus the friction factor) reaches within about 2 percent of the fully developed value. In *laminar flow,* the hydrodynamic and thermal entry lengths are given approximately as [see Kays and Crawford (1993) and Shah and Bhatti (1987)]

$$L_{h,\ \text{laminar}} \approx 0.05\ \text{Re}\ D \tag{8–11}$$

$$L_{t,\ \text{laminar}} \approx 0.05\ \text{Re}\ \text{Pr}\ D = \text{Pr}\ L_{h,\ \text{laminar}} \tag{8–12}$$

For $\text{Re} = 20$, the hydrodynamic entry length is about the size of the diameter, but it increases linearly with velocity. In the limiting case of $\text{Re} = 2300$, the hydrodynamic entry length is $115D$. When the flow is simultaneously hydrodynamically and thermally developing, a longer thermal entrance length is required than for hydrodynamically fully developed flow. Meyer and Everts (2018a) suggest replacing the coefficient of 0.05 used in Eq. 8–12 with 0.12 for simultaneously hydrodynamically and thermally developing forced convection laminar flow.

In *turbulent flow,* the intense mixing during random fluctuations usually overshadows the effects of molecular diffusion, and therefore the hydrodynamic and thermal entry lengths are of about the same size and independent of the Prandtl number.

The entry length is much shorter in turbulent flow, as expected, and its dependence on the Reynolds number is weaker. In many tube flows of practical interest, the entrance effects become insignificant beyond a tube length of 10 diameters, and the hydrodynamic and thermal entry lengths are approximately taken to be

$$L_{h,\ \text{turbulent}} \approx L_{t,\ \text{turbulent}} \approx 10D \tag{8–13}$$

The variation of local Nusselt number along a tube in turbulent flow for both constant surface temperature and constant surface heat flux is given

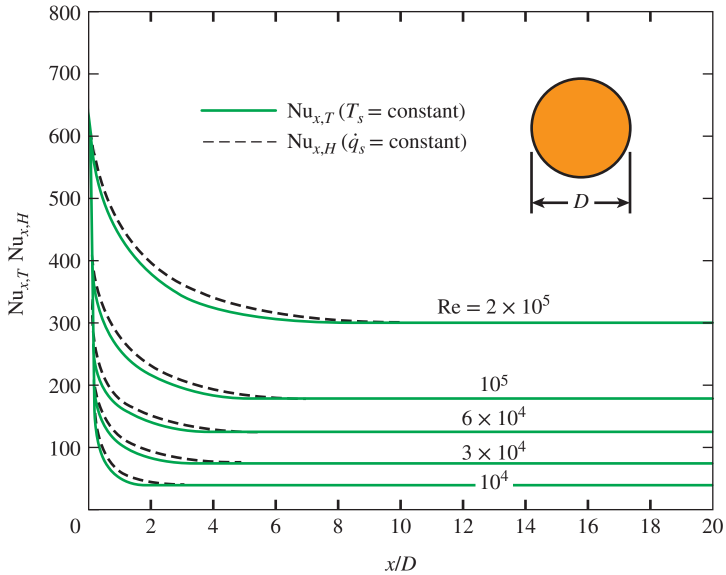


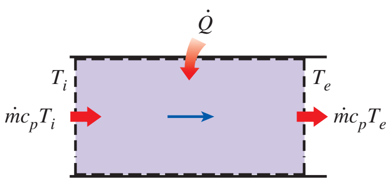
FIGURE 8–9
Variation of local Nusselt number along a tube in turbulent flow for both constant surface temperature and constant surface heat flux
Source: Deissler, 1953.

in Fig. 8–9 for the range of Reynolds numbers encountered in heat transfer equipment. We make these important observations from this figure:

- The Nusselt numbers and thus the convection heat transfer coefficients are much higher in the entrance region.
- The Nusselt number reaches a constant value at a distance of less than 10 diameters, and thus the flow can be assumed to be fully developed for $x > 10D$.
- The Nusselt numbers for the constant surface temperature and constant surface heat flux conditions are identical in the fully developed regions and nearly identical in the entrance regions. Therefore, Nusselt number is insensitive to the type of thermal boundary condition, and the turbulent flow correlations can be used for either type of boundary condition.

Precise correlations for the friction and heat transfer coefficients for the entrance regions are available in the literature. However, the tubes used in practice in forced convection are usually several times the length of either entrance region, and thus the flow through the tubes is often assumed to be fully developed for the entire length of the tube. This simplistic approach gives *reasonable* results for the rate of heat transfer for long tubes and *conservative* results for short ones.

It should be noted that the above observations are only valid for turbulent flow. In laminar flow Nusselt number values are much lower than turbulent flow values, the distance for the Nusselt number to reach a constant value is typically much longer, and the flow is sensitive to the thermal boundary conditions imposed on the flow.



Energy balance:
$\dot{Q} = \dot{m}c_p(T_e - T_i)$

FIGURE 8–10
The heat transfer to a fluid flowing in a tube is equal to the increase in the energy of the fluid.

8–4 ▪ GENERAL THERMAL ANALYSIS

In the absence of any work interactions (such as electric resistance heating), the conservation of energy equation for the steady flow of a fluid in a tube can be expressed as (Fig. 8–10)

$$\dot{Q} = \dot{m}c_p(T_e - T_i) \quad \text{(W)} \tag{8–14}$$

where T_i and T_e are the mean fluid temperatures at the inlet and exit of the tube, respectively, and $\dot{Q}$ is the rate of heat transfer to or from the fluid. Note that the temperature of a fluid flowing in a tube remains constant in the absence of any energy interactions through the wall of the tube.

The thermal conditions at the surface can usually be approximated with reasonable accuracy to be *constant surface temperature* (T_s = constant) or *constant surface heat flux* ($\dot{q}_s$ = constant). For example, the constant surface temperature condition is realized when a phase change process such as boiling or condensation occurs at the outer surface of a tube. The constant surface heat flux condition is realized when the tube is subjected to radiation or electric resistance heating uniformly from all directions.

Surface heat flux is expressed as

$$\dot{q}_s = h_x(T_s - T_m) \qquad (\text{W/m}^2) \tag{8–15}$$

where h_x is the *local* heat transfer coefficient and T_s and T_m are the surface and the mean fluid temperatures at that location. Note that the mean fluid temperature T_m of a fluid flowing in a tube must change during heating or cooling. Therefore, when $h_x = h$ = constant, the surface temperature T_s must change when $\dot{q}_s$ = constant, and the surface heat flux $\dot{q}_s$ must change when T_s = constant. Thus we may have either T_s = constant or $\dot{q}_s$ = constant at the surface of a tube, but not both. Next we consider convection heat transfer for these two common cases.

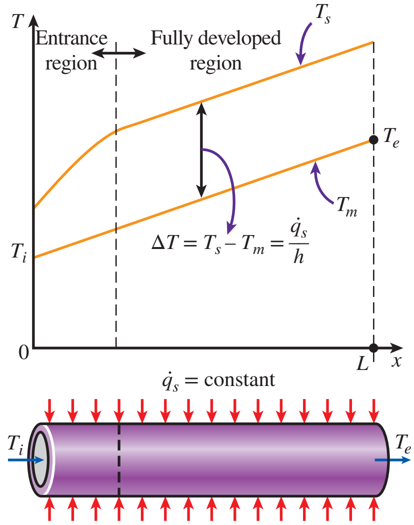


FIGURE 8–11
Variation of the *tube surface* and the *mean fluid* temperatures along the tube for the case of constant surface heat flux.

Constant Surface Heat Flux ($\dot{q}_s$ = constant)

In the case of $\dot{q}_s$ = constant, the rate of heat transfer can also be expressed as

$$\dot{Q} = \dot{q}_s A_s = \dot{m}c_p(T_e - T_i) \qquad (\text{W}) \tag{8–16}$$

Then the mean fluid temperature at the tube exit becomes

$$T_e = T_i + \frac{\dot{q}_s A_s}{\dot{m}c_p} \tag{8–17}$$

Note that the mean fluid temperature increases *linearly* in the flow direction in the case of constant surface heat flux, since the surface area increases linearly in the flow direction (A_s is equal to the perimeter, which is constant, times the tube length).

The surface temperature in the case of constant surface heat flux $\dot{q}_s$ can be determined from

$$\dot{q}_s = h(T_s - T_m) \quad \longrightarrow \quad T_s = T_m + \frac{\dot{q}_s}{h} \tag{8–18}$$

In the fully developed region, the surface temperature T_s will also increase linearly in the flow direction since h is constant and thus $T_s - T_m$ = constant (Fig. 8–11). Of course this is true when the fluid properties remain constant during flow.

The slope of the mean fluid temperature T_m on a T-x diagram can be determined by applying the steady-flow energy balance to a tube slice of thickness dx shown in Fig. 8–12. It gives

$$\dot{m}c_p\,dT_m = \dot{q}_s(p\,dx) \quad \longrightarrow \quad \frac{dT_m}{dx} = \frac{\dot{q}_s p}{\dot{m}c_p} = \text{constant} \tag{8–19}$$

where p is the perimeter of the tube.

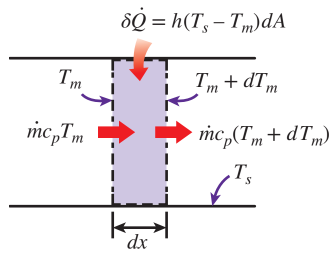


FIGURE 8–12
Energy interactions for a differential control volume in a tube.

Noting that both $\dot{q}_s$ and h are constants, the differentiation of Eq. 8–18 with respect to x gives

$$\frac{dT_m}{dx} = \frac{dT_s}{dx} \tag{8–20}$$

Also, the requirement that the dimensionless temperature profile remains unchanged in the fully developed region gives

$$\frac{\partial}{\partial x}\left(\frac{T_s - T}{T_s - T_m}\right) = 0 \quad \longrightarrow \quad \frac{1}{T_s - T_m}\left(\frac{\partial T_s}{\partial x} - \frac{\partial T}{\partial x}\right) = 0 \quad \longrightarrow \quad \frac{\partial T}{\partial x} = \frac{dT_s}{dx} \tag{8–21}$$

since $T_s - T_m =$ constant. Combining Eqs. 8–19, 8–20, and 8–21 gives

$$\frac{\partial T}{\partial x} = \frac{dT_s}{dx} = \frac{dT_m}{dx} = \frac{\dot{q}_s p}{\dot{m} c_p} = \text{constant} \tag{8–22}$$

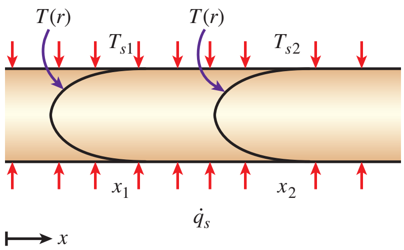


FIGURE 8–13
The shape of the temperature profile remains unchanged in the fully developed region of a tube subjected to constant surface heat flux.

Then we conclude that *in fully developed flow in a tube subjected to constant surface heat flux, the temperature gradient is independent of x, and thus the shape of the temperature profile does not change along the tube* (Fig. 8–13).

Integrating Eq. 8–22 from $x = 0$ (tube inlet where $T_m = T_i$) we obtain an expression for the variation of mean temperature along the tube

$$T_m = T_i + \frac{\dot{q}_s p}{\dot{m} c_p} x \tag{8–23}$$

We evaluate this equation at $x = L$ (tube exit where $T_m = T_e$). Then when we recognize that $A_s = pL$, Eq. 8–17 is obtained. From Eq. 8–23 we can conclude again that the mean temperature varies linearly with x along the tube for the case of constant heat flux.

For a circular tube, $p = 2\pi R$ and $\dot{m} = \rho V_{avg} A_c = \rho V_{avg}(\pi R^2)$. Then Eq. 8–22 becomes

Circular tube:
$$\frac{\partial T}{\partial x} = \frac{dT_s}{dx} = \frac{dT_m}{dx} = \frac{2\dot{q}_s}{\rho V_{avg} c_p R} = \text{constant} \tag{8–24}$$

where V_{avg} is the mean velocity of the fluid.

Constant Surface Temperature (T_s = constant)

From Newton's law of cooling, the rate of heat transfer to or from a fluid flowing in a tube can be expressed as

$$\dot{Q} = hA_s \Delta T_{avg} = hA_s(T_s - T_m)_{avg} \qquad \text{(W)} \tag{8–25}$$

where h is the average convection heat transfer coefficient, A_s is the heat transfer surface area (it is equal to πDL for a circular pipe of length L), and ΔT_{avg} is some appropriate *average* temperature difference between the fluid and the surface. Below we discuss two suitable ways of expressing ΔT_{avg}.

In the constant surface temperature (T_s = constant) case, ΔT_{avg} can be expressed *approximately* by the **arithmetic mean temperature difference** ΔT_{am} as

$$\Delta T_{avg} \approx \Delta T_{am} = \frac{\Delta T_i + \Delta T_e}{2} = \frac{(T_s - T_i) + (T_s - T_e)}{2} = T_s - \frac{T_i + T_e}{2}$$
$$= T_s - T_b \tag{8–26}$$

where $T_b = (T_i + T_e)/2$ is the *bulk mean fluid temperature,* which is the *arithmetic average* of the mean fluid temperatures at the inlet and the exit of the tube.

Note that the *arithmetic mean temperature difference* ΔT_{am} is simply the *average* of the *temperature differences* between the surface and the fluid at the inlet and the exit of the tube. Inherent in this definition is the assumption that the mean fluid temperature varies linearly along the tube, which is hardly ever the case when $T_s =$ constant. This simple approximation often gives acceptable results, but not always. Therefore, we need a better way to evaluate ΔT_{avg}.

Consider the heating of a fluid in a tube of constant cross section whose inner surface is maintained at a constant temperature of T_s. We know that the mean temperature of the fluid T_m increases in the flow direction as a result of heat transfer. The energy balance on a differential control volume shown in Fig. 8–12 gives

$$\dot{m}c_p\, dT_m = h(T_s - T_m)dA_s \tag{8–27}$$

That is, the increase in the energy of the fluid (represented by an increase in its mean temperature by dT_m) is equal to the heat transferred to the fluid from the tube surface by convection. Noting that the differential surface area is $dA_s = pdx$, where p is the perimeter of the tube, and that $dT_m = -d(T_s - T_m)$, since T_s is constant, the relation above can be rearranged as

$$\frac{d(T_s - T_m)}{T_s - T_m} = -\frac{hp}{\dot{m}c_p}dx \tag{8–28}$$

Integrating from $x = 0$ (tube inlet where $T_m = T_i$) to $x = L$ (tube exit where $T_m = T_e$) gives

$$\ln\frac{T_s - T_e}{T_s - T_i} = -\frac{hA_s}{\dot{m}c_p} \tag{8–29}$$

where $A_s = pL$ is the surface area of the tube and h is the constant *average* convection heat transfer coefficient. Taking the exponential of both sides and solving for T_e gives the following relation, which is very useful for the determination of the *mean fluid temperature at the tube exit:*

$$T_e = T_s - (T_s - T_i)\exp(-hA_s/\dot{m}c_p) \tag{8–30}$$

This relation can also be used to determine the mean fluid temperature $T_m(x)$ at any x by replacing $A_s = pL$ with px.

Note that the temperature difference between the fluid and the surface *decays exponentially* in the flow direction, and the rate of decay depends on the magnitude of the exponent $hA_s/\dot{m}c_p$, as shown in Fig. 8–14. This dimensionless parameter is called the *number of transfer units,* denoted by NTU, and is a measure of the effectiveness of the heat transfer systems. For NTU > 5, the exit temperature of the fluid becomes almost equal to the surface temperature, $T_e \approx T_s$ (Fig. 8–15). Noting that the fluid temperature can approach the surface temperature but cannot cross it, an NTU of about 5 indicates that the limit is reached for heat transfer, and the heat transfer does not increase no matter how much we extend the length of the tube. A small value of NTU, on the other hand, indicates more opportunities for heat transfer, and the heat transfer continues to increase as the tube length is increased. A large NTU and thus a large heat transfer surface area (which means a large tube) may be desirable from a heat transfer point of view, but it may be unacceptable from an economic point of view. The selection of heat transfer equipment usually reflects a compromise between heat transfer performance and cost.

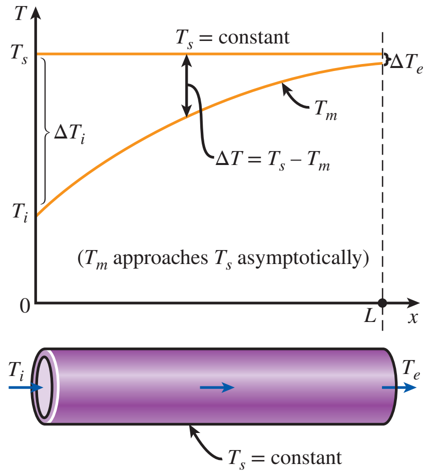
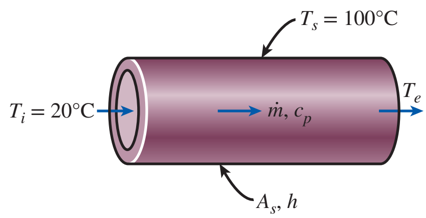


FIGURE 8–14
The variation of the *mean fluid* temperature along the tube for the case of constant temperature.



NTU = $hA_s/\dot{m}c_p$	T_e, °C
0.01	20.8
0.05	23.9
0.10	27.6
0.50	51.5
1.00	70.6
5.00	99.5
10.00	100.0

FIGURE 8–15
An NTU greater than 5 indicates that the fluid flowing in a tube will reach the surface temperature at the exit regardless of the inlet temperature.

Solving Eq. 8–29 for $\dot{m}c_p$ gives

$$\dot{m}c_p = -\frac{hA_s}{\ln[(T_s - T_e)/(T_s - T_i)]} \tag{8–31}$$

Substituting this into Eq. 8–14, we obtain

$$\dot{Q} = hA_s \Delta T_{\text{lm}} \tag{8–32}$$

where

$$\Delta T_{\text{lm}} = \frac{T_i - T_e}{\ln[(T_s - T_e)/(T_s - T_i)]} = \frac{\Delta T_e - \Delta T_i}{\ln(\Delta T_e/\Delta T_i)} \tag{8–33}$$

is the **log mean temperature difference**. Note that $\Delta T_i = T_s - T_i$ and $\Delta T_e = T_s - T_e$ are the temperature differences between the surface and the fluid at the inlet and the exit of the tube, respectively. This ΔT_{lm} relation appears to be prone to misuse, but it is practically fail-safe, since using T_i in place of T_e and vice versa in the numerator and/or the denominator will, at most, affect the sign, not the magnitude. Also, it can be used for both heating ($T_s > T_i$ and T_e) and cooling ($T_s < T_i$ and T_e) of a fluid in a tube.

The log mean temperature difference ΔT_{lm} is obtained by tracing the actual temperature profile of the fluid along the tube and is an *exact* representation of the *average temperature difference* between the fluid and the surface. It truly reflects the exponential decay of the local temperature difference. When ΔT_e differs from ΔT_i by no more than 40 percent, the error in using the arithmetic mean temperature difference is less than 1 percent. But the error increases to undesirable levels when ΔT_e differs from ΔT_i by greater amounts. Therefore, we should always use the log mean temperature difference when determining the convection heat transfer in a tube whose surface is maintained at a constant temperature T_s.

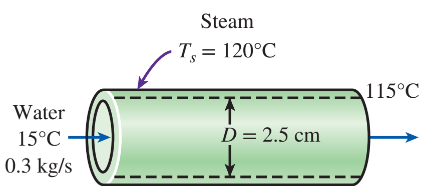


FIGURE 8–16
Schematic for Example 8–1.

EXAMPLE 8–1 Heating of Water in a Tube by Steam

Water enters a 2.5-cm-internal-diameter thin copper tube of a heat exchanger at 15°C at a rate of 0.3 kg/s, and is heated by steam condensing outside at 120°C. If the average heat transfer coefficient is 800 W/m^2·K, determine the length of the tube required in order to heat the water to 115°C (Fig. 8–16).

SOLUTION Water is heated by steam in a circular tube. The tube length required to heat the water to a specified temperature is to be determined.

Assumptions **1** Steady operating conditions exist. **2** Fluid properties are constant. **3** The convection heat transfer coefficient is constant. **4** The conduction resistance of copper tube is negligible, so the inner surface temperature of the tube is equal to the condensation temperature of steam.

Properties The specific heat of water at the bulk mean temperature of (15 + 115)/2 = 65°C is 4187 J/kg·K. The heat of condensation of steam at 120°C is 2203 kJ/kg (Table A–9).

Analysis Knowing the inlet and exit temperatures of water, the rate of heat transfer is determined to be

$$\begin{aligned}\dot{Q} = \dot{m}c_p(T_e - T_i) &= (0.3\text{ kg/s})(4.187\text{ kJ/kg·K})(115°\text{C} - 15°\text{C})\\ &= 125.6\text{ kW}\end{aligned}$$

The log mean temperature difference is

$$\Delta T_e = T_s - T_e = 120°\text{C} - 115°\text{C} = 5°\text{C}$$

$$\Delta T_i = T_s - T_i = 120°\text{C} - 15°\text{C} = 105°\text{C}$$

$$\Delta T_{\text{lm}} = \frac{\Delta T_e - T_i}{\ln(\Delta T_e/\Delta T_i)} = \frac{5 - 105}{\ln(5/105)} = 32.85°\text{C}$$

The heat transfer surface area is

$$\dot{Q} = hA_s\Delta T_{\text{lm}} \longrightarrow A_s = \frac{\dot{Q}}{h\Delta T_{\text{lm}}} = \frac{125.6 \text{ kW}}{(0.8 \text{ kW/m}^2\cdot\text{K})(32.85°\text{C})} = 4.78 \text{ m}^2$$

Then the required tube length becomes

$$A_s = \pi DL \longrightarrow L = \frac{A_s}{\pi D} = \frac{4.78 \text{ m}^2}{\pi(0.025 \text{ m})} = \mathbf{61 \text{ m}}$$

Discussion The bulk mean temperature of water during this heating process is 65°C, and thus the *arithmetic* mean temperature difference is $\Delta T_{\text{am}} = 120 - 65 = 55°\text{C}$. Using ΔT_{am} instead of ΔT_{lm} would give $L = 36$ m, which is grossly in error. This shows the importance of using the log mean temperature in calculations.

8–5 ▪ LAMINAR FLOW IN TUBES

We mentioned in Sec. 8–2 that flow in tubes is laminar for Re $\lesssim$ 2300 and that the flow is fully developed if the tube is sufficiently long (relative to the entry length) so that the entrance effects are negligible. In this section we consider steady, laminar, incompressible flow of a fluid with constant properties in the fully developed region of a straight circular pipe. We obtain the momentum equation by applying a force balance to a differential volume element, and we obtain the velocity profile by solving it. Then we use it to obtain a relation for the friction factor. An important aspect of the analysis here is that it is one of the few available for viscous flow.

In fully developed laminar flow, each fluid particle moves at a constant axial velocity along a streamline, and the velocity profile $u(r)$ remains unchanged in the flow direction. There is no motion in the radial direction, and thus the velocity component in the direction normal to the pipe axis is everywhere zero. There is no acceleration since the flow is steady and fully developed.

Now consider a ring-shaped differential volume element of radius r, thickness dr, and length dx oriented coaxially with the tube, as shown in Fig. 8–17. The volume element involves only pressure and viscous effects, and thus the pressure and shear forces must balance each other. The pressure force acting on a submerged plane surface is the product of the pressure at the centroid of the surface and the surface area. A force balance on the volume element in the flow direction gives

$$(2\pi r\, dr\, P)_x - (2\pi r\, dr\, P)_{x+dx} + (2\pi r\, dx\, \tau)_r - (2\pi r\, dx\, \tau)_{r+dr} = 0 \qquad \textbf{(8–34)}$$

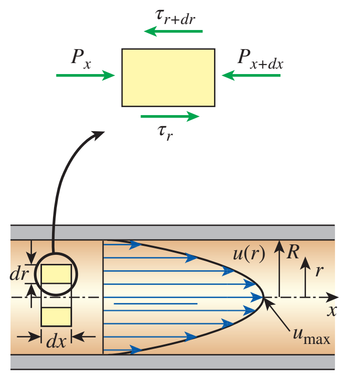
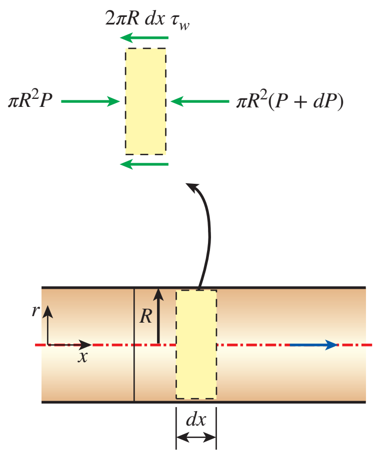


FIGURE 8–17
Free-body diagram of a ring-shaped differential fluid element of radius r, thickness dr, and length dx oriented coaxially with a horizontal tube in fully developed laminar flow.



Force balance:

$$\pi R^2 P - \pi R^2 (P + dP) - 2\pi R\, dx\, \tau_w = 0$$

Simplifying:

$$\frac{dP}{dx} = -\frac{2\tau_w}{R}$$

FIGURE 8–18
Free-body diagram of a fluid disk element of radius R and length dx in fully developed laminar flow in a horizontal tube.

which indicates that in fully developed flow in a horizontal tube, the viscous and pressure forces balance each other. Dividing by $2\pi dr dx$ and rearranging,

$$r\frac{P_{x+dx} - P_x}{dx} + \frac{(r\tau)_{r+dr} - (r\tau)_r}{dr} = 0 \tag{8–35}$$

Taking the limit as $dr, dx \longrightarrow 0$ gives

$$r\frac{dP}{dr} + \frac{d(r\tau)}{dr} = 0 \tag{8–36}$$

Substituting $\tau = -\mu(du/dr)$ and taking $\mu =$ constant gives the desired equation,

$$\frac{\mu}{r}\frac{d}{dr}\left(r\frac{du}{dr}\right) = \frac{dP}{dx} \tag{8–37}$$

The quantity du/dr is negative in pipe flow, and the negative sign is included to obtain positive values for τ. (Or, $du/dr = -du/dy$ since $y = R - r$.) The left side of Eq. 8–37 is a function of r, and the right side is a function of x. The equality must hold for any value of r and x, and an equality of the form $f(r) = g(x)$ can be satisfied only if both $f(r)$ and $g(x)$ are equal to the same constant. Thus we conclude that $dP/dx =$ constant. This can be verified by writing a force balance on a volume element of radius R and thickness dx (a slice of the tube), which gives (Fig. 8–18)

$$\frac{dP}{dx} = -\frac{2\tau_w}{R}$$

Here τ_w is constant since the viscosity and the velocity profile are constants in the fully developed region. Therefore, $dP/dx =$ constant.

Equation 8–37 is solved by rearranging and integrating it twice to give

$$u(r) = \frac{1}{4\mu}\left(\frac{dP}{dx}\right) + C_1 \ln r + C_2 \tag{8–38}$$

The velocity profile $u(r)$ is obtained by applying the boundary conditions $\partial u/\partial r = 0$ at $r = 0$ (because of symmetry about the centerline) and $u = 0$ at $r = R$ (the no-slip condition at the tube wall). We get

$$u(r) = -\frac{R^2}{4\mu}\left(\frac{dP}{dx}\right)\left(1 - \frac{r^2}{R^2}\right) \tag{8–39}$$

Therefore, the velocity profile in fully developed laminar flow in a tube is *parabolic* with a maximum at the centerline and a minimum (zero) at the tube wall. Also, the axial velocity u is positive for any r, and thus the axial pressure gradient dP/dx must be negative (i.e., pressure must decrease in the flow direction because of viscous effects).

The average velocity is determined from its definition by substituting Eq. 8–39 into Eq. 8–2 and performing the integration, yielding

$$V_{\text{avg}} = \frac{2}{R^2}\int_0^R u(r) r\, dr = \frac{-2}{R^2}\int_0^R \frac{R^2}{4\mu}\left(\frac{dP}{dx}\right)\left(1 - \frac{r^2}{R^2}\right) r\, dr = -\frac{R^2}{8\mu}\left(\frac{dP}{dx}\right) \tag{8–40}$$

Combining the last two equations, the velocity profile is rewritten as

$$u(r) = 2V_{avg}\left(1 - \frac{r^2}{R^2}\right) \quad \text{(8–41)}$$

This is a convenient form for the velocity profile since V_{avg} can be determined easily from the flow rate information.

The maximum velocity occurs at the centerline and is determined from Eq. 8–41 by substituting $r = 0$,

$$u_{max} = 2V_{avg} \quad \text{(8–42)}$$

Therefore, *the average velocity in fully developed laminar pipe flow is one-half of the maximum velocity.*

Pressure Drop

A quantity of interest in the analysis of pipe flow is the *pressure drop* ΔP since it is directly related to the power requirements of the fan or pump to maintain flow. We note that dP/dx = constant, and integrating from $x = x_1$ where the pressure is P_1 to $x = x_1 + L$ where the pressure is P_2 gives

$$\frac{dP}{dx} = \frac{P_2 - P_1}{L} \quad \text{(8–43)}$$

Substituting Eq. 8–43 into the V_{avg} expression in Eq. 8–40, the pressure drop is expressed as

Laminar flow:
$$\Delta P = P_1 - P_2 = \frac{8\mu L V_{avg}}{R^2} = \frac{32\mu L V_{avg}}{D^2} \quad \text{(8–44)}$$

The symbol Δ is typically used to indicate the difference between the final and initial values, like $\Delta y = y_2 - y_1$. But in fluid flow, ΔP is used to designate pressure drop, and thus it is $P_1 - P_2$. A pressure drop due to viscous effects represents an irreversible pressure loss, and it is called **pressure loss** ΔP_L to emphasize that it is a *loss* (just like the head loss h_L, which as we shall see is proportional to ΔP).

Note from Eq. 8–44 that the pressure drop is proportional to the viscosity μ of the fluid, and ΔP would be zero if there were no friction. Therefore, the drop of pressure from P_1 to P_2 in this case is due entirely to viscous effects, and Eq. 8–44 represents the pressure loss ΔP_L when a fluid of viscosity μ flows through a pipe of constant diameter D and length L at average velocity V_{avg}.

In practice, it is convenient to express the pressure loss for all types of fully developed internal flows (laminar or turbulent flows, circular or noncircular pipes, smooth or rough surfaces, horizontal or inclined pipes) as (Fig. 8–19)

Pressure loss:
$$\Delta P_L = f\frac{L}{D}\frac{\rho V_{avg}^2}{2} \quad \text{(8–45)}$$

where $\rho V_{avg}^2/2$ is the *dynamic pressure* and f is the **Darcy friction factor**,

$$f = \frac{8\tau_w}{\rho V_{avg}^2}$$

It is also called the **Darcy–Weisbach friction factor**, named after the Frenchman Henry Darcy (1803–1858) and the German Julius Weisbach (1806–1871), the two engineers who provided the greatest contribution in its

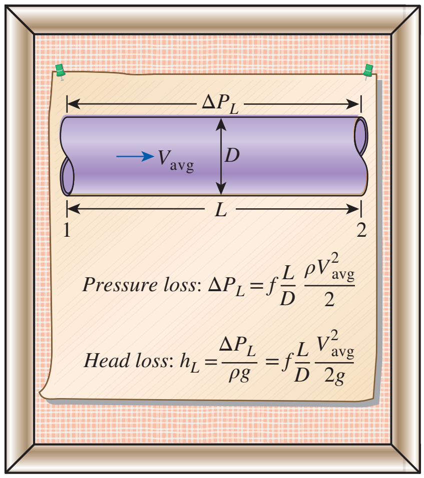


FIGURE 8–19
The relation for pressure loss (and head loss) is one of the most general relations in fluid mechanics, and it is valid for laminar or turbulent flows, circular or noncircular tubes, and pipes with smooth or rough surfaces.

development. It should not be confused with the *friction coefficient* C_f [also called the *Fanning friction factor,* named after the American engineer John Fanning (1837–1911)], which is defined as $C_f = 2\tau_w/(\rho V_{avg}^2) = f/4$.

Setting Eqs. 8–44 and 8–45 equal to each other and solving for f gives the friction factor for fully developed laminar flow in a circular tube,

Circular tube, laminar:
$$f = \frac{64\mu}{\rho D V_{avg}} = \frac{64}{\text{Re}} \tag{8–46}$$

This equation shows that *in laminar flow, the friction factor is a function of the Reynolds number only and is independent of the roughness of the pipe surface.*

In the analysis of piping systems, pressure losses are commonly expressed in terms of the *equivalent fluid column height,* called the **head loss** h_L. Noting from fluid statics that $\Delta P = \rho g h$ and thus a pressure difference of ΔP corresponds to a fluid height of $h = \Delta P/\rho g$, the *pipe head loss* is obtained by dividing ΔP_L by ρg to give

$$h_L = \frac{\Delta P_L}{\rho g} = f\frac{L}{D}\frac{V_{avg}^2}{2g}$$

The head loss h_L represents *the additional height that the fluid needs to be raised by a pump in order to overcome the frictional losses in the pipe.* The head loss is caused by viscosity, and it is directly related to the wall shear stress. Equation 8–45 is valid for both laminar and turbulent flows in both circular and noncircular tubes, but Eq. 8–46 is valid only for fully developed laminar flow in circular pipes.

Once the pressure loss (or head loss) is known, the required pumping power *to overcome the pressure loss* is determined from

$$\dot{W}_{pump,\,L} = \dot{V}\Delta P_L = \dot{V}\rho g h_L = \dot{m} g h_L \tag{8–47}$$

where $\dot{V}$ is the volume flow rate and $\dot{m}$ is the mass flow rate.

The average velocity for laminar flow in a horizontal tube is, from Eq. 8–44,

Horizontal tube:
$$V_{avg} = \frac{(P_1 - P_2)R^2}{8\mu L} = \frac{(P_1 - P_2)D^2}{32\mu L} = \frac{\Delta P\, D^2}{32\mu L}$$

Then the volume flow rate for laminar flow through a horizontal tube of diameter D and length L becomes

$$\dot{V} = V_{avg}A_c = \frac{(P_1 - P_2)R^2}{8\mu L}\pi R^2 = \frac{(P_1 - P_2)\pi D^4}{128\mu L} = \frac{\Delta P\, \pi D^4}{128\mu L} \tag{8–48}$$

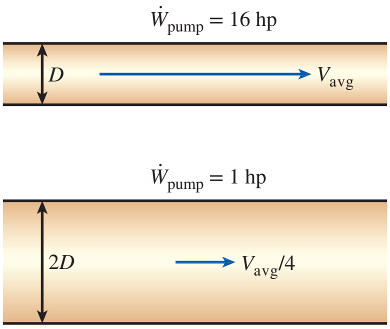


FIGURE 8–20
The pumping power requirement for a laminar flow piping system can be reduced by a factor of 16 by doubling the tube diameter.

This equation is known as **Poiseuille's law**, and this flow is called *Hagen–Poiseuille flow* in honor of the works of G. Hagen (1797–1884) and J. Poiseuille (1799–1869) on the subject. Note from Eq. 8–48 that *for a specified flow rate, the pressure drop and thus the required pumping power is proportional to the length of the pipe and the viscosity of the fluid, but it is inversely proportional to the fourth power of the radius (or diameter) of the pipe.* Therefore, the pumping power requirement for a piping system can be reduced by a factor of 16 by doubling the tube diameter (Fig. 8–20). Of course the benefits of the reduction in the energy costs must be weighed against the increased cost of construction due to using a larger-diameter tube.

The pressure drop ΔP equals the pressure loss ΔP_L in the case of a horizontal tube, but this is not the case for inclined pipes or pipes with variable cross-sectional area because of the changes in elevation and velocity.

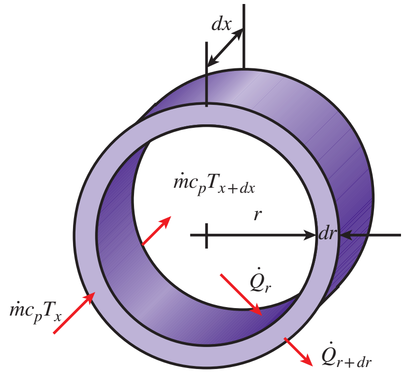


FIGURE 8–21
The differential volume element used in the derivation of energy balance relation.

Temperature Profile and the Nusselt Number

In the previous analysis, we have obtained the velocity profile for fully developed flow in a circular tube from a force balance applied on a volume element and determined the friction factor and the pressure drop. Next we obtain the energy equation by applying the energy balance on a differential volume element and solve it to obtain the temperature profile for the constant surface temperature and the constant surface heat flux cases.

Reconsider steady laminar flow of a fluid in a circular tube of radius R. The fluid properties ρ, k, and c_p are constant, and the work done by viscous forces is negligible. The fluid flows along the x-axis with velocity u. The flow is fully developed so that u is independent of x and thus $u = u(r)$. Noting that energy is transferred by mass in the x-direction and by conduction in the r-direction (heat conduction in the x-direction is assumed to be negligible), the steady-flow energy balance for a cylindrical shell element of thickness dr and length dx can be expressed as (Fig. 8–21)

$$\dot{m}c_pT_x - \dot{m}c_pT_{x+dx} + \dot{Q}_r - \dot{Q}_{r+dr} = 0 \tag{8–49}$$

where $\dot{m} = \rho u A_c = \rho u(2\pi r dr)$. Substituting and dividing by $2\pi r dr dx$ gives, after rearranging,

$$\rho c_p u \frac{T_{x+dx} - T_x}{dx} = -\frac{1}{2\pi r dx}\frac{\dot{Q}_{r+dr} - \dot{Q}_r}{dr} \tag{8–50}$$

or

$$u\frac{\partial T}{\partial x} = -\frac{1}{2\rho c_p \pi r dx}\frac{\partial \dot{Q}}{\partial r} \tag{8–51}$$

But from Fourier's law of heat conduction in the radial direction,

$$\frac{\partial \dot{Q}}{\partial r} = \frac{\partial}{\partial r}\left(-2\pi r k dx \frac{\partial T}{\partial r}\right) = -2\pi k dx \frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right) \tag{8–52}$$

Substituting and using $\alpha = k/\rho c_p$ gives

$$u\frac{\partial T}{\partial x} = \frac{\alpha}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right) \tag{8–53}$$

which states that *the rate of net energy transfer to the control volume by mass flow is equal to the net rate of heat conduction in the radial direction.*

Constant Surface Heat Flux

For fully developed flow in a circular tube subjected to constant surface heat flux, we have, from Eq. 8–24,

$$\frac{\partial T}{\partial x} = \frac{dT_s}{dx} = \frac{dT_m}{dx} = \frac{2\dot{q}_s}{\rho V_{\text{avg}} c_p R} = \text{constant} \tag{8–54}$$

If heat conduction in the x-direction were considered in the derivation of Eq. 8–53, it would give an additional term $\alpha\partial^2 T/\partial x^2$, which would be equal to zero since $\partial T/\partial x =$ constant and thus $T = T(r)$. Therefore, the assumption that there is no axial heat conduction is satisfied exactly in this case.

Substituting Eq. 8–54 and the relation for velocity profile (Eq. 8–41) into Eq. 8–53 gives

$$\frac{4\dot{q}_s}{kR}\left(1-\frac{r^2}{R^2}\right)=\frac{1}{r}\frac{d}{dr}\left(r\frac{dT}{dr}\right) \tag{8–55}$$

which is a second-order ordinary differential equation. Its general solution is obtained by separating the variables and integrating twice to be

$$T=\frac{\dot{q}_s}{kR}\left(r^2-\frac{r^4}{4R^2}\right)+C_1\ln r+C_2 \tag{8–56}$$

The desired solution to the problem is obtained by applying the boundary conditions $\partial T/\partial r=0$ at $r=0$ (because of symmetry) and $T=T_s$ at $r=R$. We get

$$T=T_s-\frac{\dot{q}_sR}{k}\left(\frac{3}{4}-\frac{r^2}{R^2}+\frac{r^4}{4R^4}\right) \tag{8–57}$$

The mean temperature T_m is determined by substituting the velocity and temperature profile relations (Eqs. 8–41 and 8–57) into Eq. 8–4 and performing the integration. It gives

$$T_m=T_s-\frac{11}{24}\frac{\dot{q}_sR}{k} \tag{8–58}$$

Combining this relation with $\dot{q}_s=h(T_s-T_m)$ gives

$$h=\frac{24}{11}\frac{k}{R}=\frac{48}{11}\frac{k}{D}=4.36\frac{k}{D} \tag{8–59}$$

or

Circular tube, laminar ($\dot{q}_s$ = constant): $\quad \text{Nu}=\frac{hD}{k}=4.36$ **(8–60)**

Therefore, for fully developed laminar flow in a circular tube subjected to constant surface heat flux, the Nusselt number is a constant. There is no dependence on the Reynolds or the Prandtl numbers.

Constant Surface Temperature

A similar analysis can be performed for fully developed laminar flow in a circular tube for the case of constant surface temperature T_s. The solution procedure in this case is more complex than the constant surface heat flux case, but the resulting Nusselt number relation obtained is equally simple. This problem has been solved iteratively as reported by Kays and Crawford (1993) and more recently by Shokouhmand and Hooman (2003) using an exact analytical procedure. The resulting Nusselt number relation obtained is (Fig. 8–22):

Circular tube, laminar (T_s = constant): $\quad \text{Nu}=\frac{hD}{k}=3.66$ **(8–61)**

Comparison of Eqs. 8–60 and 8–61 shows that the Nusselt number for the case of constant surface heat flux is 16 percent higher than the case of constant surface temperature for the fully developed laminar pipe flow. This shows that laminar flow is sensitive to the applied surface thermal boundary condition and for applications requiring higher rates of heat transfer, whenever possible; the constant surface heat flux boundary condition should be used. This is contrary to the results shown in Fig. 8–9 for the turbulent flow, which showed

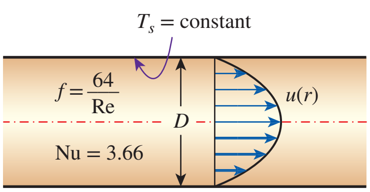


FIGURE 8–22
In laminar flow in a tube with constant surface temperature, both the *friction factor* and the *heat transfer coefficient* remain constant in the fully developed region.

no sensitivity to the different surface thermal boundary conditions in the fully developed region.

The thermal conductivity k for use in the Nu relations above should be evaluated at the bulk mean fluid temperature, which is the arithmetic average of the mean fluid temperatures at the inlet and the exit of the tube. For laminar flow, the effect of *surface roughness* on the friction factor and the heat transfer coefficient is negligible.

Laminar Flow in Noncircular Tubes

The friction factor f and the Nusselt number relations are given in Table 8–1 for *fully developed laminar flow* in tubes of various cross sections. The Reynolds and Nusselt numbers for flow in these tubes are based on the hydraulic diameter $D_h = 4A_c/p$, where A_c is the cross-sectional area of the tube and p is its perimeter. Once the Nusselt number is available, the convection heat transfer coefficient is determined from $h = k\text{Nu}/D_h$.

TABLE 8–1

Nusselt number and friction factor for fully developed laminar flow in tubes of various cross sections ($D_h = 4A_c/p$, $\text{Re} = V_{avg}D_h/\nu$, and $\text{Nu} = hD_h/k$)

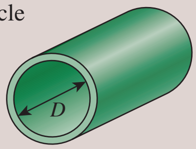
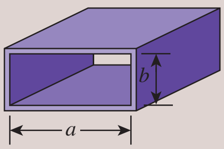
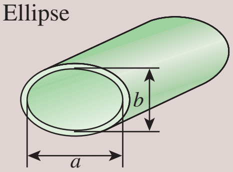
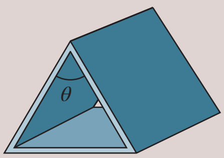
		Nusselt Number		
Tube Geometry	a/b or $\theta°$	T_s = Const.	$\dot{q}_s$ = Const.	Friction Factor f
Circle 	—	3.66	4.36	64.00/Re
Rectangle 	a/b			
	1	2.98	3.61	56.92/Re
	2	3.39	4.12	62.20/Re
	3	3.96	4.79	68.36/Re
	4	4.44	5.33	72.92/Re
	6	5.14	6.05	78.80/Re
	8	5.60	6.49	82.32/Re
	∞	7.54	8.24	96.00/Re
Ellipse 	a/b			
	1	3.66	4.36	64.00/Re
	2	3.74	4.56	67.28/Re
	4	3.79	4.88	72.96/Re
	8	3.72	5.09	76.60/Re
	16	3.65	5.18	78.16/Re
Isosceles triangle 	θ			
	10°	1.61	2.45	50.80/Re
	30°	2.26	2.91	52.28/Re
	60°	2.47	3.11	53.32/Re
	90°	2.34	2.98	52.60/Re
	120°	2.00	2.68	50.96/Re

FIGURE 8–23
Leo Graetz (1856–1941), a German physicist, was born at Breslau (then in Germany, now called Wroclaw and in Poland). His scientific work was first concerned with the fields of heat conduction, radiation, friction, and elasticity. He was one of the first to investigate the propagation of electromagnetic energy. The dimensionless **Graetz number** describing heat transfer is named after him.
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Developing Laminar Flow in the Entrance Region

The heat transfer analysis presented so far has been for hydrodynamically and thermally developed (fully developed) laminar flow in a circular tube where conditions given by Eqs. 8–7 and 8–8 were valid. However, in the entrance region, the energy equation Eq. 8–53 is no longer valid since in this region there is motion in the radial direction and the velocity profile $u(r)$ changes in the flow direction. In addition, for example for the case of constant surface heat flux, the axial temperature gradient $\partial T/\partial x$ can no longer be simplified through Eq. 8–54. Therefore, the solution to the energy equation in the entrance region is more complicated than the fully developed region, and the energy equation is solved numerically. The local values of Nusselt number are typically presented either graphically or in tabular form in terms of the inverse of a dimensionless parameter called the Graetz number (Fig. 8–23) which is defined as $\text{Gz} = (D/x)\text{Re Pr}$. As shown in Fig. 8–24, the fully developed conditions for both cases of constant surface heat flux and constant surface temperature are reached for $1/\text{Gz} = (x/D)/\text{Re Pr} \approx 0.05$, which is consistent with the results given by Eq. 8–12 for laminar thermal entry length. Therefore, when the inverse of the Graetz number is greater than 0.05, the local Nusselt numbers approach their fully developed values of 4.36 for constant surface heat flux and 3.66 for constant surface temperature. Excellent treatments of this subject are contained in Shah and London (1978) and Shah and Bhatti (1987). Two different solutions are obtained in the literature. The simplest case is for hydrodynamically developed flow and thermally developing flow. The more complicated case is for hydrodynamically and thermally developing flow. In this case the solution is a function of the Prandtl number, and for each case the value of the Prandtl number must be specified a priori.

There are a limited number of empirical correlations available in the literature for the average Nusselt number under constant surface temperature boundary conditions. For example, for a circular tube of length L subjected to constant surface temperature, the average Nusselt number for the *thermal entrance region* can be determined from (Edwards et al., 1979)

Entry region, laminar:
$$\text{Nu} = 3.66 + \frac{0.065(D/L)\text{ Re Pr}}{1 + 0.04[(D/L)\text{ Re Pr}]^{2/3}} \tag{8–62}$$

Note that the average Nusselt number is larger at the entrance region, as expected, and it approaches asymptotically to the fully developed value of 3.66 as $L \to \infty$. This relation assumes that the flow is hydrodynamically developed when the fluid enters the heating section, but it can also be used for flow developing hydrodynamically when $\text{Pr} \geq 5$.

When the difference between the surface and the fluid temperatures is large, it may be necessary to account for the variation of viscosity with temperature.

The average Nusselt number for hydrodynamically and thermally developing laminar flow in a circular tube in that case can be determined from (Sieder and Tate, 1936)

Entry region, laminar:
$$\text{Nu} = 1.86\left(\frac{\text{Re Pr}D}{L}\right)^{1/3}\left(\frac{\mu_b}{\mu_s}\right)^{0.14} \tag{8–63}$$

The preceding equation is recommended for $0.60 \leq \text{Pr} \leq 5$ and $0.0044 \leq (\mu_b/\mu_s) \leq 9.75$. Note that the term $(D/L)\text{RePr}$ in both Eqs. 8–62 and 8–63 is the Graetz number for a circular tube of length L. All properties appearing in Eqs. 8–62 and 8–63 should be evaluated at the bulk mean fluid temperature, $T_b = (T_i + T_e)/2$, except for μ_s, which is evaluated at the surface temperature.

The average Nusselt number for the thermal entrance region of flow between *isothermal parallel plates* of length L is expressed as (Edwards et al., 1979)

Entry region, laminar:

$$\text{Nu} = 7.54 + \frac{0.03(D_h/L)\text{Re Pr}}{1 + 0.016[(D_h/L)\text{Re Pr}]^{2/3}} \tag{8–64}$$

where D_h is the hydraulic diameter, which is twice the spacing of the plates. This relation can be used for $\text{Re} \leq 2800$.

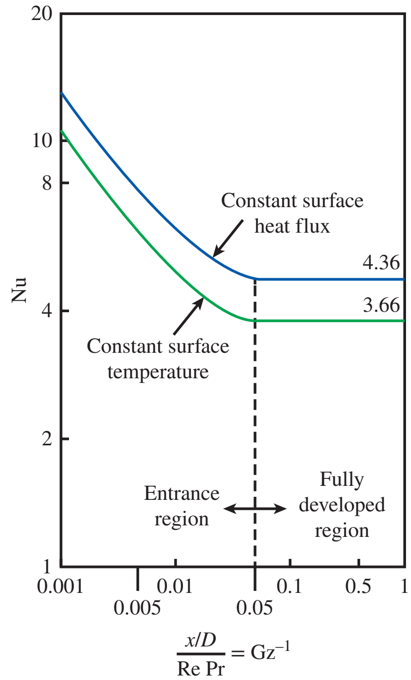


FIGURE 8–24
Local Nusselt numbers in the entry and fully developed regions for laminar flow in a circular tube for hydrodynamically developed and thermally developing flow.

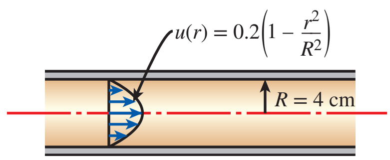


FIGURE 8–25
Schematic for Example 8–2.

EXAMPLE 8–2 Average Velocity and Temperature in Laminar Tube Flow

The velocity and temperature profiles for a fluid flowing in a circular tube of inner radius $R = 4$ cm (Fig. 8–25) are given as

$$u(r) = 0.2[1 - (r/R)^2] \quad \text{(in m/s)}$$

$$T(r) = 250 + 200(r/R)^3 \quad \text{(in K)}$$

Determine the average flow velocity and the average fluid temperature in the tube.

SOLUTION Using the given velocity and temperatures profiles in a tube, the average flow velocity and the average fluid temperature are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Properties are constant.

Analysis On the basis of the conservation of mass principle, the average velocity in a circular tube of inner radius R is expressed as

$$V_{\text{avg}} = \frac{2}{R^2}\int_0^R u(r)r\,dr$$

Substituting the known quantities and performing the integration, the average velocity is determined to be

$$V_{\text{avg}} = \frac{2}{R^2}\int_0^R 0.2\left(r - \frac{r^3}{R^2}\right)dr = \frac{2 \times 0.2}{R^2}\left(\frac{r^2}{2} - \frac{r^4}{4R^2}\right)_0^R = \frac{2 \times 0.2}{R^2}\left(\frac{R^2}{4}\right) = \mathbf{0.1\ m/s}$$

On the basis of the conservation of energy principle, the average (or mean) fluid temperature at a cross section is expressed as

$$T_m = \frac{2}{V_{\text{avg}}R^2}\int_0^R T(r)u(r)r\,dr$$

Substituting the known quantities and performing the integration, the average temperature is determined to be

$$T_m = \frac{2}{V_{avg}R^2}\int_0^R 0.2\left(1-\frac{r^2}{R^2}\right)\left(250+200\frac{r^3}{R^3}\right)rdr$$

$$= \frac{2\times 0.2}{V_{avg}R^2}\left(250\frac{r^2}{2}-250\frac{r^4}{4R^2}+200\frac{r^5}{5R^3}-200\frac{r^7}{7R^5}\right)_0^R$$

$$= \frac{2\times 0.2}{V_{avg}R^2}\left(\frac{250R^2}{2}-\frac{250R^2}{4}+\frac{200R^2}{5}-\frac{200R^2}{7}\right) = \frac{2\times 0.2\times 73.93}{0.1} = \mathbf{295.7\ K}$$

Discussion The velocity profile for laminar flow is expressed as $u(r) = 2V_{avg}[1 - (r/R)^2]$. Comparing this with the given profile $u(r) = 0.2[1 - (r/R)^2]$, we could obtain the same value of average velocity directly by observation. Also, the parabolic velocity profile indicates that this is a fully developed laminar flow.

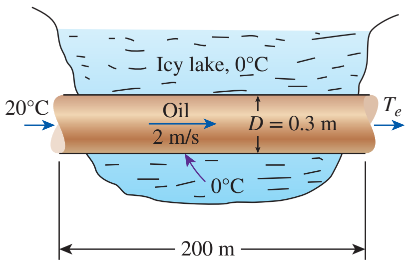


FIGURE 8–26
Schematic for Example 8–3.

EXAMPLE 8–3 Flow of Oil in a Pipeline Through a Lake

Consider the flow of oil at 20°C in a 30-cm-diameter pipeline at an average velocity of 2 m/s (Fig. 8–26). A 200-m-long section of the horizontal pipeline passes through icy waters of a lake at 0°C. Measurements indicate that the surface temperature of the pipe is very nearly 0°C. Disregarding the thermal resistance of the pipe material, determine (*a*) the temperature of the oil when the pipe leaves the lake, (*b*) the rate of heat transfer from the oil, and (*c*) the pumping power required to overcome the pressure losses and to maintain the flow of the oil in the pipe.

SOLUTION Oil flows in a pipeline that passes through icy waters of a lake at 0°C. The exit temperature of the oil, the rate of heat loss, and the pumping power needed to overcome pressure losses are to be determined.

Assumptions **1** Steady operating conditions exist. **2** The surface temperature of the pipe is very nearly 0°C. **3** The thermal resistance of the pipe is negligible. **4** The inner surfaces of the pipeline are smooth. **5** The flow is hydrodynamically developed when the pipeline reaches the lake.

Properties We do not know the exit temperature of the oil, and thus we cannot determine the bulk mean temperature, which is the temperature at which the properties of oil are to be evaluated. The mean temperature of the oil at the inlet is 20°C, and we expect this temperature to drop somewhat as a result of heat loss to the icy waters of the lake. We evaluate the properties of the oil at the inlet temperature, but we will repeat the calculations, if necessary, using properties at the evaluated bulk mean temperature. At 20°C we read (Table A–13)

$$\rho = 888.1\ \text{kg/m}^3 \qquad \nu = 9.429\times 10^{-4}\ \text{m}^2\text{/s}$$
$$k = 0.145\ \text{W/m·K} \qquad c_p = 1880\ \text{J/kg·K} \qquad \text{Pr} = 10{,}863$$

Analysis (*a*) The Reynolds number is

$$\text{Re} = \frac{V_{avg}D}{\nu} = \frac{(2\ \text{m/s})(0.3\ \text{m})}{9.429\times 10^{-4}\ \text{m}^2\text{/s}} = 636$$

which is less than the critical Reynolds number of 2300. Therefore, the flow is laminar, and the thermal entry length in this case is roughly

$$L_t \approx 0.05 \text{ Re Pr } D = 0.05 \times 636 \times 10{,}863 \times (0.3 \text{ m}) \approx 103{,}600 \text{ m}$$

which is much greater than the total length of the pipe. This is typical of fluids with high Prandtl numbers. Therefore, we assume thermally developing flow and determine the Nusselt number from

$$\begin{aligned}
\text{Nu} = \frac{hD}{k} &= 3.66 + \frac{0.065(D/L) \text{ Re Pr}}{1 + 0.04\,[(D/L) \text{ Re Pr}]^{2/3}} \\
&= 3.66 + \frac{0.065(0.3/200) \times 636 \times 10{,}863}{1 + 0.04\,[(0.3/200) \times 636 \times 10{,}863]^{2/3}} \\
&= 33.7
\end{aligned}$$

Note that this Nusselt number is considerably higher than the fully developed value of 3.66. Then,

$$h = \frac{k}{D}\text{Nu} = \frac{0.145 \text{ W/m·K}}{0.3 \text{ m}}(33.7) = 16.3 \text{ W/m}^2\text{·K}$$

Also,

$$\begin{aligned}
A_s &= \pi DL = \pi(0.3 \text{ m})(200 \text{ m}) = 188.5 \text{ m}^2 \\
\dot{m} &= \rho A_c V_{\text{avg}} = (888.1 \text{ kg/m}^3)[\tfrac{1}{4}\pi(0.3 \text{ m})^2](2 \text{ m/s}) = 125.6 \text{ kg/s}
\end{aligned}$$

Next we determine the exit temperature of oil,

$$\begin{aligned}
T_e &= T_s - (T_s - T_i) \exp(-hA_s/\dot{m}c_p) \\
&= 0°\text{C} - [(0 - 20)°\text{C}] \exp\left[-\frac{(16.3 \text{ W/m}^2\text{·K})(188.5 \text{ m}^2)}{(125.6 \text{ kg/s})(1881 \text{ J/kg·K})}\right] \\
&= \mathbf{19.74°C}
\end{aligned}$$

Thus, the mean temperature of oil drops by a mere 0.26°C as it crosses the lake. This makes the bulk mean oil temperature 19.87°C, which is practically identical to the inlet temperature of 20°C. Therefore, we do not need to reevaluate the properties.

(*b*) The log mean temperature difference and the rate of heat loss from the oil are

$$\Delta T_{\text{lm}} = \frac{T_i - T_e}{\ln \dfrac{T_s - T_e}{T_s - T_i}} = \frac{20 - 19.74}{\ln \dfrac{0 - 19.74}{0 - 20}} = -19.87°\text{C}$$

$$\dot{Q} = hA_s\,\Delta T_{\text{lm}} = (16.3 \text{ W/m}^2\text{·K})(188.5 \text{ m}^2)(-19.87°\text{C}) = \mathbf{-6.11 \times 10^4 \text{ W}}$$

Therefore, the oil will lose heat at a rate of 61.1 kW as it flows through the pipe in the icy waters of the lake. Note that ΔT_{lm} is identical to the arithmetic mean temperature in this case, since $\Delta T_i \approx \Delta T_e$.

(*c*) The laminar flow of oil is hydrodynamically developed. Therefore, the friction factor can be determined from

$$f = \frac{64}{\text{Re}} = \frac{64}{636} = 0.1006$$

Then the pressure drop in the pipe and the required pumping power become

$$\Delta P = f\frac{L}{D}\frac{\rho V_{\text{avg}}^2}{2} = 0.1006\frac{200 \text{ m}}{0.3 \text{ m}}\frac{(888.1 \text{ kg/m}^3)(2 \text{ m/s})^2}{2} = 1.19 \times 10^5 \text{ N/m}^2$$

$$\dot{W}_{\text{pump}} = \frac{\dot{m}\Delta P}{\rho} = \frac{(125.6 \text{ kg/s})(1.19 \times 10^5 \text{ N/m}^2)}{888.1 \text{ kg/m}^3} = \mathbf{16.8\ kW}$$

Discussion We need a 16.8-kW pump just to overcome the friction in the pipe as the oil flows in the 200-m-long pipe through the lake.

8–6 ▪ TURBULENT FLOW IN TUBES

We mentioned earlier that flow in smooth tubes is usually fully turbulent for $\text{Re} > 10{,}000$. Turbulent flow is commonly used in practice because of the higher heat transfer coefficients associated with it. Most correlations for the friction and heat transfer coefficients in turbulent flow are based on experimental studies because of the difficulty in dealing with turbulent flow theoretically.

For *smooth* tubes, the friction factor in turbulent flow can be determined from the explicit *first Petukhov equation* (Petukhov, 1970), given as

$$\textit{Smooth tubes:} \quad f = (0.790 \ln \text{Re} - 1.64)^{-2} \qquad 3000 < \text{Re} < 5 \times 10^6 \tag{8–65}$$

The Nusselt number in turbulent flow is related to the friction factor through the *Chilton–Colburn analogy,* expressed as

$$\text{Nu} = 0.125 f \text{RePr}^{1/3} \tag{8–66}$$

Once the friction factor is available, this equation can be used conveniently to evaluate the Nusselt number for both smooth and rough tubes.

For fully developed turbulent flow in *smooth tubes,* a simple relation for the Nusselt number can be obtained by substituting the simple power law relation $f = 0.184 \text{ Re}^{-0.2}$ for the friction factor into Eq. 8–66. It gives

$$\text{Nu} = 0.023 \text{ Re}^{0.8} \text{ Pr}^{1/3} \qquad \begin{pmatrix} 0.7 \le \text{Pr} \le 160 \\ \text{Re} > 10{,}000 \end{pmatrix} \tag{8–67}$$

which is known as the *Colburn equation.* The accuracy of this equation can be improved by modifying it as

$$\text{Nu} = 0.023 \text{ Re}^{0.8} \text{Pr}^n \tag{8–68}$$

where $n = 0.4$ for *heating* and 0.3 for *cooling* of the fluid flowing through the tube. This equation is known as the *Dittus–Boelter equation* (Dittus and Boelter, 1930), and it is preferred over the Colburn equation.

The preceding equations can be used when the temperature difference between the fluid and wall surface is not large by evaluating all fluid properties at the *bulk mean fluid temperature* $T_b = (T_i + T_e)/2$. When the variation in properties is large due to a large temperature difference, the following equation due to Sieder and Tate (1936) can be used:

$$\text{Nu} = 0.027 \text{ Re}^{0.8} \text{Pr}^{1/3} \left(\frac{\mu_b}{\mu_s}\right)^{0.14} \qquad \begin{pmatrix} 0.7 \le \text{Pr} \le 16{,}700 \\ \text{Re} \ge 10{,}000 \end{pmatrix} \tag{8–69}$$

Here all properties are evaluated at T_b except μ_s, which is evaluated at T_s.

The preceding Nusselt number relations are fairly simple, but they may give errors as large as 25 percent. This error can be reduced considerably to less than 10 percent by using more complex but accurate relations, such as the *second Petukhov equation,* expressed as

$$\mathrm{Nu} = \frac{(f/8)\mathrm{Re}\,\mathrm{Pr}}{1.07 + 12.7(f/8)^{0.5}(\mathrm{Pr}^{2/3} - 1)} \quad \begin{pmatrix} 0.5 \le \mathrm{Pr} \le 2000 \\ 10^4 < \mathrm{Re} < 5 \times 10^6 \end{pmatrix} \tag{8–70}$$

The accuracy of this relation at lower Reynolds numbers is improved by modifying it as (Gnielinski, 1976)

$$\mathrm{Nu} = \frac{(f/8)(\mathrm{Re} - 1000)\mathrm{Pr}}{1 + 12.7(f/8)^{0.5}(\mathrm{Pr}^{2/3} - 1)} \quad \begin{pmatrix} 0.5 \le \mathrm{Pr} \le 2000 \\ 3 \times 10^3 < \mathrm{Re} < 5 \times 10^6 \end{pmatrix} \tag{8–71}$$

where the friction factor f can be determined from an appropriate relation such as the first Petukhov equation. Gnielinski's equation should be preferred in calculations. Again properties should be evaluated at the bulk mean fluid temperature.

The preceding relations are not very sensitive to the *thermal conditions* at the tube surfaces, and they can be used for both $T_s =$ constant and $\dot{q}_s =$ constant cases. Despite their simplicity, the correlations already presented give sufficiently accurate results for most engineering purposes. They can also be used to obtain rough estimates of the friction factor and the heat transfer coefficients in the transition region.

The relations given so far do not apply to liquid metals because of their very low Prandtl numbers. For liquid metals ($0.004 < \mathrm{Pr} < 0.01$), the following relations are recommended by Sleicher and Rouse (1975) for $10^4 < \mathrm{Re} < 10^6$:

$$\textit{Liquid metals, } T_s = \text{constant:} \quad \mathrm{Nu} = 4.8 + 0.0156\,\mathrm{Re}^{0.85}\,\mathrm{Pr}_s^{0.93} \tag{8–72}$$

$$\textit{Liquid metals, } \dot{q}_s = \text{constant:} \quad \mathrm{Nu} = 6.3 + 0.0167\,\mathrm{Re}^{0.85}\,\mathrm{Pr}_s^{0.93} \tag{8–73}$$

where the subscript s indicates that the Prandtl number is to be evaluated at the surface temperature.

Fully Developed Transitional Flow Heat Transfer

As mentioned in Section 8–2, there is a regime of fluid flow which is neither fully laminar nor fully turbulent. Generally, internal flows with Reynolds numbers less than 2300 are considered fully laminar. As the Reynolds number rises, it contains increasingly more turbulent motion until Re ~ 4000; at this point, it is mostly turbulent. By the time the Reynolds number is increased to ~10,000, it is normally fully turbulent.

The methods to handle fully laminar and fully turbulent heat transfer have already been discussed, but in some cases the flow is in this transitional zone. Fortunately, the methods for handling fully developed turbulent flow can easily be adapted to deal with flow in this region. A simple approach is discussed in detail in Abraham et al. (2011). The recommendation is to continue to use Gnielinski's (1976) correlation (Eq. 8–71) for fully developed turbulent flow along with f values determined from the following expressions for two common flow geometries, the round tube and the parallel-plate channel. These correlations can be applied over the Reynolds number ranges listed and are reasonably accurate for any thermal boundary condition, including the uniform wall temperature and uniform wall heat flux cases. A slightly more accurate approach is available for transitional flows in round tubes (Abraham et al., 2009) but for most engineering applications, the following expressions are suitable.

Smooth round pipe:

$$f = 3.03 \times 10^{-12}\,\text{Re}^3 - 3.67 \times 10^{-8}\,\text{Re}^2 + 1.46 \times 10^{-4}\,\text{Re} - 0.151 \tag{8–74}$$

which is valid for $2300 < \text{Re} < 4500$

Smooth, parallel-plate channel:

$$f = -6.38 \times 10^{-13}\,\text{Re}^3 + 1.17 \times 10^{-8}\,\text{Re}^2 - 6.69 \times 10^{-5}\,\text{Re} + 0.147 \tag{8–75}$$

which is valid for $2300 < \text{Re} < 8000$

The calculation of Nusselt number or heat transfer coefficient in the transition region was further studied in detail by Gnielinski (2013). According to his study, the equations used to calculate the Nusselt number in tubes do not agree in the transition region between laminar and turbulent flow. There is a gap at Re = 2300. He developed a linear interpolation in the transition region between the Nusselt numbers at Re = 2300 and Re = 4000. His proposed linear interpolation connects the laminar flow equation at Re = 2300 with the turbulent flow equation at Re = 4000 and avoids the gap between the equations for laminar and turbulent flows and leads to a method of calculation providing heat transfer coefficients for all flow regimes in tubes.

As discussed in the Topic of Special Interest later in this chapter, a very detailed discussion is provided for transitional flows with particular emphasis on the entrance geometry and on the development region. That discussion also handles flows which may experience buoyant motion and property variations.

Rough Surfaces

Any irregularity or roughness on the surface disturbs the laminar sublayer and affects the flow. Therefore, unlike laminar flow, the friction factor and the convection coefficient in turbulent flow are strong functions of surface roughness.

The friction factor in fully developed turbulent pipe flow depends on the Reynolds number and the **relative roughness** ε/D, which is the ratio of the mean height of roughness of the pipe to the pipe diameter. The functional form of this dependence cannot be obtained from a theoretical analysis, and all available results are obtained from painstaking experiments using artificially roughened surfaces (usually by gluing sand grains of a known size on the inner surfaces of the pipes). Most such experiments were conducted by Prandtl's student J. Nikuradse in 1933, followed by the works of others. The friction factor was calculated from the measurements of the flow rate and the pressure drop.

The experimental results obtained are presented in tabular, graphical, and functional forms obtained by curve-fitting experimental data. In 1939, Cyril F. Colebrook (1910–1997) combined the available data for transition and turbulent flow in smooth as well as rough pipes into the following implicit relation known as the **Colebrook equation**:

$$\frac{1}{\sqrt{f}} = -2.0\log\left(\frac{\varepsilon/D}{3.7} + \frac{2.51}{\text{Re}\sqrt{f}}\right) \qquad \text{(turbulent flow)} \tag{8–76}$$

We note that the logarithm in Eq. 8–76 is a base 10 rather than a natural logarithm. In 1942, the American engineer Hunter Rouse (1906–1996) verified Colebrook's equation and produced a graphical plot of f as a function of Re and the product Re$\sqrt{f}$. He also presented the laminar flow relation and a table of commercial pipe roughness. Two years later, Lewis F. Moody (1880–1953) redrew Rouse's diagram into the form commonly used today. The now famous **Moody chart** is given in the appendix as Fig. A–20. It presents the Darcy friction factor for pipe flow as a function of the Reynolds number and ε/D over a wide range. It is probably one of the most widely accepted and used charts in engineering. Although it was developed for circular pipes, it can also be used for noncircular pipes by replacing the diameter with the hydraulic diameter.

For smooth pipes, the agreement between the Petukhov and Colebrook equations is very good. The friction factor is minimum for a smooth pipe (but still not zero because of the no-slip condition), and it increases with roughness (Fig. 8–27).

Relative Roughness, ε/D	Friction Factor, f
0.0*	0.0119
0.00001	0.0119
0.0001	0.0134
0.0005	0.0172
0.001	0.0199
0.005	0.0305
0.01	0.0380
0.05	0.0716

*Smooth surface. All values are for Re = 10^6, and are calculated from Eq. 8–76.

FIGURE 8–27
The friction factor is minimum for a smooth pipe and increases with roughness.

Commercially available pipes are specified according to a **nominal diameter**. The nominal diameter does not necessarily indicate the actual inside diameter of the pipe (Table 8–2). Therefore, in calculations we should make sure to use the actual inside diameter of the pipe. For example, a 2-inch nominal size steel pipe has an actual inside diameter of 2.067 inches. Another specification is **pipe schedule**. Schedule is the thickness of the inside diameter of the pipe. Schedule 40 is usually a standard thickness for most applications, which is normally a 1/4-inch wall (Table 8–2). As the schedule number increases or decreases, so does the wall thickness. However, the outside diameter remains the same. Standard piping schedules are 10, 40, 80, 120, and 160.

TABLE 8–2

Standard sizes for Schedule 40 steel pipes

Nominal Size, in	Actual Inside Diameter, in
⅛	0.269
¼	0.364
⅜	0.493
½	0.622
¾	0.824
1	1.049
1½	1.610
2	2.067
2½	2.469
3	3.068
5	5.047
10	10.02

Commercially available pipes differ from those used in the experiments in that the roughness of pipes in the market is not uniform, and it is difficult to give a precise description of it. Equivalent roughness values for some commercial pipes are given in Table 8–3 as well as in the Moody chart. But it should be kept in mind that these values are for new pipes, and the relative roughness of pipes may increase with use as a result of corrosion, scale buildup, and precipitation. As a result, the friction factor may increase by a factor of 5 to 10. Actual operating conditions must be considered in the design of piping systems. Also, the Moody chart and its equivalent Colebrook equation involve several uncertainties (the roughness size, experimental error, curve fitting of data, etc.), and thus the results obtained should not be treated as exact. They are usually considered to be accurate to ±15 percent over the entire range in the figure.

The Colebrook equation is implicit in f, and thus the determination of the friction factor requires iteration. An approximate explicit relation for f was given by S. E. Haaland in 1983 as

$$\frac{1}{\sqrt{f}} \cong -1.8 \log\left[\frac{6.9}{\text{Re}} + \left(\frac{\varepsilon/D}{3.7}\right)^{1.11}\right] \quad \textbf{(8–77)}$$

The results obtained from this relation are within 2 percent of those obtained from the Colebrook equation. If more accurate results are desired, Eq. 8–77 can be used as a good *first guess* in a Newton iteration when using a programmable calculator or a spreadsheet to solve for f with Eq. 8–76.

In turbulent flow, wall roughness increases the heat transfer coefficient h by a factor of 2 or more (Dipprey and Sabersky, 1963). The convection heat transfer coefficient for rough tubes can be calculated approximately from

TABLE 8–3

Equivalent roughness values for new commercial pipes*

	Roughness, ε	
Material	ft	mm
Glass, plastic	0 (smooth)	
Concrete	0.003–0.03	0.9–9
Wood stave	0.0016	0.5
Rubber, smoothed	0.000033	0.01
Copper or brass tubing	0.000005	0.0015
Cast iron	0.00085	0.26
Galvanized iron	0.0005	0.15
Wrought iron	0.00015	0.046
Stainless steel	0.000007	0.002
Commercial steel	0.00015	0.045

*The uncertainty in these values can be as much as ±60 percent.

the Nusselt number relations such as Eq. 8–71 by using the friction factor determined from the Moody chart or the Colebrook equation. However, this approach is not very accurate since there is no further increase in h with f for $f > 4f_{\text{smooth}}$ (Norris, 1970), and correlations developed specifically for rough tubes should be used when more accuracy is desired.

Developing Turbulent Flow in the Entrance Region

The entry lengths for turbulent flow are typically short, often just 10 tube diameters long, and thus the Nusselt number determined for fully developed turbulent flow can be used approximately for the entire tube. This simple approach gives reasonable results for pressure drop and heat transfer for long tubes and conservative results for short ones. Correlations for the friction and heat transfer coefficients for the entrance regions are available in the literature for better accuracy.

Turbulent Flow in Noncircular Tubes

The velocity and temperature profiles in turbulent flow are nearly straight lines in the core region, and any significant velocity and temperature gradients occur in the viscous sublayer (Fig. 8–28). Despite the small thickness of the viscous sublayer (usually much less than 1 percent of the pipe diameter), the characteristics of the flow in this layer are very important since they set the stage for flow in the rest of the pipe. Therefore, pressure drop and heat transfer characteristics of turbulent flow in tubes are dominated by the very thin viscous sublayer next to the wall surface, and the shape of the core region is not of much significance. Consequently, the turbulent flow relations given above for circular tubes can also be used for noncircular tubes with reasonable accuracy by replacing the diameter D in the evaluation of the Reynolds number by the hydraulic diameter $D_h = 4A_c/p$.

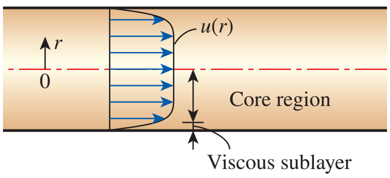


FIGURE 8–28
In turbulent flow, the velocity profile is nearly a straight line in the core region, and any significant velocity gradients occur in the viscous sublayer.

Flow Through Tube Annulus

Some simple heat transfer devices consist of two concentric tubes and are properly called *double-tube heat exchangers* (Fig. 8–29). In such devices, one fluid flows through the tube while the other flows through the annular space. The governing differential equations for both flows are identical. Therefore, steady laminar flow through an annulus can be studied analytically by using suitable boundary conditions.

Consider a concentric annulus of inner diameter D_i and outer diameter D_o. The hydraulic diameter of the annulus is

$$D_h = \frac{4A_c}{p} = \frac{4\pi(D_o^2 - D_i^2)/4}{\pi(D_o + D_i)} = D_o - D_i$$

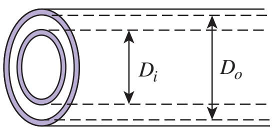


FIGURE 8–29
A double-tube heat exchanger that consists of two concentric tubes.

Annular flow is associated with two Nusselt numbers—Nu_i on the inner tube surface and Nu_o on the outer tube surface—since it may involve heat transfer on both surfaces. The Nusselt numbers for fully developed laminar flow with one surface isothermal and the other adiabatic are given in Table 8–4. When Nusselt numbers are known, the convection heat transfer coefficients for the inner and the outer surfaces are determined from

$$\text{Nu}_i = \frac{h_i D_h}{k} \quad \text{and} \quad \text{Nu}_o = \frac{h_o D_h}{k} \qquad \textbf{(8–78)}$$

For fully developed turbulent flow, the inner and outer convection heat transfer coefficients are approximately equal to each other, and the tube annulus can be treated as a noncircular duct with a hydraulic diameter of $D_h = D_o - D_i$. The Nusselt number in this case can be determined from a suitable turbulent flow relation such as the Gnielinski equation (Eq. 8–71). To improve the accuracy of Nusselt numbers obtained from these relations for annular flow, Petukhov and Roizen (1964) recommend multiplying them by the following correction factors when one of the tube walls is adiabatic and heat transfer is through the other wall:

$$F_i = 0.86\left(\frac{D_i}{D_o}\right)^{-0.16} \quad \text{(outer wall adiabatic)} \tag{8–79}$$

$$F_o = 1 - 0.14\left(\frac{D_i}{D_o}\right)^{0.6} \quad \text{(inner wall adiabatic)} \tag{8–80}$$

Gnielinski (2015), on the basis of a large amount of recent experimental data from the literature, improved his earlier heat transfer correlations for turbulent annular flow (Gnielinski, 2009) and developed a single comprehensive correlation for the calculation of the heat transfer coefficient for turbulent flow in tubes and concentric annular ducts. Only different correlations for the friction factor are required, and the appropriate hydraulic diameter must be used. Specific additional factors are needed that take into consideration the effect of the diameter ratio of the annulus and the different boundary conditions for heating and cooling. Examples include heat transfer at the inner wall with the outer wall insulated and heat transfer at the outer wall with the inner wall insulated. No experimental data were found for heat transfer from both walls to the annular flow.

TABLE 8–4

Nusselt number for fully developed laminar flow in an annulus with one surface isothermal and the other adiabatic

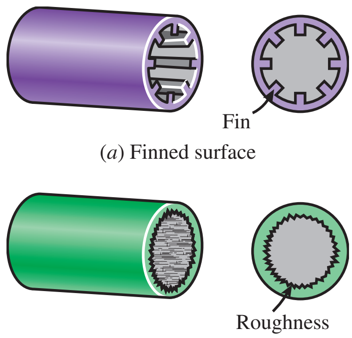
D_i/D_o	Nu_i	Nu_o
0	—	3.66
0.05	17.46	4.06
0.10	11.56	4.11
0.25	7.37	4.23
0.50	5.74	4.43
1.00	4.86	4.86

Source: Kays and Perkins, 1972

Heat Transfer Enhancement

Tubes with rough surfaces have much higher heat transfer coefficients than tubes with smooth surfaces. Therefore, tube surfaces are often intentionally *roughened, corrugated,* or *finned* in order to *enhance* the convection heat transfer coefficient and thus the convection heat transfer rate (Fig. 8–30). Heat transfer in turbulent flow in a tube has been increased by as much as 400 percent by roughening the surface. Roughening the surface, of course, also increases the friction factor and thus the power requirement for the pump or the fan.

The convection heat transfer coefficient can also be increased by inducing pulsating flow by pulse generators, by inducing swirl by inserting a twisted tape into the tube, or by inducing secondary flows by coiling the tube.



(*a*) Finned surface

(*b*) Roughened surface

FIGURE 8–30
Tube surfaces are often *roughened, corrugated,* or *finned* in order to *enhance* convection heat transfer.

EXAMPLE 8–4 Pressure Drop in a Water Tube

Water at 60°F ($\rho = 62.36$ lbm/ft^3 and $\mu = 7.536 \times 10^{-4}$ lbm/ft·s) is flowing steadily in a 2-in-internal-diameter horizontal tube made of stainless steel at a rate of 0.2 ft^3/s (Fig. 8–31). Determine the pressure drop and the required pumping power input for flow through a 200-ft-long section of the tube.

SOLUTION The flow rate through a specified water tube is given. The pressure drop and the pumping power requirements are to be determined.

Assumptions **1** The flow is steady and incompressible. **2** The entrance effects are negligible, and thus the flow is fully developed. **3** The tube involves no components such as

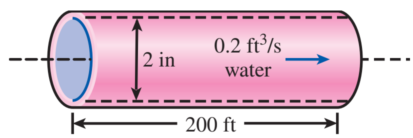


FIGURE 8–31
Schematic for Example 8–4.

bends, valves, and connectors. **4** The piping section involves no work devices such as a pump or a turbine.

Properties The density and dynamic viscosity of water are given to be $\rho = 62.36$ lbm/ft^3 and $\mu = 7.536 \times 10^{-4}$ lbm/ft·s. For stainless steel, $\varepsilon = 0.000007$ ft (Table 8–3).

Analysis First we calculate the mean velocity and the Reynolds number to determine the flow regime:

$$V = \frac{\dot{V}}{A_c} = \frac{\dot{V}}{\pi D^2/4} = \frac{0.2 \text{ ft}^3/\text{s}}{\pi(2/12 \text{ ft})^2/4} = 9.17 \text{ ft/s}$$

$$\text{Re} = \frac{\rho V D}{\mu} = \frac{(62.36 \text{ lbm/ft}^3)(9.17 \text{ ft/s})(2/12 \text{ ft})}{7.536 \times 10^{-4} \text{ lbm/ft·s}} = 126{,}400$$

Since Re is greater than 10,000, the flow is turbulent. The relative roughness of the tube is

$$\varepsilon/D = \frac{0.000007 \text{ ft}}{2/12 \text{ ft}} = 0.000042$$

The friction factor corresponding to this relative roughness and Reynolds number can simply be determined from the Moody chart. To avoid the reading error, we determine it from the Colebrook equation:

$$\frac{1}{\sqrt{f}} = -2.0 \log\left(\frac{\varepsilon/D}{3.7} + \frac{2.51}{\text{Re}\sqrt{f}}\right) \rightarrow \frac{1}{\sqrt{f}} = -2.0 \log\left(\frac{0.000042}{3.7} + \frac{2.51}{126{,}400\sqrt{f}}\right)$$

Using an equation solver or an iterative scheme, the friction factor is determined to be $f = 0.0174$. Then the pressure drop and the required power input become

$$\Delta P = f\frac{L}{D}\frac{\rho V^2}{2} = 0.0174\frac{200 \text{ ft}}{2/12 \text{ ft}}\frac{(62.36 \text{ lbm/ft}^3)(9.17 \text{ ft/s})^2}{2}\left(\frac{1 \text{ lbf}}{32.174 \text{ lbm·ft/s}^2}\right)$$

$$= \mathbf{1700 \text{ lbf/ft}^2 = 11.8 \text{ psi}}$$

$$\dot{W}_{\text{pump}} = \dot{V}\Delta P = (0.2 \text{ ft}^3/\text{s})(1700 \text{ lbf/ft}^2)\left(\frac{1 \text{ W}}{0.73756 \text{ lbf·ft/s}}\right) = \mathbf{461 \text{ W}}$$

Therefore, power input in the amount of 461 W is needed to overcome the frictional losses in the tube.

Discussion The friction factor could also be determined easily from the explicit Haaland relation. It would give $f = 0.0172$, which is sufficiently close to 0.0174. Also, the friction factor corresponding to $\varepsilon = 0$ in this case is 0.0170, which indicates that this stainless-steel pipe can be assumed to be smooth with negligible error.

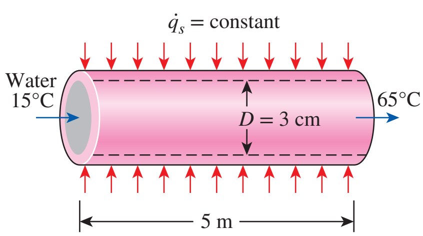


FIGURE 8–32
Schematic for Example 8–5.

EXAMPLE 8–5 Heating of Water by Resistance Heaters with Turbulent Flow in a Tube

Water is to be heated from 15°C to 65°C as it flows through a 3-cm-internal-diameter 5-m-long tube (Fig. 8–32). The tube is equipped with an electric resistance heater that provides uniform heating throughout the surface of the tube. The outer surface of the heater is well insulated, so in steady operation, all the heat generated in the heater is transferred to the water in the tube. If the system is to provide hot water at a rate of 10 L/min, determine the power rating of the resistance heater. Also estimate the inner surface temperature of the tube at the exit.

SOLUTION Water is to be heated in a tube equipped with an electric resistance heater on its surface. The power rating of the heater and the inner surface temperature at the exit are to be determined.

Assumptions **1** Steady flow conditions exist. **2** The surface heat flux is uniform. **3** The inner surfaces of the tube are smooth.

Properties The properties of water at the bulk mean temperature of $T_b = (T_i + T_e)/2 = (15 + 65)/2 = 40°C$ are (Table A–9)

$$\rho = 992.1 \text{ kg/m}^3 \qquad c_p = 4179 \text{ J/kg·K}$$

$$k = 0.631 \text{ W/m·K} \qquad \text{Pr} = 4.32$$

$$\nu = \mu/\rho = 0.658 \times 10^{-6} \text{ m}^2\text{/s}$$

Analysis The cross-sectional and heat transfer surface areas are

$$A_c = \tfrac{1}{4}\pi D^2 = \tfrac{1}{4}\pi(0.03 \text{ m})^2 = 7.069 \times 10^{-4} \text{ m}^2$$

$$A_s = \pi DL = \pi(0.03 \text{ m})(5 \text{ m}) = 0.471 \text{ m}^2$$

The volume flow rate of water is given as $\dot{V} = 10$ L/min $= 0.01$ m^3/min. Then the mass flow rate becomes

$$\dot{m} = \rho\,\dot{V} = (992.1 \text{ kg/m}^3)(0.01 \text{ m}^3\text{/min}) = 9.921 \text{ kg/min} = 0.1654 \text{ kg/s}$$

To heat the water at this mass flow rate from 15°C to 65°C, heat must be supplied to the water at a rate of

$$\begin{aligned}\dot{Q} &= \dot{m}c_p(T_e - T_i)\\ &= (0.1654 \text{ kg/s})(4.179 \text{ kJ/kg·K})(65 - 15)°\text{C}\\ &= 34.6 \text{ kJ/s} = 34.6 \text{ kW}\end{aligned}$$

All of this energy must come from the resistance heater. Therefore, the power rating of the heater must be **34.6 kW**.

The surface temperature T_s of the tube at any location can be determined from

$$\dot{q}_s = h(T_s - T_m) \rightarrow T_s = T_m + \frac{\dot{q}_s}{h}$$

where h is the heat transfer coefficient and T_m is the mean temperature of the fluid at that location. The surface heat flux is constant in this case, and its value can be determined from

$$\dot{q}_s = \frac{\dot{Q}}{A_s} = \frac{34.6 \text{ kW}}{0.471 \text{ m}^2} = 73.46 \text{ kW/m}^2$$

To determine the heat transfer coefficient, we first need to find the mean velocity of water and the Reynolds number:

$$V_{\text{avg}} = \frac{\dot{V}}{A_c} = \frac{0.010 \text{ m}^3\text{/min}}{7.069 \times 10^{-4} \text{ m}^2} = 14.15 \text{ m/min} = 0.236 \text{ m/s}$$

$$\text{Re} = \frac{V_{\text{avg}}D}{\nu} = \frac{(0.236 \text{ m/s})(0.03 \text{ m})}{0.658 \times 10^{-6} \text{ m}^2\text{/s}} = 10{,}760$$

which is greater than 10,000. Therefore, the flow is turbulent, and the entry length is roughly

$$L_h \approx L_t \approx 10D = 10 \times 0.03 = 0.3 \text{ m}$$

which is much shorter than the total length of the tube. Therefore, we can assume fully developed turbulent flow in the entire tube and determine the Nusselt number from

$$\text{Nu} = \frac{hD}{k} = 0.023 \text{ Re}^{0.8} \text{ Pr}^{0.4} = 0.023\,(10{,}760)^{0.8}\,(4.32)^{0.4} = 69.4$$

Then,

$$h = \frac{k}{D}\text{Nu} = \frac{0.631 \text{ W/m·K}}{0.03 \text{ m}}(69.4) = 1460 \text{ W/m}^2\text{·K}$$

and the inner surface temperature of the tube at the exit becomes

$$T_s = T_m + \frac{\dot{q}_s}{h} = 65°\text{C} + \frac{73{,}460 \text{ W/m}^2}{1460 \text{ W/m}^2\text{·K}} = \mathbf{115°C}$$

Discussion Note that the inner surface temperature of the tube will be 50°C higher than the mean water temperature at the tube exit. This temperature difference of 50°C between the water and the surface will remain constant throughout the fully developed flow region.

EXAMPLE 8–6 Heating of Water by Resistance Heaters with Transitional Flow in a Tube

Repeat Example 8–5 for a volume flow rate of 2.5 L/min (one-fourth of the initial value).

SOLUTION Same as Example 8–5 with a volume flow rate which is one-fourth of the initial value.

Assumptions Same as Example 8–5.

Properties Same as Example 8–5.

Analysis The cross-sectional and heat transfer surface areas are the same as in Example 8–5.

With the water volume flow rate of 2.5 L/min (one-fourth of the initial value), the mass flow rate becomes 0.04134 kg/s, and the velocity for this mass flow rate is 0.059 m/s.

To heat the water at this mass flow rate from 15°C to 65°C, heat must be supplied to the water from the resistance heater at a rate of **8.64 kW** (one-fourth of the initial value).

The surface heat flux is constant in this case, and its value is 18.34 kW/m^2 (one-fourth of the initial value).

Next, the Reynolds number and the convection heat transfer coefficient will be found.

$$\text{Re} = \frac{V_{\text{avg}}D}{\nu} = \frac{(0.059 \text{ m/s})(0.03 \text{ m})}{0.658 \times 10^{-6} \text{ m}^2\text{/s}} = 2690$$

which is between 2300 and 4000. Therefore, the flow is in the transitional region. With the Reynolds number known, it is possible to obtain the friction factor and the Nusselt number. The friction factor for transitional flow in a smooth tube with Re = 2690 is obtained from

$$f = 3.03 \times 10^{-12} \text{ Re}^3 - 3.67 \times 10^{-8} \text{ Re}^2 + 1.46 \times 10^{-4} \text{ Re} - 0.151 = 0.0352$$

To calculate the transitional Nusselt number, as it was noted earlier, the most accurate way is to use Gnielinski's (2013) proposed linear interpolation in the transition region between the Nusselt numbers at Re = 2300 and Re = 4000. However, for this problem, we will use Gnielinski's (1976) fully developed turbulent flow heat transfer correlation, which is slightly less accurate since the lower Reynolds number limit of this equation is slightly higher than the Reynolds number of 2690 of this problem. The Nusselt number in the transitional flow with Re = 2690, Pr = 4.32, and $f = 0.0352$ is obtained from

$$\text{Nu} = \frac{(f/8)(\text{Re} - 1000)\text{Pr}}{1 + 12.7(f/8)^{0.5}(\text{Pr}^{2/3} - 1)} = 13.4$$

Then, from the Nusselt number, the convective heat transfer coefficient can be found,

$$h = \frac{k}{D}\text{Nu} = \frac{0.631 \text{ W/m·K}}{0.03 \text{ m}}(13.4) = 281.8 \text{ W/m}^2\text{·K}$$

and the inner surface temperature of the tube at the exit becomes

$$T_s = T_m + \frac{\dot{q}_s}{h} = 65°\text{C} + \frac{18{,}340 \text{ W/m}^2}{281.8 \text{ W/m}^2\text{·K}} = \mathbf{130.1°C}$$

Discussion Note that the inner surface temperature of the tube will be 65°C higher than the mean water temperature at the tube exit. This temperature difference of 65°C between the water and the surface will remain constant throughout the fully developed flow region. Also note that in comparison to Example 8–5, the convective heat transfer coefficient is much lower. This reflects the lower volume flow rate of water, which gives rise to a higher inner surface temperature at the exit.

PtD *EXAMPLE 8–7* Pipe Insulation for Thermal Burn Prevention

A 10-m-long metal pipe ($k_{\text{pipe}} = 15$ W/m·K) with an inner diameter of 5 cm and an outer diameter of 6 cm is used for transporting hot saturated water vapor at a flow rate of 0.05 kg/s (Fig. 8–33). The water vapor enters and exits the pipe at 350°C and 290°C, respectively. To protect workers from thermal burns, the pipe is covered with a 2.25-cm-thick layer of insulation ($k_{\text{ins}} = 0.95$ W/m·K) to ensure that the outer surface temperature $T_{s,o}$ is below 45°C. Determine whether or not the thickness of the insulation is sufficient to alleviate the risk of thermal burns.

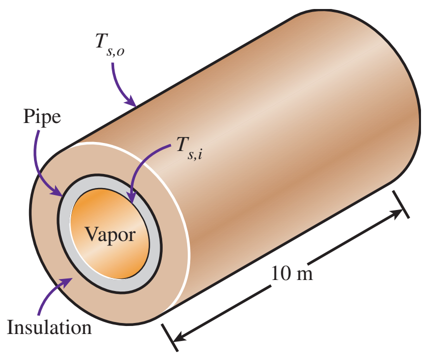


FIGURE 8–33
Schematic for Example 8–7.

SOLUTION In this example, the concepts of PtD are applied in conjunction with the concepts of internal forced convection and steady one-dimensional heat conduction. The inner pipe surface temperature $T_{s,i}$ is determined using the concept of internal forced convection. Having determined the inner surface temperature, we determine the outer surface temperature $T_{s,o}$ using one-dimensional heat conduction through the pipe wall and insulation.

Assumptions **1** Steady operating conditions exist. **2** Radiation effects are negligible. **3** Convection effects on the outer pipe surface are negligible. **4** One-dimensional heat conduction through pipe wall and insulation. **5** The thermal conductivities of pipe wall and insulation are constant. **6** Thermal resistance at the interface is negligible. **7** The surface temperatures are uniform. **8** The inner surfaces of the tube are smooth.

Properties The properties of saturated water vapor at $T_b = (T_i + T_e)/2 = 320°\text{C}$ are $c_p = 7900$ J/kg·K, $k = 0.0836$ W/m·K, $\mu = 2.084 \times 10^{-5}$ kg/m·s, and Pr = 1.97 (Table A–9).

The thermal conductivities of the pipe and the insulation are given to be $k_{pipe} = 15$ W/m·K and $k_{ins} = 0.95$ W/m·K, respectively.

Analysis The Reynolds number of the saturated water vapor flow in the pipe is

$$\text{Re} = \frac{4\dot{m}}{\pi D_i \mu} = \frac{4(0.05 \text{ kg/s})}{\pi(0.05 \text{ m})(2.084 \times 10^{-5} \text{ kg/m·s})} = 61{,}069 > 10{,}000$$

Therefore, the flow is turbulent, and the entry lengths in this case are roughly

$$L_h \approx L_t \approx 10D = 10(0.05 \text{ m}) = 0.5 \text{ m (assume fully developed turbulent flow)}$$

The Nusselt number can be determined from the Gnielinski correlation:

$$\text{Nu} = \frac{(f/8)(\text{Re} - 1000)\text{Pr}}{1 + 12.7(f/8)^{0.5}(\text{Pr}^{2/3} - 1)} = \frac{(0.02003/8)(61{,}096 - 1000)(1.97)}{1 + 12.7(0.02003/8)^{0.5}(1.97^{2/3} - 1)} = 217.45$$

where

$$f = (0.790 \ln \text{Re} - 1.64)^{-2} = 0.02003$$

Thus, the convection heat transfer coefficient for the saturated water vapor flow inside the pipe is

$$h = \frac{k}{D_i}\text{Nu} = \frac{0.0836 \text{ W/m·K}}{0.05 \text{ m}}(217.45) = 363.58 \text{ W/m}^2\text{·K}$$

The inner pipe surface temperature is

$$T_e = T_{s,i} - (T_{s,i} - T_i)\exp\left(-\frac{hA_s}{\dot{m}c_p}\right) \longrightarrow T_{s,i} = 271.52°\text{C}$$

where

$$A_s = \pi(0.05 \text{ m})(10 \text{ m}) = 1.571 \text{ m}^2$$

From Chap. 3, the thermal resistances for the pipe wall and the insulation are

$$R_{\text{pipe}} = \frac{\ln(D_{\text{interface}}/D_i)}{2\pi k_{\text{pipe}} L} = \frac{\ln(0.06/0.05)}{2\pi(15 \text{ W/m·K})(10 \text{ m})} = 1.9345 \times 10^{-4} \text{ K/W}$$

$$R_{\text{ins}} = \frac{\ln(D_o/D_{\text{interface}})}{2\pi k_{\text{ins}} L} = \frac{\ln(0.105/0.06)}{2\pi(0.95 \text{ W/m·K})(10 \text{ m})} = 9.3753 \times 10^{-3} \text{ K/W}$$

where $D_o = 0.06 \text{ m} + 2(0.0225 \text{ m}) = 0.105 \text{ m}$

The total thermal resistance and the rate of heat transfer are

$$R_{\text{total}} = R_{\text{pipe}} + R_{\text{ins}} = 9.5688 \times 10^{-3} \text{ K/W and } \dot{Q} = \frac{T_{s,i} - T_{s,o}}{R_{\text{total}}} = \dot{m}c_p(T_i - T_e)$$

Thus, the outer surface temperature is

$$\begin{aligned} T_{s,o} &= T_{s,i} - R_{\text{total}}\dot{m}c_p(T_i - T_e) \\ &= 271.52°\text{C} - (9.5688 \times 10^{-3} \text{ K/W})(0.05 \text{ kg/s})(7900 \text{ J/kg·K})(350 - 290)°\text{C} \\ &= \mathbf{44.7°C} \end{aligned}$$

Discussion The insulation thickness of 2.25 cm is just barely sufficient to keep the outer surface temperature below 45°C. To ensure that the outer surface is a few degrees below 45°C, the insulation thickness should be increased slightly to 2.3 cm, which would make $T_{s,o} = 41$°C.

EXAMPLE 8–8 Heat Loss from the Ducts of a Heating System

Hot air at atmospheric pressure and 80°C enters an 8-m-long uninsulated square duct of cross section 0.2 m × 0.2 m that passes through the attic of a house at a rate of 0.15 m^3/s (Fig. 8–34). The duct is observed to be nearly isothermal at 60°C. Determine the exit temperature of the air and the rate of heat loss from the duct to the attic space.

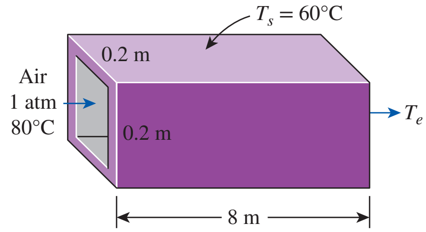


FIGURE 8–34
Schematic for Example 8–8.

SOLUTION Heat loss from uninsulated square ducts of a heating system in the attic is considered. The exit temperature and the rate of heat loss are to be determined.

Assumptions **1** Steady operating conditions exist. **2** The inner surfaces of the duct are smooth. **3** Air is an ideal gas.

Properties We do not know the exit temperature of the air in the duct, and thus we cannot determine the bulk mean temperature of air, which is the temperature at which the properties are to be determined. The temperature of air at the inlet is 80°C, and we expect this temperature to drop somewhat as a result of heat loss through the duct whose surface is at 60°C. At 80°C and 1 atm, we read (Table A–15)

$$\rho = 0.9994 \text{ kg/m}^3 \qquad c_p = 1008 \text{ J/kg·K}$$

$$k = 0.02953 \text{ W/m·K} \qquad \text{Pr} = 0.7154$$

$$\nu = 2.097 \times 10^{-5} \text{ m}^2\text{/s}$$

Analysis The characteristic length (which is the hydraulic diameter), the mean velocity, and the Reynolds number in this case are

$$D_h = \frac{4A_c}{p} = \frac{4a^2}{4a} = a = 0.2 \text{ m}$$

$$V_{\text{avg}} = \frac{\dot{V}}{A_c} = \frac{0.15 \text{ m}^3\text{/s}}{(0.2 \text{ m})^2} = 3.75 \text{ m/s}$$

$$\text{Re} = \frac{V_{\text{avg}} D_h}{\nu} = \frac{(3.75 \text{ m/s})(0.2 \text{ m})}{2.097 \times 10^{-5} \text{ m}^2\text{/s}} = 35{,}765$$

which is greater than 10,000. Therefore, the flow is turbulent, and the entry lengths in this case are roughly

$$L_h \approx L_t \approx 10D = 10 \times 0.2 \text{ m} = 2 \text{ m}$$

which is much shorter than the total length of the duct. Therefore, we can assume fully developed turbulent flow in the entire duct and determine the Nusselt number from

$$\text{Nu} = \frac{hD_h}{k} = 0.023\,\text{Re}^{0.8}\,\text{Pr}^{0.3} = 0.023\,(35{,}765)^{0.8}\,(0.7154)^{0.3} = 91.4$$

Then,

$$h = \frac{k}{D_h}\text{Nu} = \frac{0.02953 \text{ W/m·K}}{0.2 \text{ m}}(91.4) = 13.5 \text{ W/m}^2\text{·K}$$

$$A_s = 4aL = 4 \times (0.2 \text{ m})(8 \text{ m}) = 6.4 \text{ m}^2$$

$$\dot{m} = \rho\dot{V} = (0.9994 \text{ kg/m}^3)(0.15 \text{ m}^3\text{/s}) = 0.150 \text{ kg/s}$$

Next, we determine the exit temperature of air from

$$\begin{aligned} T_e &= T_s - (T_s - T_i)\exp(-hA_s/\dot{m}c_p) \\ &= 60^\circ\text{C} - [(60 - 80^\circ\text{C})]\exp\left[-\frac{(13.5\ \text{W/m}^2\cdot\text{K})(6.4\ \text{m}^2)}{(0.150\ \text{kg/s})(1008\ \text{J/kg}\cdot\text{K})}\right] \\ &= \mathbf{71.3^\circ C} \end{aligned}$$

Then the log mean temperature difference and the rate of heat loss from the air become

$$\Delta T_{\text{lm}} = \frac{T_i - T_e}{\ln \dfrac{T_s - T_e}{T_s - T_i}} = \frac{80 - 71.3}{\ln \dfrac{60 - 71.3}{60 - 80}} = -15.2^\circ\text{C}$$

$$\dot{Q} = hA_s\,\Delta T_{\text{lm}} = (13.5\ \text{W/m}^2\cdot\text{K})(6.4\ \text{m}^2)(-15.2^\circ\text{C}) = \mathbf{-1313\ W}$$

Therefore, air will lose heat at a rate of 1313 W as it flows through the duct in the attic.

Discussion The average fluid temperature is $(80 + 71.3)/2 = 75.7^\circ$C, which is sufficiently close to 80°C, at which we evaluated the properties of air. Therefore, it is not necessary to reevaluate the properties at this temperature and to repeat the calculations.

TOPIC OF SPECIAL INTEREST *

Transitional Flow in Tubes

An important design problem in industrial heat exchangers arises when flow inside the tubes falls into the transition region. In practical engineering design, the usual recommendation is to avoid design and operation in this region; however, this is not always feasible under design constraints. The usually cited transitional Reynolds number range of about 2300 (onset of turbulence) to 10,000 (fully turbulent condition) applies, strictly speaking, to a very steady and uniform entry flow with a rounded entrance. If the flow has a disturbed entrance typical of heat exchangers, in which there is a sudden contraction and possibly even a re-entrant entrance, the transitional Reynolds number range will be much different.

Ghajar and coworkers in a series of papers (listed in the references) have experimentally investigated the inlet configuration effects on the fully developed and developing transitional pressure drop under isothermal and heating conditions, as well as developing and fully developed transitional forced and mixed convection heat transfer in circular tubes. Based on their experimental data, they have developed practical and easy-to-use correlations for the friction coefficient and the Nusselt number in the transition region between laminar and turbulent flows. This section provides a brief summary of their work in the transition region.

*This section can be skipped without a loss of continuity.

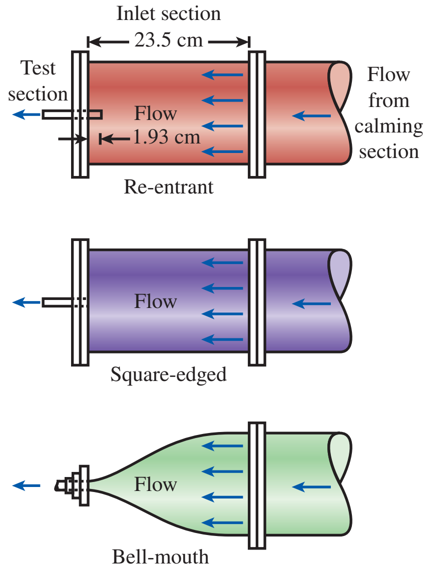


FIGURE 8–35
Schematic of the three different inlet configurations.

Pressure Drop in the Transition Region

Pressure drops are measured in circular tubes for fully developed flows in the transition regime for three types of inlet configurations shown in Fig. 8–35: re-entrant (tube extends beyond tubesheet face into head of distributor), square-edged (tube end is flush with tubesheet face), and bell-mouth (a tapered entrance of tube from tubesheet face) under isothermal and heating conditions, respectively. The widely used expressions for the *friction factor* f (also called the *Darcy friction factor*) or the *friction coefficient* C_f (also called the *Fanning friction factor*) in laminar and turbulent flows with heating are

$$f_{\text{lam}} = 4C_{f,\text{ lam}} = 4\left(\frac{16}{\text{Re}}\right)\left(\frac{\mu_b}{\mu_s}\right)^m \tag{8–81}$$

$$f_{\text{turb}} = 4C_{f,\text{ turb}} = 4\left(\frac{0.0791}{\text{Re}^{0.25}}\right)\left(\frac{\mu_b}{\mu_s}\right)^m \tag{8–82}$$

where the factors at the end account for the wall temperature effect on viscosity. The exponent m for laminar flows depends on a number of factors, while for turbulent flows the most typically quoted value for heating is -0.25. The transition friction factor is given as (Tam and Ghajar, 1997)

$$f_{\text{trans}} = 4C_{f,\text{ trans}} = 4\left[1 + \left(\frac{\text{Re}}{A}\right)^B\right]^C\left(\frac{\mu_b}{\mu_s}\right)^m \tag{8–83}$$

where

$$m = m_1 - m_2\text{Gr}^{m_3}\,\text{Pr}^{m_4} \tag{8–84}$$

and the Grashof number (Gr), which is a dimensionless number representing the ratio of the buoyancy force to the viscous force, is defined as $\text{Gr} = g\beta D^3\,(T_s - T_b)/\nu^2$ (see Chap. 9 for more details). All properties appearing in the dimensionless numbers C_f, f, Re, Pr, and Gr are evaluated at the bulk fluid temperature T_b. The values of the empirical constants in Eqs. 8–83 and 8–84 are listed in Table 8–5. The range of application of Eq. 8–83 for the transition friction factor is as follows:

Re-entrant: $2700 \le \text{Re} \le 5500$, $16 \le \text{Pr} \le 35$, $7410 \le \text{Gr} \le 158{,}300$, $1.13 \le \mu_b/\mu_s \le 2.13$

Square-edged: $3500 \le \text{Re} \le 6900$, $12 \le \text{Pr} \le 29$, $6800 \le \text{Gr} \le 104{,}500$, $1.11 \le \mu_b/\mu_s \le 1.89$

Bell-mouth: $5900 \le \text{Re} \le 9600$, $8 \le \text{Pr} \le 15$, $11{,}900 \le \text{Gr} \le 353{,}000$, $1.05 \le \mu_b/\mu_s \le 1.47$

TABLE 8–5

Constants for transition friction coefficient correlation

Inlet Geometry	A	B	C	m_1	m_2	m_3	m_4
Re-entrant	5840	–0.0145	–6.23	–1.10	0.460	–0.133	4.10
Square-edged	4230	–0.1600	–6.57	–1.13	0.396	–0.160	5.10
Bell-mouth	5340	–0.0990	–6.32	–2.58	0.420	–0.410	2.46

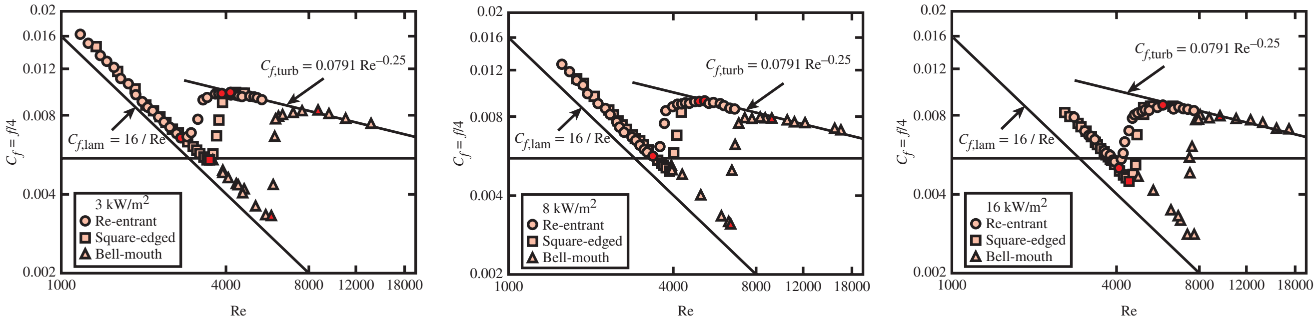


FIGURE 8–36

Fully developed friction coeffficients for three different inlet configurations and heat fluxes (filled symbols designate the start and end of the transition region for each inlet).

Source: Tam and Ghajar, 1997.

These correlations captured about 82 percent of measured data within an error band of $\pm 10\%$, and 98 percent of measured data within $\pm 20\%$. For laminar flows with heating, Tam and Ghajar (1997) give the following constants for determining the exponent m in Eq. 8–81: $m_1 = 1.65$, $m_2 = 0.013, m_3 = 0.170$, and $m_4 = 0.840$, which is applicable over the following range of parameters:

$$1100 \le \mathrm{Re} \le 7400,\ 6 \le \mathrm{Pr} \le 36,\ 17{,}100 \le \mathrm{Gr} \le 95{,}600,$$
$$\text{and } 1.25 \le \mu_b/\mu_s \le 2.40.$$

The fully developed friction coefficient results for the three different inlet configurations shown in Fig. 8–36 clearly establish the influence of heating rate on the beginning and end of the transition regions for each inlet configuration. In the laminar and transition regions, heating seems to have a significant influence on the value of the friction coefficient. However, in the turbulent region, heating did not affect the magnitude of the friction coefficient. The significant influence of heating on the values of friction coefficient in the laminar and transition regions is directly due to the effect of secondary flow.

The isothermal friction coefficients for the three inlet types showed that the range of the Reynolds number values at which transition flow exists is strongly inlet-geometry dependent. Furthermore, heating caused an increase in the laminar and turbulent friction coefficients and an increase in the lower and upper limits of the isothermal transition regime boundaries. The friction coefficient transition Reynolds number ranges for the isothermal and nonisothermal (three different heating rates) and the three different inlets used in their study are summarized in Table 8–6.

TABLE 8–6

Transition Reynolds numbers for friction coefficient

Heat Flux	Re-entrant	Square-edged	Bell-mouth
0 kW/m² (isothermal)	2870 < Re < 3500	3100 < Re < 3700	5100 < Re < 6100
3 kW/m²	3060 < Re < 3890	3500 < Re < 4180	5930 < Re < 8730
8 kW/m²	3350 < Re < 4960	3860 < Re < 5200	6480 < Re < 9110
16 kW/m²	4090 < Re < 5940	4450 < Re < 6430	7320 < Re < 9560

Figure 8–37 shows the influence of inlet configuration on the beginning and end of the isothermal fully developed friction coefficients in the transition region.

Note that the isothermal fully developed friction coefficients in the laminar, turbulent, and transition regions can be obtained easily from Eqs. 8–81, 8–82, and 8–83, respectively, by setting the viscosity ratio correction to unity (i.e., set $m = 0$).

The results presented so far were based on the experimental work of Tam and Ghajar (1997) for fully developed isothermal and nonisothermal friction factors. Ghajar and coworkers conducted similar experiments in the entrance (developing) region for square-edged and re-entrant inlets for laminar and transitional flows (Tam et al., 2013). The developing friction factor is referred to as the *apparent friction factor*. The apparent friction factor accounts for the combined effects of flow acceleration (variation in momentum flux) and surface shear forces.

For laminar developing friction factor in tubes with square-edged and re-entrant inlets, they recommended the following correlation:

$$f_{\text{app,lam}} = \left[\frac{1}{\text{Re}}\left(64 + \frac{0.01256}{4.836 \times 10^{-5} + 0.0609\left(\dfrac{x/D}{\text{Re}}\right)^{1.28}}\right)\right]\left(\frac{\mu_b}{\mu_s}\right)^m \quad \textbf{(8–85)}$$

where $m = -5.06 + 0.84 \times \text{Pr}^{0.23}\ \text{Gr}^{0.09}$ for nonisothermal flow and $m = 0$ for isothermal flow.

The range of application of Eq. 8–85 is: $897 \leq \text{Re} \leq 2189, 7141 \leq \text{Gr} \leq 18{,}224, 1.27 \leq \mu_b/\mu_s \leq 1.56, 39 \leq \text{Pr} \leq 47$, and $3 \leq x/D \leq 200$.

For transitional developing friction factor in tubes, they recommended the following correlation:

$$f_{\text{app,trans}} = \left\{\left(\frac{64}{\text{Re}}\right)[(1 + (0.0049\,\text{Re}^{0.75})^a)^{1/a} + b]\left[1 + \left(\frac{c}{x/D}\right)\right]\right\}\left(\frac{\mu_b}{\mu_s}\right)^m \quad \textbf{(8–86)}$$

where the exponent m is obtained separately for each inlet for nonisothermal flow from Table 8–7 and is set to $m = 0$ for isothermal flow. The constants in Eq. 8–86 and its range of application for each inlet are also given in Table 8–7.

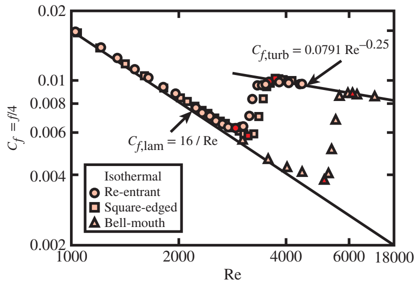


FIGURE 8–37

Influence of different inlet configurations on the isothermal fully developed friction coefficients (filled symbols designate the start and end of the transition region for each inlet).

Source: Tam and Ghajar, 1997.

TABLE 8–7

Constants and ranges of independent variables for transition friction factor correlation

Inlet Geometry	Constants and Ranges of Independent Variables
Re-entrant	$a = 0.52, b = -3.47, c = 4.8, m = -1.8 + 0.46\ \text{Gr}^{-0.13}\ \text{Pr}^{0.41}$ $1883 < \text{Re} < 3262, 19.1 < \text{Pr} < 46.5, 4560 < \text{Gr} < 24{,}339,$ $1.12 < \mu_b/\mu_s < 1.54, 3 < x/D < 200$
Square-edged	$a = 0.50, b = -4.0, c = 3.0, m = -1.13 + 0.48\ \text{Gr}^{-0.15}\ \text{Pr}^{0.55}$ $2084 < \text{Re} < 3980, 19.6 < \text{Pr} < 47.3, 6169 < \text{Gr} < 35{,}892,$ $1.10 < \mu_b/\mu_s < 1.54, 3 < x/D < 200$

EXAMPLE 8–9 Nonisothermal Fully Developed Friction Coefficient in the Transition Region

A tube with a bell-mouth inlet configuration is subjected to 8 kW/m^2 uniform wall heat flux. The tube has an inside diameter of 0.0158 m and a flow rate of 1.32×10^{-4} m^3/s. The liquid flowing inside the tube is an ethylene glycol–distilled water mixture with a mass fraction of 0.34. The properties of the ethylene glycol–distilled water mixture at the location of interest are Pr = 11.6, $\nu = 1.39 \times 10^{-6}$ m^2/s, and $\mu_b/\mu_s = 1.14$. Determine the fully developed friction coefficient at a location along the tube where the Grashof number is Gr = 60,800. What would the answer be if a square-edged inlet were used instead?

SOLUTION A liquid mixture flowing in a tube is subjected to uniform wall heat flux. The friction coefficients are to be determined for the bell-mouth and square-edged inlet cases.

Assumptions Steady operating conditions exist.

Properties The properties of the ethylene glycol–distilled water mixture are given to be Pr = 11.6, $\nu = 1.39 \times 10^{-6}$ m^2/s, and $\mu_b/\mu_s = 1.14$.

Analysis For the calculation of the nonisothermal fully developed friction coefficient, it is necessary to determine the flow regime before making any decision regarding which friction coefficient relation to use. The Reynolds number at the specified location is

$$\text{Re} = \frac{(\dot{V}/A_c)D}{\nu} = \frac{[(1.32 \times 10^{-4}\ \text{m}^3/\text{s})/(1.961 \times 10^{-4}\ \text{m}^2)](0.0158\ \text{m})}{1.39 \times 10^{-6}\ \text{m}^2/\text{s}} = 7651$$

since

$$A_c = \pi D^2/4 = \pi(0.0158\ \text{m})^2/4 = 1.961 \times 10^{-4}\ \text{m}^2$$

From Table 8–6, we see that for a bell-mouth inlet and a heat flux of 8 kW/m^2, the flow is in the transition region. Therefore, Eq. 8–83 applies. Reading the constants A, B, and C and m_1, m_2, m_3, and m_4 from Table 8–5, the friction coefficient is determined to be

$$C_{f,\ \text{trans}} = \left[1 + \left(\frac{\text{Re}}{A}\right)^B\right]^C \left(\frac{\mu_b}{\mu_s}\right)^m$$
$$= \left[1 + \left(\frac{7651}{5340}\right)^{-0.099}\right]^{-6.32} (1.14)^{-2.58 - 0.42 \times 60{,}800^{-0.41} \times 11.6^{2.46}} = \mathbf{0.010}$$

Square-Edged Inlet Case For this inlet shape, the Reynolds number of the flow is the same as that of the bell-mouth inlet (Re = 7651). However, it is necessary to check the type of flow regime for this particular inlet with 8 kW/m^2 of heating. From Table 8–6, the transition Reynolds number range for this case is 3860 < Re < 5200, which means that the flow in this case is turbulent, and Eq. 8–82 is the appropriate equation to use. It gives

$$C_{f,\ \text{turb}} = \left(\frac{0.0791}{\text{Re}^{0.25}}\right)\left(\frac{\mu_b}{\mu_s}\right)^m = \left(\frac{0.0791}{7651^{0.25}}\right)(1.14)^{-0.25} = \mathbf{0.0082}$$

Discussion Note that the friction factors f can be determined by multiplying the friction coefficient values by 4.

Heat Transfer in the Transition Region

Ghajar and coworkers also experimentally investigated the inlet configuration effects on heat transfer in the transition region between laminar and turbulent flows in tubes for the same three inlet configurations shown in Fig. 8–35. They proposed some prediction methods for this regime to bridge between laminar methods and turbulent methods, applicable to forced and mixed convection in the entrance and fully developed regions for the three types of inlet configurations, which are presented next. For a detailed discussion on this topic, refer to Tam and Ghajar (2006). The local heat transfer coefficient in transition flow is obtained from the transition Nusselt number, $\mathrm{Nu}_{\mathrm{trans}}$, which is calculated as follows at a distance x from the entrance:

$$\mathrm{Nu}_{\mathrm{trans}} = \mathrm{Nu}_{\mathrm{lam}} + \{\exp[(a - \mathrm{Re})/b] + \mathrm{Nu}_{\mathrm{turb}}^{c}\}^{c} \qquad \textbf{(8–87)}$$

where $\mathrm{Nu}_{\mathrm{lam}}$ is the laminar flow Nusselt number for entrance region laminar flows with natural convection effects,

$$\mathrm{Nu}_{\mathrm{lam}} = 1.24\left[\left(\frac{\mathrm{Re}\,\mathrm{Pr}D}{x}\right) + 0.025(\mathrm{Gr}\,\mathrm{Pr})^{0.75}\right]^{1/3}\left(\frac{\mu_b}{\mu_s}\right)^{0.14} \qquad \textbf{(8–88)}$$

and $\mathrm{Nu}_{\mathrm{turb}}$ is the turbulent flow Nusselt number with developing flow effects,

$$\mathrm{Nu}_{\mathrm{turb}} = 0.023\mathrm{Re}^{0.8}\,\mathrm{Pr}^{0.385}\left(\frac{x}{D}\right)^{-0.0054}\left(\frac{\mu_b}{\mu_s}\right)^{0.14} \qquad \textbf{(8–89)}$$

The physical properties appearing in the dimensionless numbers Nu, Re, Pr, and Gr all are evaluated at the bulk fluid temperature T_b. The values of the empirical constants a, b, and c in Eq. 8–87 depend on the inlet configuration and are given in Table 8–8. The viscosity ratio accounts for the temperature effect on the process. The range of application of the heat transfer method based on their database of 1290 points (441 points for re-entrant inlet, 416 for square-edged inlet, and 433 points for bell-mouth inlet) is as follows:

Re-entrant: $3 \le x/D \le 192$, $1700 \le \mathrm{Re} \le 9100$, $5 \le \mathrm{Pr} \le 51$, $4000 \le \mathrm{Gr} < 210{,}000$, $1.2 < \mu_b/\mu_s < 2.2$

Square-edged: $3 \le x/D \le 192$, $1600 \le \mathrm{Re} \le 10{,}700$, $5 \le \mathrm{Pr} \le 55$, $4000 \le \mathrm{Gr} \le 250{,}000$, $1.2 \le \mu_b/\mu_s \le 2.6$

Bell-mouth: $3 \le x/D \le 192$, $3300 \le \mathrm{Re} \le 11{,}100$, $13 \le \mathrm{Pr} \le 77$, $6000 \le \mathrm{Gr} \le 110{,}000$, $1.2 \le \mu_b/\mu_s \le 3.1$

These correlations capture about 70 percent of measured data within an error band of $\pm 10\%$, and 97 percent of measured data within $\pm 20\%$, which is remarkable for transition flows. The individual expressions above for $\mathrm{Nu}_{\mathrm{lam}}$ and $\mathrm{Nu}_{\mathrm{turb}}$ can be used alone for developing flows in those respective regimes. The lower and upper limits of the heat transfer transition Reynolds number ranges for the three different inlets are summarized in Table 8–9. The results shown in this table indicate that the re-entrant inlet configuration causes the earliest transition from laminar flow into the transition regime (at about 2000), while the bell-mouth entrance retards this regime change (at about 3500). The square-edged entrance falls in between (at about 2400), which is close to the often-quoted value of 2300 in most textbooks.

TABLE 8–8

Constants for transition heat transfer correlation

Inlet Geometry	a	b	c
Re-entrant	1766	276	–0.955
Square-edged	2617	207	–0.950
Bell-mouth	6628	237	–0.980

TABLE 8–9

The lower and upper limits of the heat transfer transition Reynolds numbers

Inlet Geometry	Lower Limit	Upper Limit
Re-entrant	$Re_{lower} = 2157 - 0.65[192 - (x/D)]$	$Re_{upper} = 8475 - 9.28[192 - (x/D)]$
Square-edged	$Re_{lower} = 2524 - 0.82[192 - (x/D)]$	$Re_{upper} = 8791 - 7.69[192 - (x/D)]$
Bell-mouth	$Re_{lower} = 3787 - 1.80[192 - (x/D)]$	$Re_{upper} = 10{,}481 - 5.47[192 - (x/D)]$

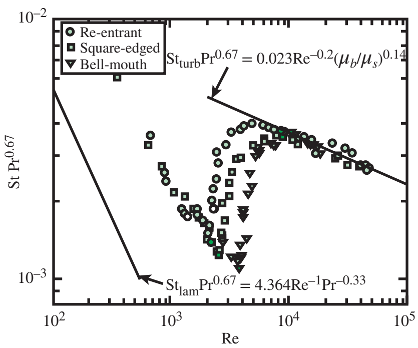


FIGURE 8–38
Influence of different inlets on the heat transfer transition region at $x/D = 192$ (filled symbols designate the start and end of the transition region for each inlet) between limits of Dittus–Boelter correlation ($Nu = 0.023\ Re^{0.8}\ Pr^n$) for fully developed turbulent flow (using $n = 1/3$ for heating) and $Nu = 4.364$ for fully developed laminar flow with a uniform heat flux boundary condition. Note buoyancy effect on the laminar flow data, giving the much larger mixed convection heat transfer coefficient.

Source: Ghajar and Tam, 1994.

Figure 8–38 clearly shows the influence of inlet configuration on the beginning and end of the heat transfer transition region. This figure plots the local average peripheral heat transfer coefficients in terms of the Colburn j factor ($j_H = St\ Pr^{0.67}$) versus local Reynolds number for all flow regimes at the length-to-diameter ratio of 192, and St is the Stanton number, which is also a dimensionless heat transfer coefficient (see Chapter 6 for more details), defined as $St = Nu/(Re\ Pr)$. The filled symbols in Fig. 8–38 represent the start and end of the heat transfer transition region for each inlet configuration. Note the large influence of natural convection superimposed on the forced convective laminar-flow heat transfer process ($Nu = 4.364$ for a fully developed laminar flow with a uniform heat-flux boundary condition without buoyancy effects), yielding a mixed convection value of about $Nu = 14.5$. Equation 8–88 includes this buoyancy effect through the Grashof number.

In a subsequent study, Tam and Ghajar (1998) experimentally investigated the behavior of local heat transfer coefficients in the transition region for a tube with a bell-mouth inlet. This type of inlet is used in some heat exchangers mainly to avoid the presence of eddies, which are believed to be one of the causes for erosion in the tube inlet region. For the bell-mouth inlet, the variation of the local heat transfer coefficient with length in the transition and turbulent flow regions is very unusual. For this inlet geometry, the boundary layer along the tube wall is at first laminar and then changes through a transition to the turbulent condition, causing a dip in the Nu versus x/D curve. In their experiments with a fixed inside diameter of 15.84 mm, the length of the dip in the transition region was much longer ($100 < x/D < 175$) than in the turbulent region ($x/D < 25$). The presence of the dip in the transition region has a significant influence on both the local and the average heat transfer coefficients. This is particularly important for heat transfer calculations in short tube heat exchangers with a bell-mouth inlet. Figure 8–39 shows the variation of local Nusselt number along the tube length in the transition region for the three inlet configurations at comparable Reynolds numbers.

As mentioned earlier, finned tubes (Fig. 8–30) enhance convection heat transfer. Single-phase liquid flow in internally enhanced tubes is becoming more important in commercial heating, ventilating, and air conditioning (HVAC) applications. One kind of internally enhanced tube is the spiral micro-fin tube (Fig. 8–40). The tube side roughening increases heat transfer surface area, resulting in high-efficiency heat exchangers. The increase in the surface area causes low flow rates in the heat exchanger tubes, resulting in flow at Reynolds numbers that are between laminar and turbulent, that is, in the transition region. Owing to the high efficiency requirements, it is

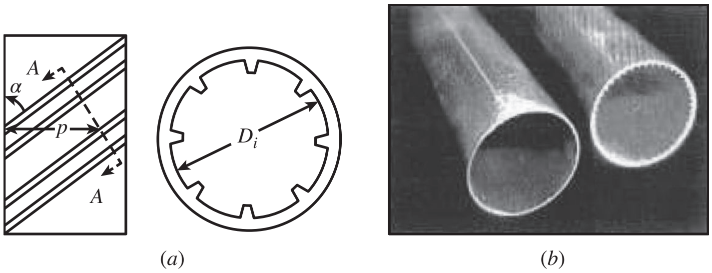


FIGURE 8–40
(*a*) Sectional view of the micro-fin tube; (*b*) The plain and micro-fin tubes.
(a) Source: Tam et al., 2012. (b) ©Afshin Ghajar

likely that more HVAC units will operate in the transition region where the understanding of the pressure drop and heat transfer is limited. To respond to some of these challenges, Ghajar and coworkers extended their plain-tube studies presented earlier to finned tubes. They conducted pressure drop (friction factor) and heat transfer experiments in the transition region in three different micro-fin tubes and a plain tube with square-edged and re-entrant inlets (Tam et al., 2012). Table 8–10 shows the specifications of the plain and micro-fin tubes used in the studies of Tam et al. (2012).

Figure 8–41 represents the fully developed friction factor characteristics for the plain and different spiral-angle micro-fin tubes and different inlet configurations under the isothermal boundary condition. The plain tube behavior is similar to what was presented earlier (Fig. 8–37). However, for all the micro-fin tubes, a parallel shift from the classic laminar equation ($C_{f,\,\text{lam}} = 16/\text{Re}$) is observed regardless of the fin geometry and the type of inlet. In the transition region, it can be observed that the transition range of the micro-fin tubes is much wider than the plain tube transition range. The friction factor for all the micro-fin tubes goes through a steep increase followed by a relatively constant C_f section and then a parallel shift from the classical Blasius turbulent friction factor correlation ($C_{f,\,\text{turb}} = 0.0791\ \text{Re}^{-0.25}$). For the end of the transition for micro-fin tube, it is defined as the first point where the friction factor data become parallel to the Blasius correlation. Beyond this point, the flow is considered turbulent. With a higher spiral angle in the micro-fin tube, the increase of friction factor can be observed in the transition and turbulent regions. The

TABLE 8–10

Specifications of the test tubes

Tube Type	Outer Dia., D_o (mm)	Inner Dia., D_i (mm)	Spiral Angle, α	Fin Height, e (mm)	Number of Start, N_s
Plain tube	15.9	14.9	—	—	—
Micro-fin tube #1	15.9	14.9	18°	0.5	25
Micro-fin tube #2	15.9	14.9	25°	0.5	25
Micro-fin tube #3	15.9	14.9	35°	0.5	25

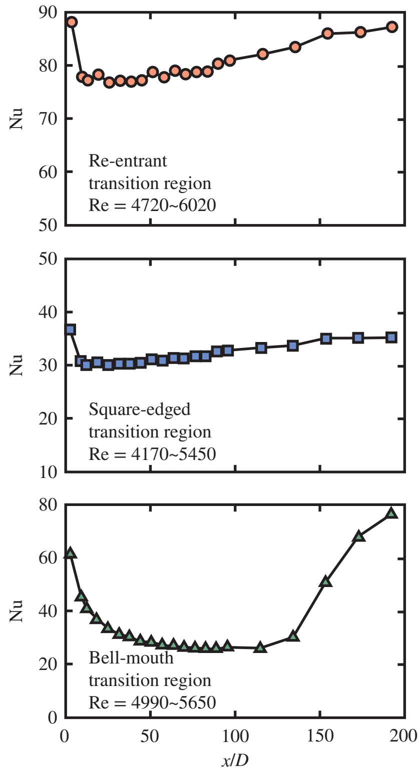


FIGURE 8–39
Variation of local Nusselt number with length for the re-entrant, square-edged, and bell-mouth inlets in the transition region.
Source: Tam and Ghajar, 1998.

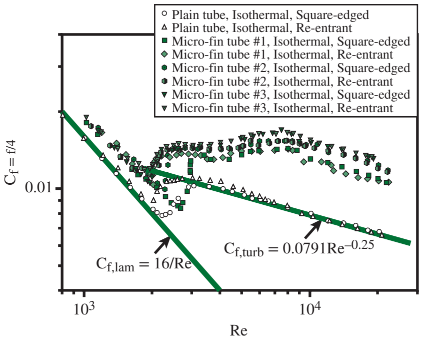


FIGURE 8–41
Fully developed friction factor characteristics for the plain and micro-fin tubes under isothermal boundary condition.
Source: Tam et al., 2012.

increase in the friction factor is caused by the stronger drag due to the larger spiral angle on the tube wall. From Fig. 8–41, it can be observed that the start of transition is inlet and spiral angle dependent. For all the micro-fin tubes, the square-edged inlet delays the start of transition. The micro-fin tube #3 with a larger spiral angle has an earlier start of transition. Moreover, it is also observed that a larger spiral angle in the micro-fin tube advances the end of transition for both inlet configurations.

Figure 8–42 shows that the heat transfer behavior in the upper transition region for the plain and micro-fin tubes is very different. For the plain tube, the heat transfer initially experienced an abrupt change in the lower transition region; this was followed by a moderate change in the upper transition, and finally the transition ended and the heat transfer data followed the Sieder and Tate correlation (1936) at a Reynolds number around 8000. The plain tube behavior is similar to what was presented earlier (Fig. 8–38). However, for the micro-fin tube, such a characteristic cannot be seen, and the abrupt change was observed in the entire transition region. When the Reynolds number is around 8000, a parallel shift of heat transfer data from the Sieder and Tate equation was observed. This increase in the Colburn j factor ($j_H = \text{St}\,\text{Pr}^{0.67}$) and the parallel shift from the Sieder and Tate correlation in the turbulent region is due to the swirling motion induced by the micro-fin. The larger spiral angle leads to a higher Colburn j factor. It is because the larger spiral angle increases the fluid mixing between the bulk fluid and the tube wall, and therefore the heat transfer is enhanced. Regarding the start and end of transition of heat transfer, it can be observed from Fig. 8–42 that the start of transition is defined as the critical point for the sudden change from the laminar region (the line parallel to Nu = 4.364) to the transition region. For the end of the transition for micro-fin tube, it is defined as the first point where the heat transfer data landed on a line parallel to the correlation proposed by Sieder and Tate (1936). As seen from Fig. 8–42, for the heat transfer, the transition is inlet dependent and the start and end of transition is spiral angle dependent. The delay of transition is obvious for smaller spiral angles, while an early transition occurred when larger spiral angle tubes were used.

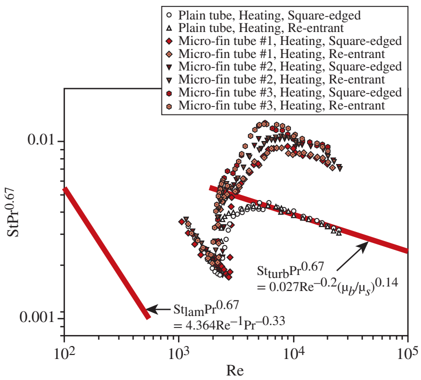


FIGURE 8–42
Fully developed heat transfer characteristics for the plain and micro-fin tubes under uniform wall heat flux boundary condition.
Source: Tam et al., 2012.

EXAMPLE 8–10 Heat Transfer in the Transition Region

An ethylene glycol–distilled water mixture with a mass fraction of 0.6 and a flow rate of 2.6×10^{-4} m^3/s flows inside a tube with an inside diameter of 0.0158 m subjected to uniform wall heat flux. For this flow, determine the Nusselt number at the location $x/D = 90$ if the inlet configuration of the tube is: (*a*) re-entrant, (*b*) square-edged, and (*c*) bell-mouth. At this location, the local Grashof number is Gr = 51,770. The properties of an ethylene glycol–distilled water mixture at the location of interest are Pr = 29.2, $\nu = 3.12 \times 10^{-6}$ m^2/s, and $\mu_b/\mu_s = 1.77$.

SOLUTION A liquid mixture flowing in a tube is subjected to uniform wall heat flux. The Nusselt number at a specified location is to be determined for three different tube inlet configurations.

Assumptions Steady operating conditions exist.

Properties The properties of the ethylene glycol–distilled water mixture are given to be Pr = 29.2, $\nu = 3.12 \times 10^{-6}$ m²/s, and $\mu_b/\mu_s = 1.77$.

Analysis For a tube with a known diameter and volume flow rate, the type of flow regime is determined before making any decision regarding which Nusselt number correlation to use. The Reynolds number at the specified location is

$$\text{Re} = \frac{(\dot{V}/A_c)D}{\nu}\frac{[(2.6 \times 10^{-4}\,\text{m}^3/\text{s})/(1.961 \times 10^{-4}\,\text{m}^2)](0.0158\text{ m})}{3.12 \times 10^{-6}\,\text{m}^2/\text{s}} = 6714$$

since

$$A_c = \pi D^2/4 = \pi(0.0158\text{ m})^2/4 = 1.961 \times 10^{-4}\text{ m}^2$$

Therefore, the flow regime is in the transition region for all three inlet configurations (thus use the information given in Table 8–9 with $x/D = 90$), and therefore Eq. 8–87 should be used with the constants *a, b, c* found in Table 8–8. However, Nu_{lam} and Nu_{turb} are the inputs to Eq. 8–87, and they need to be evaluated first from Eqs. 8–88 and 8–89, respectively. It should be mentioned that the correlations for Nu_{lam} and Nu_{turb} have no inlet dependency.

From Eq. 8–88:

$$\text{Nu}_{\text{lam}} = 1.24\left[\left(\frac{\text{Re Pr}D}{x}\right) + 0.025\,(\text{Gr Pr})^{0.75}\right]^{1/3}\left(\frac{\mu_b}{\mu_s}\right)^{0.14}$$

$$= 1.24\left[\left(\frac{(6714)(29.2)}{90}\right) + 0.025[(51{,}770)(29.2)]^{0.75}\right]^{1/3}(1.77)^{0.14} = 19.9$$

From Eq. 8–89:

$$\text{Nu}_{\text{turb}} = 0.023\ \text{Re}^{0.8}\text{Pr}^{0.385}\left(\frac{x}{D}\right)^{-0.0054}\left(\frac{\mu_b}{\mu_s}\right)^{0.14}$$

$$= 0.023(6714)^{0.8}(29.2)^{0.385}(90)^{-0.0054}(1.77)^{0.14} = 102.7$$

Then the transition Nusselt number can be determined from Eq. 8–87,

$$\text{Nu}_{\text{trans}} = \text{Nu}_{\text{lam}} + \{\exp[(a - \text{Re})/b] + \text{Nu}_{\text{turb}}^{c}\}^{c}$$

Case 1: For re-entrant inlet:

$$\text{Nu}_{\text{trans}} = 19.9 + \{\exp[(1766 - 6714)/276] + 102.7^{-0.955}\}^{-0.955} = \mathbf{88.2}$$

Case 2: For square-edged inlet:

$$\text{Nu}_{\text{trans}} = 19.9 + \{\exp[(2617 - 6714)/207] + 102.7^{-0.950}\}^{-0.950} = \mathbf{85.3}$$

Case 3: For bell-mouth inlet:

$$\text{Nu}_{\text{trans}} = 19.9 + \{\exp[(6628 - 6714)/237] + 102.7^{-0.980}\}^{-0.980} = \mathbf{21.3}$$

Discussion It is worth mentioning that, for the re-entrant and square-edged inlets, the flow behaves normally. For the bell-mouth inlet, the Nusselt number is low in comparison to the other two inlets. This is because of the unusual behavior of the bell-mouth inlet noted earlier (see Fig. 8–39); i.e., the boundary layer along the tube wall is at first laminar and then changes through a transition region to the turbulent condition.

Analogies Between Momentum and Heat Transfer in the Transition Region

Meyer and Everts (2018b) experimentally investigated the pressure drop and heat transfer characteristics of developing and fully developed flow in smooth tubes in the laminar, transitional, quasi-turbulent, and turbulent flow regimes. The Reynolds number was varied between 700 and 10,000 to ensure that the transitional and quasi-turbulent flow regimes, as well as sufficient parts of the laminar and turbulent flow regimes, were covered. It was found that the boundaries of the different flow regimes were the same for pressure drop and heat transfer, and that a direct relationship between pressure drop and heat transfer existed in the transitional flow regime as well. In the laminar flow regime, the relationship between pressure drop and heat transfer was a function of Grashof number (thus free convection effects), while it was a function of Reynolds number in the other three flow regimes. Correlations to predict the average friction factors as a function of average Nusselt number, as well as correlations to predict the average Nusselt number for developing and fully developed flow in all flow regimes, were also developed.

Pressure Drop in the Transition Region in Mini and Micro Tubes

Due to rapid advancement in fabrication techniques, the miniaturization of devices and components is ever increasing in many applications. Whether it is in the application of miniature heat exchangers, fuel cells, pumps, compressors, turbines, sensors, or artificial blood vessels, a sound understanding of fluid flow in microscale channels and tubes is required. To better understand the flow behavior in small tubes, Ghajar and coworkers extended their isothermal transitional pressure drop studies in standard-size tubes to mini- and microtubes. They performed a systematic and careful experimental study of friction factor in the transition region for single-phase water flow in 12 stainless steel tubes with diameters ranging from 2083 μm to 337 μm (Fig. 8–43). The pressure-drop measurements were carefully performed by paying particular attention to the sensitivity of the pressure-sensing diaphragms used in the pressure transducer. Experimental results indicated that the start and end of the transition region were influenced by the tube diameter (Table 8–11). The friction factor profile was not significantly affected for

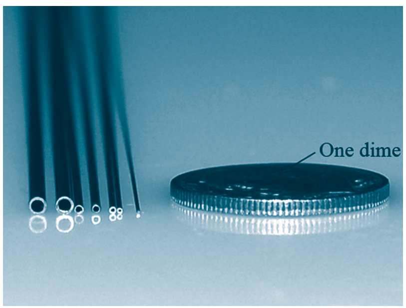


FIGURE 8–43
Representative mini/micro stainless steel tubes used in the experiments of Ghajar et al. (2010).
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TABLE 8–11

Summary of transition Reynolds number ranges for various stainless steel tube sizes

Tube I.D. [μm]	Transition Range	Tube I.D. [μm]	Transition Range
2083	1500 < Re < 4000	732	2200 < Re < 3000
1600	1700 < Re < 4000	667	2200 < Re < 3000
1372	1900 < Re < 4000	559	1900 < Re < 2500
1067	2000 < Re < 4000	508	1700 < Re < 2100
991	2000 < Re < 4000	413	1500 < Re < 1900
838	2200 < Re < 4000	337	1300 < Re < 1700

Source: Ghajar et al., 2010.

the tube diameters between 2083 μm and 1372 μm (Fig. 8–44). However, the influence of the tube diameter on the friction factor profile became noticeable as the diameter decreased from 838 μm to 337 μm (Fig. 8–45). The Reynolds number range for transition flow became narrower with decreasing tube diameter (Table 8–11). The results show that a decrease in tube diameter and an increase in relative roughness (ε/D), which is approximately within the range of $0.01 < \varepsilon/D < 0.04$, influence the friction factor, even in the laminar region. In addition, these factors cause the onset of transition from laminar flow to occur at lower Reynolds numbers.

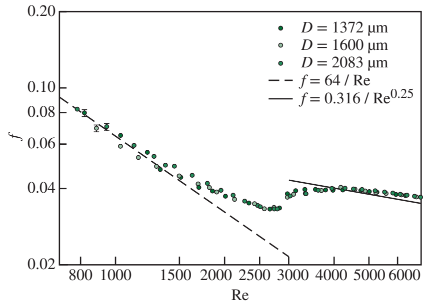

FIGURE 8–44

Transition region of stainless steel tubes with diameters from 2083 to 1372 μm.

Source: Ghajar et al., 2010.

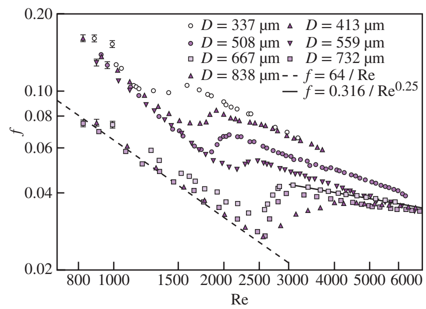

FIGURE 8–45

Transition region of stainless steel tubes with diameters from 838 to 337 μm.

Source: Ghajar et al., 2010.

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SUMMARY

Internal flow is characterized by the fluid being completely confined by the inner surfaces of the tube. The mean or average velocity and temperature for a circular tube of radius R are expressed as

$$V_{\text{avg}} = \frac{2}{R^2}\int_0^R u(r)rdr \quad \text{and} \quad T_m = \frac{2}{V_{\text{avg}}R^2}\int_0^R u(r)T(r)rdr$$

The Reynolds number for internal flow and the hydraulic diameter are defined as

$$\text{Re} = \frac{\rho V_{\text{avg}} D}{\mu} = \frac{V_{\text{avg}} D}{\nu} \quad \text{and} \quad D_h = \frac{4A_c}{p}$$

The flow in a tube is laminar for $\text{Re} < 2300$, turbulent for about $\text{Re} > 10{,}000$, and transitional in between.

The length of the region from the tube inlet to the point at which the flow becomes fully developed is the *hydrodynamic entry length* L_h. The region beyond the entrance region in which the velocity profile is fully developed is the *hydrodynamically fully developed region.* The length of the region of flow over which the thermal boundary layer develops and reaches the tube center is the *thermal entry length* L_t. The region in which the flow is both hydrodynamically and thermally developed is the *fully developed flow region.* The entry lengths are given by

$$L_{h,\text{ laminar}} \approx 0.05\,\text{Re}\,D$$
$$L_{t,\text{ laminar}} \approx 0.05\,\text{Re}\,\text{Pr}\,D = \text{Pr}\,L_{h,\text{ laminar}}$$
$$L_{h,\text{ turbulent}} \approx L_{t,\text{ turbulent}} = 10D$$

For $\dot{q}_s = \text{constant}$, the rate of heat transfer is expressed as

$$\dot{Q} = \dot{q}_s A_s = \dot{m}c_p(T_e - T_i)$$

For $T_s = \text{constant}$, we have

$$\dot{Q} = hA_s\Delta T_{\text{lm}} = \dot{m}c_p(T_e - T_i)$$
$$T_e = T_s - (T_s - T_i)\exp(-hA_s/\dot{m}c_p)$$
$$\Delta T_{\text{lm}} = \frac{T_i - T_e}{\ln[(T_s - T_e)/(T_s - T_i)]} = \frac{\Delta T_e - \Delta T_i}{\ln(\Delta T_e/\Delta T_i)}$$

The irreversible pressure loss due to frictional effects and the required pumping power to overcome this loss for a volume flow rate of $\dot{V}$ are

$$\Delta P_L = f\frac{L}{D}\frac{\rho V_{\text{avg}}^2}{2} \quad \text{and} \quad \dot{W}_{\text{pump}} = \dot{V}\Delta P_L$$

For *fully developed laminar flow* in a circular pipe, we have:

$$u(r) = 2V_{\text{avg}}\left(1 - \frac{r^2}{R^2}\right) = u_{\max}\left(1 - \frac{r^2}{R^2}\right)$$
$$f = \frac{64\mu}{\rho D V_{\text{avg}}} = \frac{64}{\text{Re}}$$
$$\dot{V} = V_{\text{avg}}A_c = \frac{\Delta P R^2}{8\mu L}\pi R^2 = \frac{\pi R^4 \Delta P}{8\mu L} = \frac{\pi R^4 \Delta P}{128\mu L}$$

Circular tube, laminar ($\dot{q}_s = \text{constant}$): $\text{Nu} = \dfrac{hD}{k} = 4.36$

Circular tube, laminar ($T_s = \text{constant}$): $\text{Nu} = \dfrac{hD}{k} = 3.66$

For *developing laminar flow* in the entrance region with constant surface temperature, we have

Circular tube: $\text{Nu} = 3.66 + \dfrac{0.065(D/L)\text{Re Pr}}{1 + 0.04[(D/L)\text{Re Pr}]^{2/3}}$

Circular tube: $\text{Nu} = 1.86\left(\dfrac{\text{Re Pr}D}{L}\right)^{1/3}\left(\dfrac{\mu_b}{\mu_s}\right)^{0.14}$

Parallel plates: $\text{Nu} = 7.54 + \dfrac{0.03(D_h/L)\text{Re Pr}}{1 + 0.016[(D_h/L)\text{Re Pr}]^{2/3}}$

For *fully developed turbulent flow with smooth surfaces,* we have

$$f = (0.790 \ln \text{Re} - 1.64)^{-2} \qquad 10^4 < \text{Re} < 10^6$$
$$\text{Nu} = 0.125 f\,\text{Re}\,\text{Pr}^{1/3}$$
$$\text{Nu} = 0.023\,\text{Re}^{0.8}\,\text{Pr}^{1/3} \quad \begin{pmatrix} 0.7 \le \text{Pr} \le 160 \\ \text{Re} > 10{,}000 \end{pmatrix}$$

$\text{Nu} = 0.023\,\text{Re}^{0.8}\,\text{Pr}^n$ with $n = 0.4$ for *heating* and 0.3 for *cooling* of fluid

$$\text{Nu} = \frac{(f/8)(\text{Re} - 1000)\text{Pr}}{1 + 12.7(f/8)^{0.5}(\text{Pr}^{2/3} - 1)} \quad \begin{pmatrix} 0.5 \le \text{Pr} \le 2000 \\ 3 \times 10^3 < \text{Re} < 5 \times 10^6 \end{pmatrix}$$

The fluid properties are evaluated at the *bulk mean fluid temperature* $T_b = (T_i + T_e)/2$. For liquid metal flow in the range of $10^4 < \text{Re} < 10^6$ we have:

$T_s = \text{constant}$: $\quad \text{Nu} = 4.8 + 0.0156\,\text{Re}^{0.85}\,\text{Pr}_s^{0.93}$

$\dot{q}_s = \text{constant}$: $\quad \text{Nu} = 6.3 + 0.0167\,\text{Re}^{0.85}\,\text{Pr}_s^{0.93}$

For *fully developed turbulent flow with rough surfaces,* the friction factor f is determined from the Moody chart or

$$\frac{1}{\sqrt{f}} = -2.0 \log\left(\frac{\varepsilon/D}{3.7} + \frac{2.51}{\mathrm{Re}\sqrt{f}}\right) \approx -1.8 \log\left[\frac{6.9}{\mathrm{Re}} + \left(\frac{\varepsilon/D}{3.7}\right)^{1.11}\right]$$

For a *concentric annulus,* the hydraulic diameter is $D_h = D_o - D_i$, and the Nusselt numbers are expressed as

$$\mathrm{Nu}_i = \frac{h_i D_h}{k} \quad \text{and} \quad \mathrm{Nu}_o = \frac{h_o D_h}{k}$$

where the values for the Nusselt numbers are given in Table 8–4.

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PROBLEMS*

General Flow Analysis

8–1C Which fluid at room temperature requires a larger pump to move at a specified velocity in a given tube: water or engine oil? Why?

8–2C What do the average velocity V_{avg} and the mean temperature T_m represent in flow through circular tubes of constant diameter?

8–3C What is the generally accepted value of the Reynolds number above which the flow in smooth pipes is turbulent?

8–4C What is hydraulic diameter? How is it defined? What is it equal to for a circular tube of diameter D?

8–5C What fluid property is responsible for the development of the velocity boundary layer? For what kinds of fluids will there be no velocity boundary layer in a pipe?

*Problems designated by a "C" are concept questions, and students are encouraged to answer them all. Problems designated by an "E" are in English units, and the SI users can ignore them. Problems with the icon are comprehensive in nature and are intended to be solved with appropriate software. Problems with the PtD icon are Prevention through Design problems. Problems with the C&S icon are Codes and Standards problems.

8–6C In the fully developed region of flow in a circular tube, will the velocity profile change in the flow direction? How about the temperature profile?

8–7C Consider laminar flow in a circular tube. Will the friction factor be higher near the inlet of the tube or near the exit? Why? What would your response be if the flow were turbulent?

8–8C How does the friction factor f vary along the flow direction in the fully developed region in (*a*) laminar flow and (*b*) turbulent flow?

8–9C How does surface roughness affect the pressure drop in a tube if the flow is turbulent? What would your response be if the flow were laminar?

8–10C How is the hydrodynamic entry length defined for flow in a tube? Is the entry length longer in laminar or turbulent flow?

8–11C Consider the flow of oil in a tube. How will the hydrodynamic and thermal entry lengths compare if the flow is laminar? How would they compare if the flow were turbulent?

8–12C Consider the flow of mercury (a liquid metal) in a tube. How will the hydrodynamic and thermal entry lengths compare if the flow is laminar? How would they compare if the flow were turbulent?

8–13C How is the thermal entry length defined for flow in a tube? In what region is the flow in a tube fully developed?

8–14C Consider laminar forced convection in a circular tube. Will the heat flux be higher near the inlet of the tube or near the exit? Why?

8–15C Consider turbulent forced convection in a circular tube. Will the heat flux be higher near the inlet of the tube or near the exit? Why?

8–16C What does the logarithmic mean temperature difference represent for flow in a tube whose surface temperature is constant? Why do we use the logarithmic mean temperature instead of the arithmetic mean temperature?

8–17C Consider fluid flow in a tube whose surface temperature remains constant. What is the appropriate temperature difference for use in Newton's law of cooling with an average heat transfer coefficient?

8–18C What is the physical significance of the number of transfer units NTU $= hA_s/\dot{m}c_p$? What do small and large NTU values tell us about a heat transfer system?

8–19 Consider the velocity and temperature profiles for a fluid flow in a tube with a diameter of 50 mm that can be expressed as

$$u(r) = 0.05[1 - (r/R)^2]$$
$$T(r) = 400 + 80\,(r/R)^2 - 30\,(r/R)^3$$

with units in m/s and K, respectively. Determine the average velocity and the mean (average) temperature from the given velocity and temperature profiles.

8–20 Consider the velocity and temperature profiles for airflow in a tube with a diameter of 8 cm that can be expressed as

$$u(r) = 0.2[(1 - (r/R)^2)]$$
$$T(r) = 250 + 200\,(r/R)^3$$

with units in m/s and K, respectively. If the convection heat transfer coefficient is 100 W/m²·K, determine the mass flow rate and surface heat flux using the given velocity and temperature profiles. Evaluate the air properties at 20°C and 1 atm.

8–21 C&S Liquid water flows in a circular tube at a mass flow rate of 7 g/s. The water enters the tube at 5°C, and the average convection heat transfer coefficient for the internal flow is 20 W/m²·K. The tube is 3 m long and has an inner diameter of 25 mm. The tube surface is maintained at a constant temperature. The inner surface of the tube is lined with polyvinylidene chloride (PVDC) lining. According to the ASME Code for Process Piping (ASME B31.3-2014, Table A323.4.3), the recommended maximum temperature for PVDC lining is 79°C. If the water exits the tube at 15°C, determine the heat rate transferred to the water. Would the inner surface temperature of the tube exceed the recommended maximum temperature for PVDC lining?

8–22 C&S In a heating system, liquid water flows in a circular tube at a mass flow rate of 3.5 g/s. The water enters the tube at 5°C, where it is heated at a rate of 1 kW. The tube surface is maintained at a constant temperature. The effectiveness of the system to heat water in the tube has been determined to have a number of transfer units of NTU = 2. According to the service restrictions of the ASME Boiler and Pressure Vessel Code (ASME BPVC.IV-2015, HG-101), hot water heaters should not be operating at temperatures exceeding 120°C at or near the heater outlet. The inner surface of the tube is lined with polyvinylidene chloride (PVDC) lining. According to the ASME Code for Process Piping (ASME B31.3-2014, Table A323.4.3), the recommended maximum temperature for PVDC lining is 79°C. To comply with both ASME codes, determine (*a*) Whether the water exiting the tube is at a temperature below 120°C. (*b*) Whether the inner surface temperature of the tube exceeds 79°C. Evaluate the fluid properties at 40°C. Is this an appropriate temperature to evaluate the fluid properties?

8–23 Air enters an 18-cm-diameter, 12-m-long underwater duct at 50°C and 1 atm at a mean velocity of 7 m/s and is cooled by the water outside. If the average heat transfer coefficient is 65 W/m²·K and the tube temperature is nearly equal to the water temperature of 10°C, determine the exit temperature of air and the rate of heat transfer. Evaluate air properties at a bulk mean temperature of 30°C. Is this a good assumption?

8–24 Combustion gases passing through a 5-cm-internal-diameter circular tube are used to vaporize wastewater at atmospheric pressure. Hot gases enter the tube at 115 kPa and 250°C at a mean velocity of 5 m/s and leave at 150°C. If the average heat transfer coefficient is 120 W/m²·K and the inner surface temperature of the tube is 110°C, determine (*a*) the tube length and (*b*) the rate of evaporation of water. Use air properties for the combustion gases.

8–25 Repeat Prob. 8–24 for a heat transfer coefficient of 40 W/m²·K.

8–26 Cooling water available at 10°C is used to condense steam at 30°C in the condenser of a power plant at a rate of 0.15 kg/s by circulating the cooling water through a bank of 5-m-long, 1.2-cm-internal-diameter thin copper tubes. Water enters the tubes at a mean velocity of 4 m/s and leaves at a temperature of 24°C. The tubes are nearly isothermal at 30°C. Determine the average heat transfer coefficient between the water, the tubes, and the number of tubes needed to achieve the indicated heat transfer rate in the condenser.

8–27 Repeat Prob. 8–26 for steam condensing at a rate of 0.60 kg/s.

8–28 Reconsider Prob. 8–26. Using appropriate software, investigate the effect of the cooling water average (mean) velocity on the number of tubes needed to achieve the indicated heat transfer rate in the condenser. By varying the cooling water average velocity from 0.3 to 6 m/s, plot the number of tubes as a function of the water average velocity.

8–29 Inside a condenser, there is a bank of seven copper tubes with cooling water flowing in them. Steam condenses at a rate of 0.6 kg/s on the outer surfaces of the tubes that are at a constant temperature of 68°C. Each copper tube is 5 m long and has an inner diameter of 25 mm. Cooling water enters each tube at 5°C and exits at 60°C. Determine the average heat transfer coefficient of the cooling water flowing inside each tube and the cooling water mean velocity needed to achieve the indicated heat transfer rate in the condenser.

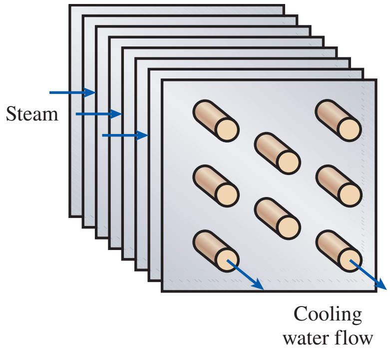

FIGURE P8–29

8–30 Reconsider Prob. 8–29. Using appropriate software, evaluate the effect of the cooling water mean velocity on the rate of steam condensation in the condenser. By varying the cooling water mean velocity for $0 < V_{avg} \leq 2$ m/s, plot the steam condensation rate as a function of the water mean velocity.

8–31 In a gas-fired boiler, water is being boiled at 120°C by hot air flowing through a 5-m-long, 5-cm-diameter tube submerged in water. Hot air enters the tube at 1 atm and 300°C at a mean velocity of 7 m/s and leaves at 150°C. If the surface temperature of the tube is 120°C, determine the average convection heat transfer coefficient of the air and the rate of water evaporation, in kg/h.

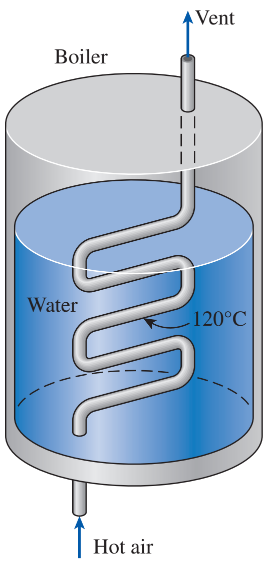

FIGURE P8–31

8–32 Reconsider Prob. 8–31. Using appropriate software, evaluate the effect of the tube length on the average convection heat transfer coefficient of air. By varying the tube length from 3 to 18 m, plot the average convection heat transfer coefficient as a function of the tube length.

8–33 C&S Liquid water flows in fully developed conditions through a circular tube at a mass flow rate of 3.5 g/s. The water enters the tube at 5°C, and the average convection heat transfer coefficient for the internal flow is 20 W/m²·K. The tube is 3 m long and has an inner diameter of 25 mm. The tube surface is subjected to a constant heat flux at a rate of 300 W. The inner surface of the tube is lined with polyvinylidene chloride (PVDC) lining. According to the ASME Code for Process Piping (ASME B31.3-2014, Table A323.4.3), the recommended maximum temperature for PVDC lining is 79°C. Would the inner surface temperature of the tube exceed the recommended maximum temperature for PVDC lining? If so, determine the axial location along the tube where the tube's inner surface temperature reaches 79°C. Evaluate the fluid properties at 15°C. Is this an appropriate temperature at which to evaluate the fluid properties?

Laminar, Transitional, and Turbulent Flow in Tubes

8–34C Someone claims that the volume flow rate in a circular pipe with laminar flow can be determined by measuring the velocity at the centerline in the fully developed region, multiplying it by the cross-sectional area, and dividing the result by 2. Do you agree? Explain.

8–35C Someone claims that the average velocity in a circular pipe in fully developed laminar flow can be determined by simply measuring the velocity at $R/2$ (midway between the wall surface and the centerline). Do you agree? Explain.

8–36C How is the friction factor for flow in a tube related to the pressure drop? How is the pressure drop related to the pumping power requirement for a given mass flow rate?

8–37C Someone claims that in fully developed turbulent flow in a tube, the shear stress is a maximum at the tube surface. Do you agree with this claim? Explain.

8–38C Consider fully developed flow in a circular pipe with negligible entrance effects. If the length of the pipe is doubled, the pressure drop will (*a*) double, (*b*) more than double, (*c*) less than double, (*d*) reduce by half, or (*e*) remain constant.

8–39C Consider fully developed laminar flow in a circular pipe. If the diameter of the pipe is reduced by half while the flow rate and the pipe length are held constant, the pressure drop will (*a*) double, (*b*) triple, (*c*) quadruple, (*d*) increase by a factor of 8, or (*e*) increase by a factor of 16.

8–40C Consider fully developed laminar flow in a circular pipe. If the viscosity of the fluid is reduced by half by heating while the flow rate is held constant, how will the pressure drop change?

8–41 In fully developed laminar flow in a circular pipe, the velocity at $R/2$ (midway between the wall surface and the centerline) is measured to be 6 m/s. Determine the velocity at the center of the pipe. *Answer:* 8 m/s

8–42 The velocity profile in fully developed laminar flow in a circular pipe of inner radius $R = 10$ cm, in m/s, is given by $u(r) = 4(1 - r^2/R^2)$. Determine the mean and maximum velocities in the pipe, and determine the volume flow rate.

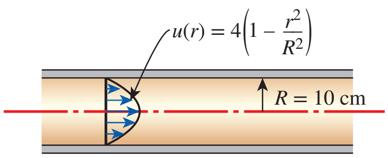

FIGURE P8–42

8–43 In fully developed laminar flow inside a circular pipe, the velocities at $r = 0.5R$ (midway between the wall surface and the centerline) are measured to be 3, 6, and 9 m/s. (*a*) Determine the maximum velocity for each of the measured midway velocities. (*b*) By varying r/R for $-1 \le r/R \le 1$, plot the velocity profile for each of the measured midway velocities with r/R as the y-axis and $V(r/R)$ as the x-axis.

8–44 Water is flowing in fully developed conditions through a 3-cm-diameter smooth tube with a mass flow rate of 0.02 kg/s at 15°C. Determine (*a*) the maximum velocity of the flow in the tube and (*b*) the pressure gradient for the flow.

8–45 Water at 15°C is flowing through a 5-cm-diameter smooth tube with a length of 200 m. Determine the Darcy friction factor and pressure loss associated with the tube for (*a*) mass flow rate of 0.02 kg/s and (*b*) mass flow rate of 0.5 kg/s.

8–46 Water at 10°C ($\rho = 999.7$ kg/m^3 and $\mu = 1.307 \times 10^{-3}$ kg/m·s) is flowing in a 0.20-cm-diameter, 15-m-long pipe steadily at an average velocity of 1.2 m/s. Determine (*a*) the pressure drop and (*b*) the pumping power requirement to overcome this pressure drop. Assume flow is fully developed. Is this a good assumption? *Answers:* (*a*) 188 kPa, (*b*) 0.71 W

8–47 Determine the hydrodynamic and thermal entry lengths for water, engine oil, and liquid mercury flowing through a 2.5-cm-diameter smooth tube with mass flow rate of 0.01 kg/s and temperature of 100°C.

8–48E Determine the average velocity and hydrodynamic and thermal entry lengths for water, engine oil, and liquid mercury flowing through a standard 2-in Schedule 40 pipe with a mass flow rate of 0.1 lbm/s and a temperature of 100°F.

8–49 An engineer is to design an experimental apparatus that consists of a 25-mm-diameter smooth tube, where different fluids at 100°C are to flow through in fully developed laminar flow conditions. For hydrodynamically and thermally fully developed laminar flow of water, engine oil, and liquid mercury, determine (*a*) the minimum tube length and (*b*) the required pumping power to overcome the pressure loss in the tube at the largest allowable flow rate.

8–50 Consider a 25-mm-diameter and 15-m-long smooth tube that is used for heating fluids. The wall is heated electrically to provide a constant surface heat flux along the entire tube. Fluids enter the tube at 50°C and exit at 150°C. If the mass flow rate is maintained at 0.01 kg/s, determine the convection heat transfer coefficients at the tube outlet for water, engine oil, and liquid mercury.

8–51 C&S Liquid water flows in an ASTM B75 copper tube at a mass flow rate of 3.6 g/s. The water enters the tube at 40°C, and the tube surface is subjected to a constant heat flux at a rate of 1.8 kW. The tube is circular with an inner diameter of 25 mm and a length of 3 m. The maximum use temperature for ASTM B75 copper tube is 204°C (ASME Code for Process Piping, B31.3-2014, Table A-1M). Would the surface temperature of the tube exceed the maximum use temperature for the copper tube? If so, determine the axial location along the tube where the tube's surface temperature reaches 204°C. Evaluate the fluid properties at 100°C. Is this an appropriate temperature at which to evaluate the fluid properties?

8–52 Consider a 25-mm-diameter and 15-m-long smooth tube that is maintained at a constant surface temperature. Fluids enter the tube at 50°C with a mass flow rate of 0.01 kg/s. Determine the tube surface temperatures necessary to heat water, engine oil, and liquid mercury to the desired outlet temperature of 150°C.

8–53 C&S In a heating system, liquid water flows in a circular tube with a length of 3 m and an inner diameter of 12.5 mm. The water enters the tube at 15°C, where it is heated at a rate of 1.5 kW. The tube surface is maintained at a constant temperature. The flow is laminar, and it experiences a pressure loss of 5 Pa in the tube. According to the service restrictions of the ASME Boiler and Pressure Vessel Code (ASME BPVC.IV-2015, HG-101), hot water heaters should not be operating at temperatures exceeding 120°C at or near the heater outlet. The tube's inner surface is lined with polyvinylidene fluoride (PVDF) lining. According to the ASME Code for Process Piping (ASME B31.3-2014, Table A323.4.3), the recommended maximum temperature for PVDF lining is 135°C. To comply with both ASME codes, determine (*a*) whether the water exiting the tube is at a temperature below 120°C, and (*b*) whether the inner surface temperature of the tube exceeds 135°C. Evaluate the fluid properties at 80°C. Is this an appropriate temperature at which to evaluate the fluid properties?

8–54E In a chemical process plant, liquid isobutane at 50°F is being transported through a 30-ft-long standard 3/4-in Schedule 40 cast iron pipe with a mass flow rate of 0.4 lbm/s. Accuracy of the results is an important issue in this problem; therefore, use the most appropriate equation to determine (*a*) the pressure loss and (*b*) the pumping power required to overcome the pressure loss. Assume flow is fully developed. Is this a good assumption?

8–55 Water at 15°C is flowing through a 200-m-long standard 1-in Schedule 40 cast iron pipe with a mass flow rate of 0.5 kg/s. If accuracy is an important issue, use the appropriate equation to determine (*a*) the pressure loss and (*b*) the pumping power required to overcome the pressure loss. Assume flow is fully developed. Is this a good assumption?

8–56 Reconsider Prob. 8–55. Using appropriate software, investigate the effect of the pipe roughness on the pumping power required to overcome the pressure loss. By varying the pipe roughness for $0.1 \leq \varepsilon \leq 0.26$ mm, plot the pumping power as a function of the pipe roughness.

8–57 Water at 15°C ($\rho = 999.1$ kg/m^3 and $\mu = 1.138 \times 10^{-3}$ kg/m·s) is flowing in a 4-cm-diameter and 25-m-long horizontal pipe made of stainless steel steadily at a rate of 7 L/s. Determine (*a*) the pressure drop and (*b*) the pumping power requirement to overcome this pressure drop. Assume flow is fully developed. Is this a good assumption?

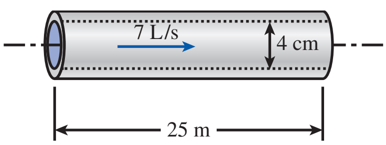

FIGURE P8–57

8–58 Reconsider Prob. 8–57. Using appropriate software, investigate the effect of the pipe diameter on the pumping power requirement to overcome the pressure loss. By varying the pipe diameter from 2 to 4 cm, plot the pumping power as a function of the pipe diameter.

8–59 Consider a fluid with mean inlet temperature T_i flowing through a tube of diameter D and length L, at a mass flow rate $\dot{m}$. The tube is subjected to a surface heat flux that can be expressed as $\dot{q}_s(x) = a + b\sin(x\pi/L)$, where a and b are constants. Determine an expression for the mean temperature of the fluid as a function of the x-coordinate.

8–60 A fluid is flowing in fully developed laminar conditions in a tube with diameter D and length L at a mass flow rate $\dot{m}$. The tube is subjected to a surface heat flux that can be expressed as $\dot{q}_s(x) = a\exp(-bx/2)$, where a and b are constants. Determine an expression for the difference in mean temperature at the tube inlet and outlet.

8–61 In a thermal system, water enters a 25-mm-diameter and 23-m-long circular tube with a mass flow rate of 0.1 kg/s at 25°C. The heat transfer from the tube surface to the water can be expressed in terms of heat flux as $\dot{q}_s(x) = ax$. The coefficient a is 400 W/m^3, and the axial distance from the tube inlet is x measured in meters. Determine (*a*) an expression for the mean temperature $T_m(x)$ of the water, (*b*) the outlet mean temperature of the water, and (*c*) the value of a uniform heat flux $\dot{q}_s$ on the tube surface that would result in the same outlet mean temperature calculated in part (*b*). Evaluate water properties at 35°C.

8–62 Consider a 10-m-long smooth rectangular tube, with $a = 50$ mm and $b = 25$ mm, that is maintained at a constant surface temperature. Liquid water enters the tube at 20°C with a mass flow rate of 0.01 kg/s. Determine the tube surface temperature necessary to heat the water to the desired outlet temperature of 80°C.

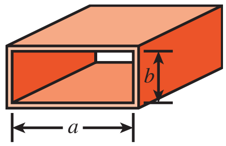

FIGURE P8–62

8–63 A 15-cm × 20-cm printed circuit board whose components are not allowed to come into direct contact with air for reliability reasons is to be cooled by passing cool air through a 20-cm-long channel of rectangular cross section 0.2 cm × 14 cm drilled into the board. The heat generated by the electronic components is conducted across the thin layer of the board to the channel, where it is removed by air that enters the channel at 15°C. The heat flux at the top surface of the channel can be considered to be uniform, and heat transfer through other surfaces is negligible. If the velocity of the air at the inlet of the channel is not to exceed 4 m/s and the surface temperature of the channel is to remain under 50°C, determine the maximum total power of the electronic components that can safely be mounted on this circuit board. As a first approximation, assume flow is fully developed in the channel. Evaluate properties of air at a bulk mean temperature of 25°C. Is this a good assumption?

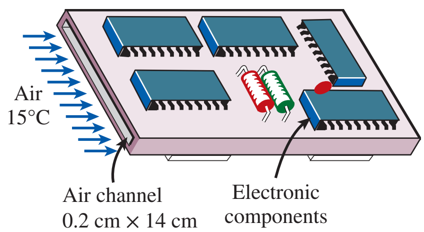

FIGURE P8–63

8–64 Repeat Prob. 8–63 by replacing air with helium, which has six times the thermal conductivity of air. Use the following properties for helium: $\rho = 0.1636$ kg/m^3, $k = 0.1502$ W/m·K, $\nu = 1.214 \times 10^{-4}$ m^2/s, $c_p = 5193$ J/kg·K, and Pr = 0.6867.

8–65 Reconsider Prob. 8–63. Using appropriate software, investigate the effects of air velocity at the inlet of the channel and the maximum surface temperature on the maximum total power dissipation of electronic components. Let the air velocity vary from 1 m/s to 20 m/s and the surface temperature from 30°C to 90°C. Plot the power dissipation as functions of air velocity and surface temperature, and discuss the results.

8–66 A computer cooled by a fan contains eight printed circuit boards (PCBs), each dissipating 12 W of power. The height of the PCBs is 12 cm and the length is 15 cm. The clearance between the tips of the components on the PCB and the back surface of the adjacent PCB is 0.3 cm. The cooling air is supplied by a 10-W fan mounted at the inlet. If the temperature rise of air as it flows through the case of the computer is not to exceed 10°C, determine (*a*) the flow rate of the air that the fan needs to deliver, (*b*) the fraction of the temperature rise of air that is due to the heat generated by the fan and its motor, and (*c*) the highest allowable inlet air temperature if the surface temperature of the components is not to exceed 70°C anywhere in the system. As a first approximation, assume flow is fully developed in the channel. Evaluate properties of air at a bulk mean temperature of 25°C. Is this a good assumption?

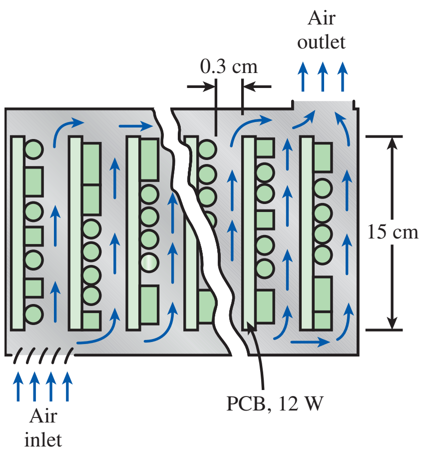

FIGURE P8–66

8–67 Liquid water is flowing between two very thin parallel 1-m-wide and 10-m-long plates with a spacing of 12.5 mm. The water enters the parallel plates at 20°C with a mass flow rate of 0.58 kg/s. The outer surface of the parallel plates is subjected to hydrogen gas (an ideal gas at 1 atm) flow width-wise in parallel over the upper and lower surfaces of the two plates. The free-stream hydrogen gas has a velocity of 5 m/s at a temperature of 155°C. Determine the outlet mean temperature of the water, the surface temperature of the parallel plates, and the total rate of heat transfer. Evaluate the properties for water at 30°C and the properties of H_2 gas at 100°C. Is this a good assumption?

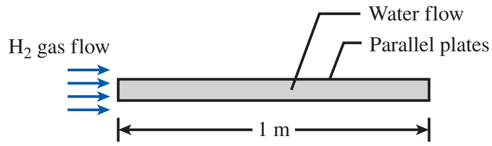

FIGURE P8–67

8–68 Reconsider Prob. 8–67. Using appropriate software, evaluate the effect of water mass flow rate on the free-stream H_2 gas velocity and the surface temperature of the parallel plates and the effect of the free-stream H_2 gas velocity on the total heat transfer rate. By varying the mass flow rate of water for $0 < \dot{m} \leq 0.58$ kg/s, plot the free-stream H_2 gas velocity and the surface temperature of the plates as a function of the mass flow rate of water. Also, plot the total heat transfer rate as a function of the free-stream H_2 gas velocity.

8–69 Consider the flow of oil at 10°C in a 40-cm-diameter pipeline at an average velocity of 0.5 m/s. A 1500-m-long section of the pipeline passes through icy waters of a lake at 0°C. Measurements indicate that the surface temperature of the pipe is very nearly 0°C. Disregarding the thermal resistance of the pipe material, determine (*a*) the temperature of the oil when the pipe leaves the lake, (*b*) the rate of heat transfer from the oil, and (*c*) the pumping power required to overcome the pressure losses and to maintain the flow of oil in the pipe.

8–70 In a manufacturing plant that produces cosmetic products, glycerin is being heated by flowing through a 25-mm-diameter and 10-m-long tube. With a mass flow rate of 0.5 kg/s, the flow of glycerin enters the tube at 25°C. The tube surface is maintained at a constant surface temperature of 140°C. Determine the outlet mean temperature and the total rate of heat transfer for the tube. Evaluate the properties for glycerin at 30°C.

8–71 Liquid glycerin is flowing through a 25-mm-diameter and 10-m-long tube. The liquid glycerin enters the tube at 20°C with a mass flow rate of 0.5 kg/s. If the outlet mean temperature is 40°C and the tube surface temperature is constant, determine the surface temperature of the tube.

8–72 C&S Liquid water flows in a thin-walled circular tube at a mass flow rate of 11 g/s. The water enters the tube at 60°C, where it is heated at a rate of 3.8 kW. The tube is circular with a length of 3 m and an inner diameter of 25 mm. The tube surface is maintained at a constant temperature. At the tube exit, a hydrogenated nitrile rubber (HNBR) o-ring is attached to the tube's outer surface. The maximum temperature permitted for the o-ring is 150°C (ASME Boiler and Pressure Vessel Code, BPVC.IV-2015, HG-360). Is the HNBR o-ring suitable for this operation? Evaluate the fluid properties at 100°C. Is this a reasonable temperature at which to evaluate the fluid properties?

8–73 Air at 20°C (1 atm) enters into a 5-mm-diameter and 10-cm-long circular tube at an average velocity of 5 m/s. The tube wall is maintained at a constant surface temperature of

160°C. Determine the convection heat transfer coefficient and the outlet mean temperature. Evaluate the air properties at 50°C.

8–74 C & S Liquid methane flows at an average velocity of 1.7 mm/s between two isothermal parallel plates that are spaced 25 mm apart. The plates are made of ASTM A240 410S stainless steel, and each plate has a length of 2 m. The inlet temperature for the liquid methane is −150°C, and the liquid exits the parallel plates at −50°C. The minimum temperature suitable for ASTM A240 410S stainless steel plate is −30°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-1M). To keep the plates from getting too cold, the plates are heated at 700 W/m^2. Determine the surface temperature of the plates. Are the ASTM A240 410S stainless steel plates suitable for this operation?

8–75 Glycerin is being heated by flowing between two parallel 1-m-wide and 10-m-long plates with 12.5-mm spacing. The glycerin enters the parallel plates with a temperature of 25°C and a mass flow rate of 0.7 kg/s. The plates have a constant surface temperature of 40°C. Determine the outlet mean temperature of the glycerin and the total rate of heat transfer. Evaluate the properties for glycerin at 30°C. Is this a good assumption?

8–76 Reconsider Prob. 8–75. Using appropriate software, evaluate the effect of the glycerin mass flow rate on the surface temperature of the parallel plates and the total rate of heat transfer necessary to keep the outlet mean temperature of the glycerin at 35°C. By varying the mass flow rate from 0.05 to 6 kg/s, plot the surface temperature of the parallel plates and the total rate of heat transfer as a function of the mass flow rate.

8–77 Glycerin is being heated by flowing between two very thin parallel 1-m-wide and 10-m-long plates with 12.5-mm spacing. The glycerin enters the parallel plates with a temperature 20°C and a mass flow rate of 0.7 kg/s. The outer surface of the parallel plates is subjected to hydrogen gas (an ideal gas at 1 atm) flow width-wise in parallel over the upper and lower surfaces of the two plates. The free-stream hydrogen gas has a velocity of 3 m/s and a temperature of 150°C. Determine the outlet mean temperature of the glycerin, the surface temperature of the parallel plates, and the total rate of heat transfer. Evaluate the properties for glycerin at 30°C and the properties of H_2 gas at 100°C. Are these good assumptions?

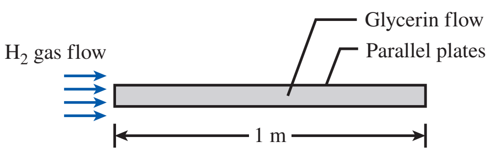

FIGURE P8–77

8–78 Reconsider Prob. 8–77. Using appropriate software, evaluate the effect of the glycerin mass flow rate on the free-stream velocity of the hydrogen gas needed to keep the outlet mean temperature of the glycerin at 40°C. By varying the mass flow rate of glycerin from 0.5 to 2.4 kg/s, plot the free-stream velocity of the hydrogen gas as a function of the mass flow rate of the glycerin.

8–79E Air flows in an isothermal tube under fully developed conditions. The inlet temperature is 60°F and the tube surface temperature is 120°F. The tube is 10 ft long, and the inner diameter is 2 in. The air mass flow rate is 18.2 lbm/h. Calculate the exit temperature of the air and the total rate of heat transfer from the tube wall to the air. Evaluate the air properties at a temperature 80°F. Is this a good assumption?

8–80 Water flows through a circular pipe. The pipe's inner diameter is 2 cm, and its length is 6 m. The average velocity of the water is 15 cm/s, and its inlet temperature is 20°C. Assuming that the heating is uniform and the flow is fully developed, what is the heat flux required to raise the water temperature by 1°C by the time it reaches the end of the pipe? What is the surface temperature of the pipe at the exit? Evaluate the properties of water at 20°C.

8–81 Air flows in a pipe under fully developed conditions with an average velocity of 1.25 m/s and a temperature of 20°C. The pipe's inner diameter is 4 cm, and its length is 4 m. The first half of the pipe is kept at a constant wall temperature of 100°C. The second half of the pipe is subjected to a constant heat flux of 200 W. Determine (*a*) the air temperature at the 2 m length, (*b*) the air temperature at the exit, (*c*) the total heat transfer to the air, and (*d*) the wall temperature at the exit of the tube. Evaluate the properties of air at 80°C.

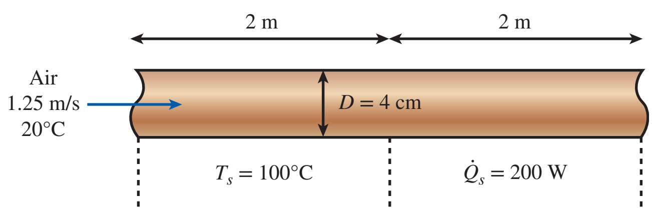

FIGURE P8–81

8–82E Air flows with an average velocity of 3 ft/s in a duct that is 1 ft wide and 1 in high and 40 ft long. The aspect ratio of the duct is very small (0.083), and the duct can be treated as a parallel-plate channel. The duct passes through a heated oven, and the temperature of the duct is approximately constant and equal to 100°F. The air enters the duct at 50°F. For fully developed flow of air in the duct, determine (*a*) the temperature of the air as it leaves the duct, (*b*) the rate of heat transfer from the duct to the air, and (c) the heat flux at the duct exit. Evaluate the properties of air at 75°F. Is this a good assumption?

8–83 Determine the convection heat transfer coefficient for the flow of (*a*) air and (*b*) water at a velocity of 5 m/s in an 8-cm-diameter and 10-m-long tube when the tube is subjected to uniform heat flux from all surfaces. Use fluid properties at 25°C.

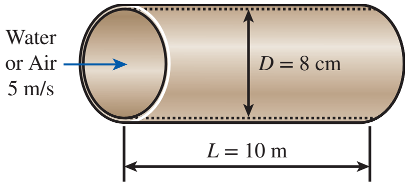

FIGURE P8–83

8–84 C&S Liquid water flows in a circular tube at a mass flow rate of 0.12 kg/s. The water enters the tube at 65°C, where it is heated at a rate of 5.5 kW. The tube is circular with a length of 3 m and an inner diameter of 25 mm. The tube surface is maintained isothermal. The inner surface of the tube is lined with polyvinylidene chloride (PVDC) lining. The recommended maximum temperature for PVDC lining is 79°C (ASME Code for Process Piping, ASME B31.3-2014, Table A323.4.3). Is the PVDC lining suitable for the tube under these conditions? Evaluate the fluid properties at 70°C. Is this an appropriate temperature at which to evaluate the fluid properties?

8–85 C&S Saturated liquid propane flows in a circular ASTM A268 TP443 stainless steel tube at a mass flow rate of 42 g/s. The liquid propane enters the tube at −50°C, and the tube surface is maintained isothermal. The tube has an inner diameter of 25 mm, and its length is 3 m. The inner surface of the tube has a relative surface roughness of 0.05. The ASME Code for Process Piping limits the minimum temperature suitable for using ASTM A268 TP443 stainless steel tube at −30°C(ASME B31.3-2014, Table A-1M). To keep the tube surface from getting too cold, the tube is heated at a rate of 1.3 kW. Determine the surface temperature of the tube. Is the ASTM A268 TP443 stainless steel tube suitable under these conditions? Evaluate the fluid properties at −40°C. Is this an appropriate temperature at which to evaluate the fluid properties?

8–86 Air at 10°C enters a 12-cm-diameter and 5-m-long pipe at a rate of 0.065 kg/s. The inner surface of the pipe has a roughness of 0.22 mm, and the pipe is nearly isothermal at 50°C. Determine the rate of heat transfer to air using the Nusselt number relation given by (*a*) Eq. 8–66 and (*b*) Eq. 8–71. Evaluate air properties at a bulk mean temperature of 20°C. Is this a good assumption?

8–87 C&S Liquid water flows in a thin-walled circular tube, where the pumping power required to overcome the turbulent flow pressure loss in the tube is 100 W. The water enters the tube at 10°C, where it is heated at a rate of 3.6 kW. The average convection heat transfer coefficient for the internal flow is 120 W/m^2·K. The tube is 3 m long and has an inner diameter of 12.5 mm. The tube surface is maintained at a constant temperature. At the tube exit, an ethylene propylene diene (EPDM) rubber o-ring is attached to the tube's outer surface. The maximum temperature permitted for the o-ring is 150°C (ASME Boiler and Pressure Vessel Code, BPVC. IV-2015, HG-360). Is the EPDM o-ring suitable for this operation? Evaluate the fluid properties at 10°C. Is this an appropriate temperature at which to evaluate the fluid properties?

8–88 A 12-m-long and 12-mm-inner-diameter pipe made of commercial steel is used to heat a liquid in an industrial process. The liquid enters the pipe with $T_i = 25°C, V = 0.8$ m/s. A uniform heat flux is maintained by an electric resistance heater wrapped around the outer surface of the pipe so that the fluid exits at 75°C. Assuming fully developed flow and taking the average fluid properties to be $\rho = 1000$ kg/m^3, $c_p = 4000$ J/kg·K, $\mu = 2 \times 10^{-3}$ kg/m·s, $k = 0.48$ W/m·K, and Pr = 10, determine:

(*a*) The required surface heat flux $\dot{q}_s$, produced by the heater
(*b*) The surface temperature at the exit, T_s
(*c*) The pressure loss through the pipe and the minimum power required to overcome the resistance to flow.

8–89 Water is to be heated from 10°C to 80°C as it flows through a 2-cm-internal-diameter, 13-m-long tube. The tube is equipped with an electric resistance heater, which provides uniform heating throughout the surface of the tube. The outer surface of the heater is well insulated, so in steady operation all the heat generated in the heater is transferred to the water in the tube. If the system is to provide hot water at a rate of 5 L/min, determine the power rating of the resistance heater. Also, estimate the inner surface temperature of the pipe at the exit.

8–90 Consider a fluid with a Prandtl number of 7 flowing through a smooth circular tube. Using the Colburn, Petukhov, and Gnielinski equations, determine the Nusselt numbers for Reynolds numbers at 3500, 10^4, and 5×10^5. Compare and discuss the results.

8–91E The hot water needs of a household are to be met by heating water at 55°F to 180°F by a parabolic solar collector at a rate of 5 lbm/s. Water flows through a 1.25-in-diameter thin aluminum tube whose outer surface is anodized black in order to maximize its solar absorption ability. The centerline of the tube coincides with the focal line of the collector, and a glass sleeve is placed outside the tube to minimize the heat losses. If solar energy is transferred to water at a net rate of 350 Btu/h per ft length of the tube, determine the required length of the parabolic collector to meet the hot water requirements of this house. Also, determine the surface temperature of the tube at the exit.

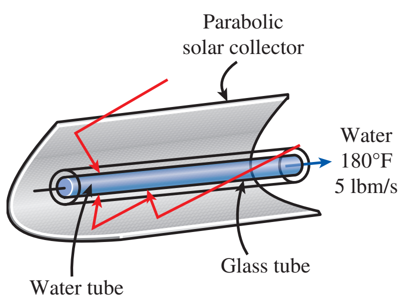

FIGURE P8–91E

8–92E Water at 60°F is heated by passing it through 0.75-in-internal-diameter thin-walled copper tubes. Heat is supplied to the water by steam that condenses outside the copper tubes at 250°F. If water is to be heated to 140°F at a rate of 0.4 lbm/s, determine (*a*) the length of the copper tube that needs to be used and (*b*) the pumping power required to overcome pressure losses. Assume the entire copper tube to be at the steam temperature of 250°F.

8–93 Air (1 atm) enters into a 5-cm-diameter circular tube at 20°C with an average velocity of 5 m/s. The tube wall is maintained at a constant surface temperature of 160°C, and the outlet mean temperature is 80°C. Estimate the length of the tube. Is the flow fully developed?

8–94 PtD In a food processing plant, hot liquid water is being transported in a pipe ($k = 15$ W/m·K, $D_i = 2.5$ cm, $D_o = 3$ cm, and $L = 10$ m.) The hot water flowing with a mass flow rate of 0.15 kg/s enters the pipe at 100°C and exits at 60°C. The plant supervisor thinks that since the hot water exits the pipe at 60°C, the pipe's outer surface temperature should be safe from thermal burn hazards. In order to prevent thermal burn upon accidental contact with skin tissue for individuals working in the vicinity of the pipe, the pipe's outer surface temperature should be kept below 45°C. Determine whether or not there is a risk of thermal burn on the pipe's outer surface. Assume the pipe outer surface temperature remains constant.

8–95 PtD Reconsider Prob. 8–94. The mass flow rate of the hot water can be varied between 0.05 and 0.5 kg/s. Using appropriate software, evaluate the effect of the hot water mass flow rate on the pipe's outer surface temperature. Plot the pipe's outer surface temperature as a function of the hot water mass flow rate for the water exit temperature of (*a*) 60°C and (*b*) 70°C. To prevent thermal burn on individuals working in the vicinity of the pipe, determine the condition of the hot water mass flow rate such that the pipe's outer surface temperature is below 45°C.

8–96 PtD A metal pipe ($k_{pipe} = 15$ W/m·K, $D_{i,\ pipe} = 5$ cm, $D_{o,\ pipe} = 6$ cm, and $L = 10$ m) situated in an engine room is used for transporting hot saturated water vapor at a flow rate of 0.03 kg/s. The water vapor enters and exits the pipe at 325°C and 290°C, respectively. Oil leakage can occur in the engine room, and when leaked oil comes in contact with hot spots above the oil's autoignition temperature, it can ignite spontaneously. To prevent any fire hazard caused by oil leakage on the hot surface of the pipe, determine the required insulation ($k_{ins} = 0.95$ W/m·K) layer thickness over the pipe for keeping the outer surface temperature below 180°C.

8–97 PtD Reconsider Prob. 8–96. Using appropriate software, evaluate the effect of the saturated water vapor mass flow rate on required insulation thickness for keeping the outer surface temperature below 180°C to prevent a fire hazard if oil leaks. By varying the mass flow rate from 0.01 to 0.1 kg/s, plot the insulation layer thickness as a function of the mass flow rate. If the flow rate of the saturated water vapor is adjusted between 0.03 and 0.1 kg/s, determine the insulation layer thickness that is suitable for this flow rate range.

8–98 PtD Exposure to high concentrations of gaseous ammonia can cause lung damage. To prevent gaseous ammonia from leaking out, ammonia is transported in its liquid state through a pipe ($k = 15$ W/m·K, $D_{i,\ pipe} = 5$ cm, $D_{o,\ pipe} = 6$ cm, and $L = 10$ m). The liquid ammonia enters the pipe at −30°C with a flow rate of 0.15 kg/s. Determine the insulation thickness for the pipe using a material with $k = 0.95$ W/m·K to keep the liquid ammonia's exit temperature at −20°C, while maintaining the insulated pipe outer surface temperature at 20°C.

8–99 PtD Reconsider Prob. 8–98. Using appropriate software, evaluate the effect of the liquid ammonia mass flow rate on the insulation thickness needed to keep the outer surface temperature at 20°C and the flow exit temperature at −20°C. By varying the mass flow rate from 0.02 to 0.2 kg/s, plot the insulation layer thickness as a function of the mass flow rate. If the flow rate of the liquid ammonia is adjusted between 0.04 and 0.2 kg/s, calculate the insulation layer thickness that is suitable for this range of flow rate.

8–100 An 8-m-long, uninsulated square duct of cross section 0.2 m × 0.2 m and relative roughness 10^{-3} passes through the attic space of a house. Hot air enters the duct at 1 atm and 80°C at a volume flow rate of 0.15 m³/s. The duct surface is nearly isothermal at 60°C. Determine the rate of heat loss from the duct to the attic space and the pressure difference between the inlet and outlet sections of the duct. Evaluate air properties at a bulk mean temperature of 80°C. Is this a good assumption?

8–101 Hot air at atmospheric pressure and 75°C enters a 10-m-long uninsulated square duct of cross section 0.15 m × 0.15 m that passes through the attic of a house at a rate of 0.2 m³/s. The duct is observed to be nearly isothermal at 70°C. Determine the exit temperature of the air and the rate of heat loss from the duct to the airspace in the attic. Evaluate air properties at a bulk mean temperature of 75°C. Is this a good assumption?

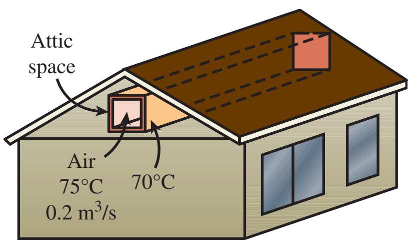

FIGURE P8–101

8–102 Reconsider Prob. 8–101. Using appropriate software, investigate the effect of the volume flow rate of air on the exit temperature of air and the rate of heat loss. Let the flow rate vary from 0.05 m^3/s to 0.15 m^3/s. Plot the exit temperature and the rate of heat loss as a function of flow rate, and discuss the results.

8–103 Hot air at 60°C leaving the furnace of a house enters a 12-m-long section of a sheet metal duct of rectangular cross section 20 cm × 20 cm at an average velocity of 4 m/s. The thermal resistance of the duct is negligible, and the outer surface of the duct, whose emissivity is 0.3, is exposed to the cold air at 10°C in the basement, with a convection heat transfer coefficient of 10 $W/m^2 \cdot K$. Taking the walls of the basement to be at 10°C also, determine (*a*) the temperature at which the hot air will leave the basement and (*b*) the rate of heat loss from the hot air in the duct to the basement. Evaluate air properties at a bulk mean temperature of 50°C. Is this a good assumption?

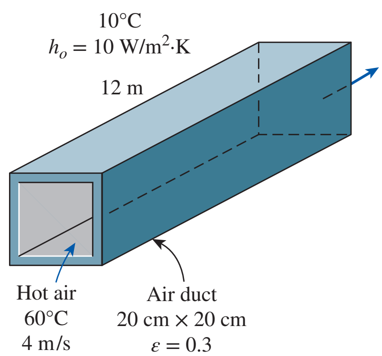

FIGURE P8–103

8–104 Reconsider Prob. 8–103. Using appropriate software, investigate the effect of air velocity and the surface emissivity on the exit temperature of air and the rate of heat loss. Let the air velocity vary from 1 m/s to 10 m/s and the emissivity from 0.1 to 1.0. Plot the exit temperature and the rate of heat loss functions of the air velocity and emissivity, and discuss the results.

8–105 The components of an electronic system dissipating 220 W are located in a 1-m-long horizontal duct whose cross section is 16 cm × 16 cm. The components in the duct are cooled by forced air, which enters at 27°C at a rate of 0.65 m^3/min. Assuming 85 percent of the heat generated inside is transferred to air flowing through the duct and the remaining 15 percent is lost through the outer surfaces of the duct, determine (*a*) the exit temperature of air and (*b*) the highest component surface temperature in the duct. As a first approximation, assume fully developed turbulent flow in the channel. Evaluate the properties of air at a bulk mean temperature of 35°C. Is this a good assumption?

8–106 Repeat Prob. 8–105 for a circular horizontal duct of 15 cm diameter.

8–107 A concentric annulus tube has inner and outer diameters of 25 mm and 100 mm, respectively. Liquid water flows at a mass flow rate of 0.05 kg/s through the annulus with the inlet and outlet mean temperatures of 20°C and 80°C, respectively. The inner tube wall is maintained with a constant surface temperature of 120°C, while the outer tube surface is insulated. Determine the length of the concentric annulus tube. Assume flow is fully developed.

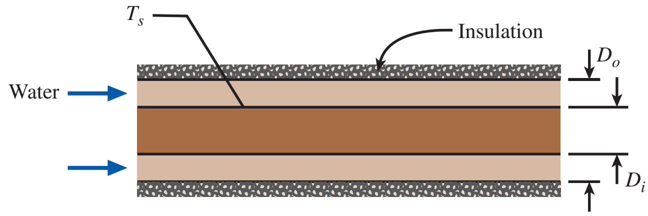

FIGURE P8–107

8–108 Liquid water flows at a mass flow rate of 0.7 kg/s through a concentric annulus tube with the inlet and outlet mean temperatures of 20°C and 80°C, respectively. The concentric annulus tube has inner and outer diameters of 10 mm and 100 mm, respectively. The inner tube wall is maintained with a constant surface temperature of 120°C, while the outer tube surface is insulated. Determine the length of the concentric annulus tube.

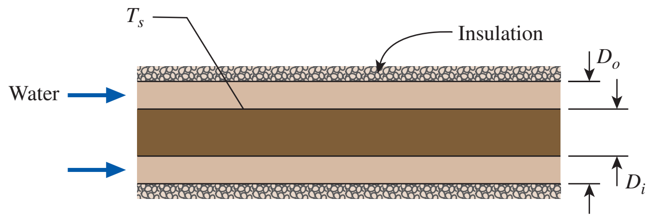

FIGURE P8–108

Special Topic: Transitional Flow

8–109 A tube with a bell-mouth inlet configuration is subjected to uniform wall heat flux of 3 kW/m^2. The tube has an inside diameter of 0.0158 m (0.622 in) and a flow rate of 1.43×10^{-4} m^3/s (2.27 gpm). The liquid flowing inside the tube is an ethylene glycol–distilled water mixture with a mass fraction of 2.27. Determine the fully developed friction coefficient at a location along the tube where the Grashof number is Gr = 16,600. The physical properties of the ethylene glycol–distilled water mixture at the location of interest are Pr = 14.85, $\nu = 1.93 \times 10^{-6}$ m^2/s, and $\mu_b/\mu_s = 1.07$.

8–110 Reconsider Prob. 8–109. Calculate the fully developed friction coefficient if the volume flow rate is increased by 50 percent while the rest of the parameters remain unchanged.

8–111E A tube with a square-edged inlet configuration is subjected to uniform wall heat flux of 8 kW/m^2. The tube has an inside diameter of 0.622 in and a flow rate of 2.16 gpm. The liquid flowing inside the tube is an ethylene glycol–distilled water mixture with a mass fraction of 2.27. Determine the friction coefficient at a location along the tube where the Grashof number is Gr = 35,450. The physical properties of the ethylene glycol–distilled water mixture at the location of interest are Pr = 13.8, $\nu = 18.4 \times 10^{-6}$ ft^2/s, and $\mu_b/\mu_s = 1.12$. Then recalculate the fully developed friction coefficient if the volume flow rate is increased by 50 percent while the rest of the parameters remain unchanged.

8–112 A 4-m-long tube is subjected to uniform wall heat flux. The tube has an inside diameter of 0.0149 m and a flow rate of 7.8×10^{-5} m^3/s. The liquid flowing inside the tube is an ethylene glycol–distilled water mixture with a mass fraction of 0.5. Determine the apparent (developing) friction factor at the location $x/D = 20$ if the inlet configuration of the tube is: (*a*) re-entrant and (*b*) square-edged. At this location, the local Grashof number is Gr = 24,000, and the properties of the mixture of distilled water and ethylene glycol are: Pr = 20.9, $\nu = 2.33 \times 10^{-6}$ m^2/s, and $\mu_b/\mu_s = 1.25$.

8–113 An ethylene glycol–distilled water mixture with a mass fraction of 0.72 and a flow rate of 2.05×10^{-4} m^3/s flows inside a tube with an inside diameter of 0.0158 m and a uniform wall heat flux boundary condition. For this flow, determine the Nusselt number at the location $x/D = 10$ for the inlet tube configuration of (*a*) bell-mouth and (*b*) re-entrant. Compare the results for parts (*a*) and (*b*). Assume the Grashof number is Gr = 60,000. The physical properties of an ethylene glycol–distilled water mixture are Pr = 33.46, $\nu = 3.45 \times 10^{-6}$ m^2/s, and $\mu_b/\mu_s = 2.0$.

8–114 Repeat Prob. 8–113 for the location $x/D = 90$.

Review Problems

8–115 The velocity profile in fully developed laminar flow in a circular pipe, in m/s, is given by $u(r) = 4(1 - 100r^2)$ where r is the radial distance from the centerline of the pipe in m. Determine (*a*) the radius of the pipe, (*b*) the mean velocity through the pipe, and (*c*) the maximum velocity in the pipe.

8–116E The velocity profile in fully developed laminar flow of water at 40°F in a 140-ft-long horizontal circular pipe, in ft/s, is given by $u(r) = 0.8(1 - 625r^2)$ where r is the radial distance from the centerline of the pipe in ft. Determine (*a*) the volume flow rate of water through the pipe, (*b*) the pressure drop across the pipe, and (*c*) the useful pumping power required to overcome this pressure drop.

8–117 Consider laminar flow of a fluid through a square channel maintained at a constant temperature. Now the mean velocity of the fluid is doubled. Determine the change in the pressure drop and the change in the rate of heat transfer between the fluid and the walls of the channel. Assume the flow regime remains unchanged. Assume fully developed flow, and disregard any changes in ΔT_{lm}.

8–118 Repeat Prob. 8–117 for turbulent flow.

8–119 A liquid hydrocarbon enters a 2.5-cm-diameter tube that is 5.0 m long. The liquid inlet temperature is 20°C and the tube wall temperature is 60°C. Average liquid properties are $c_p = 2.0$ kJ/kg·K, $\mu = 10$ mPa·s, and $\rho = 900$ kg/m^3. At a flow rate of 1200 kg/h, the liquid outlet temperature is measured to be 30°C. Estimate the liquid outlet temperature when the flow rate is reduced to 400 kg/h. *Hint:* For heat transfer in tubes, Nu $\propto$ Re$^{1/3}$ in laminar flow and Nu $\propto$ Re$^{4/5}$ in turbulent flow.

8–120 A silicon chip is cooled by passing water through microchannels etched in the back of the chip, as shown in Fig. P8–120. The channels are covered with a silicon cap. Consider a 12-mm × 12-mm-square chip in which $N = 60$ rectangular microchannels, each of width $W = 50$ μm and height $H = 200$ μm, have been etched. Water enters the microchannels at a temperature $T_i = 290$ K and a total flow rate of 0.005 kg/s. The chip and cap are maintained at a uniform temperature of 350 K. Assuming that the flow in the channels is fully developed, all the heat generated by the circuits on the top surface of the chip is transferred to the water, and using circular tube correlations, determine:

(*a*) The water outlet temperature, T_e
(*b*) The chip power dissipation, $\dot{W}_e$

Evaluate liquid water properties at a bulk mean temperature of 298 K. Is this a good assumption?

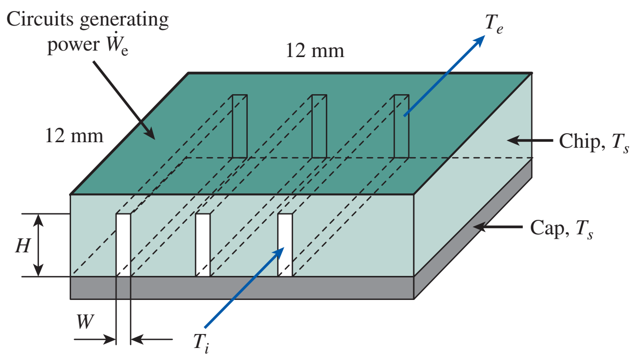

FIGURE P8–120

8–121 A fluid ($\rho = 1000$ kg/m^3, $\mu = 1.4 \times 10^{-3}$ kg/m·s, $c_p = 4.2$ kJ/kg·K, and $k = 0.58$ W/m·K) flows with an average velocity of 0.3 m/s through a 14-m-long tube with inside diameter of 0.01 m. Heat is uniformly added to the entire tube at the rate of 1500 W/m^2. Determine *(a)* the value of convection heat transfer coefficient at the exit, (*b*) the value of $T_s - T_m$, and (*c*) the value of $T_e - T_i$.

8–122 Crude oil at 22°C enters a 20-cm-diameter pipe with an average velocity of 32 cm/s. The average pipe wall temperature is 2°C. Crude oil properties are as given below. Calculate the rate of heat transfer and the pipe length if the crude oil outlet temperature is 20°C.

T °C	ρ kg/m^3	k W/m·K	μ mPa·s	c_p kJ/kg·K
2.0	900	0.145	60.0	1.80
22.0	890	0.145	20.0	1.90

8–123 Air (1 atm) enters a 5-mm-diameter circular tube at an average velocity of 5 m/s. The tube wall is maintained at a constant surface temperature. Determine the convection heat transfer coefficient for (*a*) a 10-cm-long tube and (*b*) a 50-cm-long tube. Evaluate the air properties at 50°C.

8–124 To cool a storehouse in the summer without using a conventional air-conditioning system, the owner decided to hire an engineer to design an alternative system that would make use of the water in the nearby lake. The engineer decided to flow air through a thin, smooth, 10-cm-diameter copper tube that is submerged in the lake. The water in the lake is typically at a constant temperature of 15°C and a convection heat transfer coefficient of 1000 W/m^2·K. If air (1 atm) enters the copper tube at a mean temperature of 30°C with an average velocity of 2.5 m/s, determine the necessary copper tube length so that the outlet mean temperature of the air is 20°C.

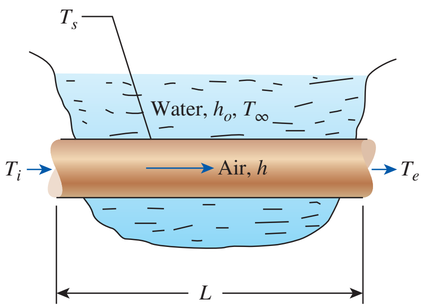

FIGURE P8–124

8–125 Liquid mercury is flowing at 0.6 kg/s through a 5-cm-diameter tube with inlet and outlet mean temperatures of 100°C and 200°C, respectively. The tube surface temperature is kept constant at 250°C. Determine the tube length using (*a*) the appropriate Nusselt number relation for liquid metals and (*b*) the Dittus-Boelter equation. (*c*) Compare the results of (*a*) and (*b*).

8–126 In the effort to find the best way to cool a smooth, thin-walled copper tube, an engineer decided to flow air either through the tube or across the outer tube surface. The tube has a diameter of 5 cm, and the surface temperature is held constant. Determine (*a*) the convection heat transfer coefficient when air is flowing through its inside at 25 m/s with a bulk mean temperature of 50°C and (*b*) the convection heat transfer coefficient when air is flowing across its outer surface at 25 m/s with a film temperature of 50°C.

8–127 Water is heated at a rate of 10 kg/s from a temperature of 15°C to 35°C by passing it through five identical tubes, each 5.0 cm in diameter, whose surface temperature is 60.0°C. Estimate (*a*) the steady rate of heat transfer and (*b*) the length of tubes needed to accomplish this task.

8–128 Repeat Prob. 8–127 for a flow rate of 20 kg/s.

8–129 Water at 1500 kg/h and 10°C enters a 10-mm-diameter smooth tube whose wall temperature is maintained at 49°C. Calculate (*a*) the tube length necessary to heat the water to 40°C, and (*b*) the water outlet temperature if the tube length is doubled. Assume average water properties to be the same as in (*a*).

8–130E The exhaust gases of an automotive engine leave the combustion chamber and enter an 8-ft-long and 3.5-in-diameter thin-walled steel exhaust pipe at 800°F and 15.5 psia at a rate of 0.05 lbm/s. The surrounding ambient air is at a temperature of 80°F, and the heat transfer coefficient on the outer surface of the exhaust pipe is 3 Btu/h·ft^2·°F. Assuming the exhaust gases to have the properties of air, determine (*a*) the velocity of the exhaust gases at the inlet of the exhaust pipe and (*b*) the temperature at which the exhaust gases will leave the pipe and enter the air.

8–131E Air is flowing through a smooth, thin-walled, 4-in-diameter copper tube that is submerged in water. The water maintains a constant temperature of 60°F and a convection heat transfer coefficient of 176 Btu/h·ft^2·R. If air (1 atm) enters the copper tube at a mean temperature of 90°F with an average velocity of 8 ft/s, determine the necessary copper tube length so that the outlet mean temperature of the air is 70°F.

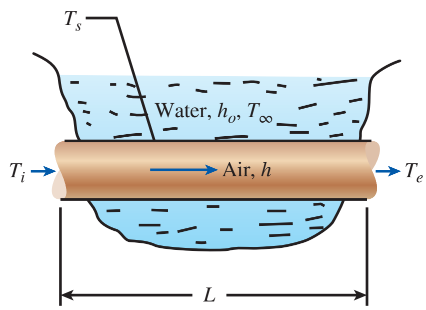

FIGURE P8–131E

8–132 Hot water at 90°C enters a 15-m section of a cast iron pipe (k = 52 W/m·K) whose inner and outer diameters are 4 and 4.6 cm, respectively, at an average velocity of 1.2 m/s. The outer surface of the pipe, whose emissivity is 0.7, is exposed to the cold air at 10°C in a basement, with a convection heat transfer coefficient of 12 W/m^2·K. Taking the walls of the basement to be at 10°C also, determine (*a*) the rate of heat loss from the water and (*b*) the temperature at which the water leaves the basement.

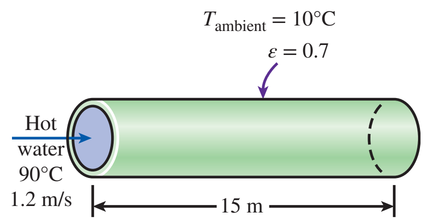

FIGURE P8–132

8–133 Repeat Prob. 8–132 for a pipe made of copper (k = 386 W/m·K) instead of cast iron.

8–134 Hot exhaust gases leaving a stationary diesel engine at 450°C enter a 15-cm-diameter pipe at an average velocity of 7.2 m/s. The surface temperature of the pipe is 180°C. Determine the pipe length if the exhaust gases are to leave the pipe at 250°C after transferring heat to water in a heat recovery unit. Use the properties of air for exhaust gases.

8–135 Geothermal steam at 165°C condenses in the shell side of a heat exchanger over the tubes through which water flows. Water enters the 4-cm-diameter, 14-m-long tubes at 20°C at a rate of 0.8 kg/s. Determine the exit temperature of water and the rate of condensation of geothermal steam.

8–136 Cold air at 5°C enters a 12-cm-diameter, 20-m-long isothermal pipe at a velocity of 2.5 m/s and leaves at 19°C. Estimate the surface temperature of the pipe.

8–137 100 kg/s of crude oil is heated from 20°C to 40°C through the tube side of a multitube heat exchanger. The crude oil flow is divided evenly among all 100 tubes in the tube bundle. The ID of each tube is 10 mm, and the inside tube-wall temperature is maintained at 100°C. Average properties of the crude oil are ρ = 950 kg/m^3, c_p = 1.9 kJ/kg·K, k = 0.25 W/m·K, μ = 12 mPa·s, and μ_w = 4 mPa·s. Estimate the rate of heat transfer and the tube length.

8–138 Liquid water enters a 10-m-long smooth rectangular tube with a = 50 mm and b = 25 mm. The surface temperature is kept constant, and water enters the tube at 20°C with a mass flow rate of 0.25 kg/s. Determine the tube surface temperature necessary to heat the water to the desired outlet temperature of 80°C.

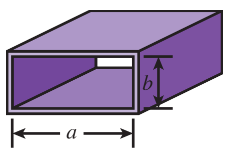

FIGURE P8–138

8–139 A geothermal district heating system involves the transport of geothermal water at 110°C from a geothermal well to a city at about the same elevation for a distance of 12 km at a rate of 1.5 m^3/s in 60-cm-diameter stainless steel pipes. The fluid pressures at the wellhead and at the arrival point in the city are to be the same. The minor losses are negligible because of the large length-to-diameter ratio and the relatively small number of components that cause minor losses. (*a*) Assuming the pump-motor efficiency to be 65 percent, determine the electric power consumption of the system for pumping. (*b*) Determine the daily cost of power consumption of the system if the unit cost of electricity is \$0.06/kWh. (*c*) The temperature of geothermal water is estimated to drop 0.5°C during this long flow. Determine if the frictional heating during flow can make up for this drop in temperature.

8–140 Repeat Prob. 8–139 for cast iron pipes of the same diameter.

8–141 A house built on a riverside is to be cooled in summer by utilizing the cool water of the river, which flows at an average temperature of 15°C. A 15-m-long section of a circular duct of 20 cm diameter passes through the water. Air enters the underwater section of the duct at 25°C at a velocity of 3 m/s. Assuming the surface of the duct to be at the temperature of the water, determine the outlet temperature of air as it leaves the underwater portion of the duct. Also, for an overall fan efficiency of 55 percent, determine the fan power input needed to overcome the flow resistance in this section of the duct.

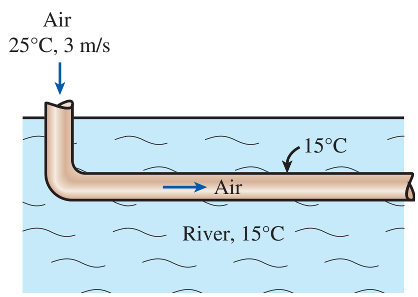

FIGURE P8–141

8–142 Repeat Prob. 8–141 assuming that a 0.25-mm-thick layer of mineral deposit (k = 3 W/m·K) formed on the inner surface of the pipe.

8–143 Oil at 15°C is to be heated by saturated steam at 1 atm in a double-pipe heat exchanger to a temperature of 25°C. The inner and outer diameters of the annular space are 3 cm and 5 cm, respectively, and oil enters with a mean velocity of 0.8 m/s. The inner tube may be assumed to be isothermal at 100°C, and the outer tube is well insulated. Assuming fully developed flow for oil, determine the tube length required to heat the oil to the indicated temperature. In reality, will you need a shorter or longer tube? Explain.

8–144E A concentric annulus tube has inner and outer diameters of 1 in and 4 in, respectively. Liquid water flows at a mass flow rate of 396 lbm/h through the annulus with the inlet and outlet mean temperatures of 68°F and 172°F, respectively. The inner tube wall is maintained with a constant surface temperature of 250°F, while the outer tube surface is insulated. Determine the length of the concentric annulus tube. Assume flow is fully developed.

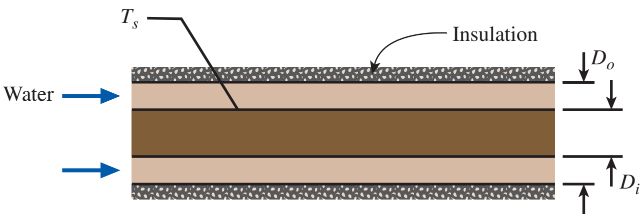

FIGURE P8–144E

Fundamentals of Engineering (FE) Exam Problems

8–145 Internal force flows are said to be fully developed once the ______ at a cross section no longer changes in the direction of flow.

(*a*) temperature distribution
(*b*) entropy distribution
(*c*) velocity distribution
(*d*) pressure distribution
(*e*) none of the above

8–146 The bulk or mixed temperature of a fluid flowing through a pipe or duct is defined as

(*a*) $T_b = \dfrac{1}{A_c}\displaystyle\int_{A_c} T\,dA_c$

(*b*) $T_b = \dfrac{1}{\dot m}\displaystyle\int_{A_c} T\rho V\,dA_c$

(*c*) $T_b = \dfrac{1}{\dot m}\displaystyle\int_{A_c} h\rho V\,dA_c$

(*d*) $T_b = \dfrac{1}{A_c}\displaystyle\int_{A_c} h\,dA_c$

(*e*) $T_b = \dfrac{1}{\dot V}\displaystyle\int_{A_c} T\rho V\,dA_c$

8–147 Water ($\mu = 9.0 \times 10^{-4}$ kg/m·s, $\rho = 1000$ kg/m^3) enters a 4-cm-diameter and 3-m-long tube whose walls are maintained at 100°C. The water enters this tube with a bulk temperature of 25°C and a volume flow rate of 3 m^3/h. The Reynolds number for this internal flow is

(*a*) 29,500 (*b*) 38,200 (*c*) 72,500
(*d*) 118,100 (*e*) 122,9000

8–148 Water enters a circular tube whose walls are maintained at constant temperature at a specified flow rate and temperature. For fully developed turbulent flow, the Nusselt number can be determined from Nu = 0.023 Re$^{0.8}$Pr$^{0.4}$. The correct temperature difference to use in Newton's law of cooling in this case is

(*a*) The difference between the inlet and outlet water bulk temperatures.
(*b*) The difference between the inlet water bulk temperature and the tube wall temperature.
(*c*) The log mean temperature difference.
(*d*) The difference between the average water bulk temperature and the tube temperature.
(*e*) None of the above.

8–149 Air ($c_p = 1007$ J/kg·K) enters a 17-cm-diameter and 4-m-long tube at 65°C at a rate of 0.08 kg/s and leaves at 15°C. The tube is observed to be nearly isothermal at 5°C. The average convection heat transfer coefficient is

(*a*) 24.5 W/m^2·K (*b*) 46.2 W/m^2·K
(*c*) 53.9 W/m^2·K (*d*) 67.6 W/m^2·K
(*e*) 90.7 W/m^2·K

8–150 Water ($c_p = 4180$ J/kg·K) enters a 12-cm-diameter and 8.5-m-long tube at 75°C at a rate of 0.35 kg/s and is cooled by a refrigerant evaporating outside at −10°C. If the average heat transfer coefficient on the inner surface is 500 W/m^2·K, the exit temperature of the water is

(*a*) 18.4°C (*b*) 25.0°C (*c*) 33.8°C
(*d*) 46.5°C (*e*) 60.2°C

8–151 Air enters a duct at 20°C at a rate of 0.08 m^3/s, and it is heated to 150°C by steam condensing outside at 200°C. The error in calculating the rate of heat transfer to the air caused by using the arithmetic mean temperature difference instead of the logarithmic mean temperature difference is

(*a*) 0% (*b*) 5.4% (*c*) 8.1%
(*d*) 10.6% (*e*) 13.3%

8–152 Engine oil at 60°C ($\mu = 0.07399$ kg/m·s, $\rho = 864$ kg/m^3) flows in a 5-cm-diameter tube with a velocity of 2.2 m/s. The pressure drop along a fully developed 6-m-long section of the tube is

(*a*) 5.4 kPa (*b*) 8.1 kPa (*c*) 12.5 kPa
(*d*) 18.4 kPa (*e*) 20.8 kPa

8–153 Engine oil flows in a 15-cm-diameter horizontal tube with a velocity of 1.3 m/s, experiencing a pressure drop of 12 kPa. The pumping power requirement to overcome this pressure drop is

(*a*) 190 W (*b*) 276 W (*c*) 407 W
(*d*) 655 W (*e*) 900 W

8–154 Water enters a 5-mm-diameter and 13-m-long tube at 15°C with a velocity of 0.3 m/s and leaves at 45°C. The tube is subjected to a uniform heat flux of 2000 W/m^2 on its surface. The temperature of the tube surface at the exit is

(*a*) 48.7°C (*b*) 49.4°C (*c*) 51.1°C
(*d*) 53.7°C (*e*) 55.2°C

(For water, use $k = 0.615$ W/m·K, $\text{Pr} = 5.42$, $\nu = 0.801 \times 10^{-6}$ m^2/s.)

8–155 Water enters a 5-mm-diameter and 13-m-long tube at 45°C with a velocity of 0.3 m/s. The tube is maintained at a constant temperature of 8°C. The exit temperature of the water is

(*a*) 4.4°C (*b*) 8.9°C (*c*) 10.6°C
(*d*) 12.0°C (*e*) 14.1°C

(For water, use $k = 0.607$ W/m·K, $\text{Pr} = 6.14$, $\nu = 0.894 \times 10^{-6}$ m^2/s, $c_p = 4180$ J/kg·K, $\rho = 997$ kg/m^3.)

8–156 Water enters a 5-mm-diameter and 13-m-long tube at 45°C with a velocity of 0.3 m/s. The tube is maintained at a constant temperature of 5°C. The required length of the tube in order for the water to exit the tube at 25°C is

(*a*) 1.55 m (*b*) 1.72 m (*c*) 1.99 m
(*d*) 2.37 m (*e*) 2.96 m

(For water, use $k = 0.623$ W/m·K, $\text{Pr} = 4.83$, $\nu = 0.724 \times 10^{-6}$ m^2/s, $c_p = 4178$ J/kg·K, $\rho = 994$ kg/m^3.)

8–157 Air enters a 7-cm-diameter and 4-m-long tube at 65°C and leaves at 15°C. The tube is observed to be nearly isothermal at 5°C. If the average convection heat transfer coefficient is 20 W/m^2·°C, the rate of heat transfer from the air is

(*a*) 491 W (*b*) 616 W (*c*) 810 W
(*d*) 907 W (*e*) 975 W

8–158 Air ($c_p = 1000$ J/kg·K) enters a 16-cm-diameter and 19-m-long underwater duct at 50°C and 1 atm at an average velocity of 7 m/s and is cooled by the water outside. If the average heat transfer coefficient is 35 W/m^2·K and the tube temperature is nearly equal to the water temperature of 5°C, the exit temperature of the air is

(*a*) 6°C (*b*) 10°C (*c*) 18°C
(*d*) 25°C (*e*) 36°C

8–159 Water enters a 2-cm-diameter and 3-m-long tube whose walls are maintained at 100°C with a bulk temperature of 25°C and a volume flow rate of 3 m^3/h. Neglecting the entrance effects and assuming turbulent flow, the Nusselt number can be determined from $\text{Nu} = 0.023\ \text{Re}^{0.8}\ \text{Pr}^{0.4}$. The convection heat transfer coefficient in this case is

(*a*) 4140 W/m^2·K (*b*) 6160 W/m^2·K
(*c*) 8180 W/m^2·K (*d*) 9410 W/m^2·K
(*e*) 2870 W/m^2·K

(For water, use $k = 0.610$ W/m·K, $\text{Pr} = 6.0$, $\mu = 9.0 \times 10^{-4}$ kg/m·s, $\rho = 1000$ kg/m^3.)

8–160 Air at 110°C enters an 18-cm-diameter and 9-m-long duct at a velocity of 4.5 m/s. The duct is observed to be nearly isothermal at 85°C. The rate of heat loss from the air in the duct is

(*a*) 760 W (*b*) 890 W (*c*) 1210 W
(*d*) 1370 W (*e*) 1400 W

(For air, use $k = 0.03095$ W/m·K, $\text{Pr} = 0.7111$, $\nu = 2.306 \times 10^{-5}$ m^2/s, $c_p = 1009$ J/kg·K.)

8–161 Air at 10°C enters an 18-m-long rectangular duct of cross section 0.15 m × 0.20 m at a velocity of 4.5 m/s. The duct is subjected to uniform radiation heating throughout the surface at a rate of 400 W/m^2. The wall temperature at the exit of the duct is

(*a*) 58.8°C (*b*) 61.9°C (*c*) 64.6°C
(*d*) 69.1°C (*e*) 75.5°C

(For air, use $k = 0.02551$ W/m·K, $\text{Pr} = 0.7296$, $\nu = 1.562 \times 10^{-5}$ m^2/s, $c_p = 1007$ J/kg·K, $\rho = 1.184$ kg/m^3.)

Design and Essay Problems

8–162 Electronic boxes such as computers are commonly cooled by a fan. Write an essay on forced air cooling of electronic boxes and on the selection of the fan for electronic devices.

8–163 Design a heat exchanger to pasteurize milk with steam in a dairy plant. Milk is to flow through a bank of 1.2-cm-internal-diameter tubes while steam condenses outside the tubes at 1 atm. Milk is to enter the tubes at 4°C, and it is to be heated to 72°C at a rate of 15 L/s. Making reasonable assumptions, you are to specify the tube length and the number of tubes and the pump for the heat exchanger.

8–164 A desktop computer is to be cooled by a fan. The electronic components of the computer consume 80 W of power under full-load conditions. The computer is to operate in environments at temperatures up to 50°C and at elevations up to 3000 m where the atmospheric pressure is 70.12 kPa. The exit temperature of air is not to exceed 60°C to meet the reliability requirements. Also, the average velocity of air is not to exceed 120 m/min at the exit of the computer case where the fan is installed; this is to keep the noise level down. Specify the flow rate of the fan that needs to be installed and the diameter of the casing of the fan.